Designing Return Policy in E-Commerce: The Role of Excess Return and Surplus Extraction

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Abstract

Return shipping insurance encourages consumers to purchase before inspecting products, but it can also raise costly returns. Because insurance shifts both purchase rates and the share of purchases that are kept, it feeds into seller pricing, and high prices associated with stricter refund rules can exclude socially valuable buyers. We study a mid-sized women's clothing retailer on Taobao that repeatedly switches between retailers' return shipping insurance (RI), provided automatically, and consumers' return shipping insurance (CI), purchased optionally. Within products, RI is associated with lower prices and higher return rates. We estimate a structural model of purchase, insurance, and return decisions under monopoly pricing across six apparel categories and evaluate return-penalty discount (RPD), which pairs a discounted price with stricter refund terms. Even when returns are made efficient, monopoly pricing leaves about 25%–30% of first-best welfare unrealized because too few consumers purchase. In baseline counterfactuals, RI attains the highest welfare for dresses, where valuable additional purchases outweigh additional returns, whereas CI dominates RI in thinner-margin categories. Full-forfeiture RPD, under which returning buyers forfeit the discounted payment, reduces returns but excludes valuable buyers. It nonetheless outperforms both insurance policies outside dresses. Partial-forfeiture RPD with restocking fees of 10%–15% improves this tradeoff and, for dresses, restores much of what full forfeiture loses, though it remains below RI. When policy cannot be tailored to category margins and return losses, partial-forfeiture RPD is a practical default.

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1 Introduction

Product returns are a central challenge in online retail. Return rates have reached 19.3% in the United States and exceed 35% in China.[1] In monetary terms, total retail returns in the United States are projected to reach $849.9 billion in 2025.[2] The problem is especially acute for apparel, whose fit and quality are difficult to assess online. Industry benchmarks place online apparel return rates in the United States between 20% and 40%.[3] These high rates arise in part because consumers must purchase before inspecting the product. The uncertainty is resolved only after delivery, when consumers learn their realized match value and decide whether or not to keep the product. Under a typical return policy, consumers are refunded the product price but bear the cost of return shipping, while the seller incurs handling costs and recover only part of the product's value. Return policy shapes both the willingness to purchase uncertain products and the return decision after purchase.

Making returns less costly creates a basic trade-off. It can encourage valuable purchases that would not occur if consumers feared paying for return shipping. On the other hand, it can induce returns with negative social value, i.e., returns whose social cost exceeds their consumer benefit. Return Shipping Insurance (RSI), which is commonly used on Chinese e-commerce platforms, makes this trade-off especially clear by reimbursing the buyer's return-shipping expense. RSI comes in two forms: coverage can be automatic for all buyers, or it can be purchased as an add-on. Automatic coverage extends the subsidy to every buyer. Optional coverage limits the subsidy to those who buy insurance, but it draws in buyers who already expect to return. Because profit depends on whether the product is ultimately kept or returned, insurance policies also affect the pricing incentives of the sellers. Their welfare consequences, therefore, depend on product margins and return costs.

These observations lead to three research questions: (i) How do automatic and optional return-shipping insurance affect purchases, insurance selection, returns, and seller pricing? (ii) When is automatic coverage superior to optional insurance, and how does this ranking vary across apparel categories? (iii) Can alternative policies that better balance purchases and returns create greater social value than the existing insurance policies?

We formally refer to automatic coverage as *Retailers' Return Shipping Insurance (RI)* and optional coverage as *Consumers' Return Shipping Insurance (CI)*. In our model, a monopolistic seller sets prices, anticipating consumers' purchase and return decisions. Consumers know their expected value of a product when deciding whether to buy and learn their final match value after delivery, when they decide whether to keep or return. Under either policy, consumers receive a full refund on the price of the product for returns, and insurance reimburses return-shipping fees. The seller pays production cost on every sale, earns the price

[1] U.S. figure: National Retail Federation (2025), "2025 Retail Returns Landscape," `https://nrf.com/research/2025-retail-returns-landscape` (accessed July 28, 2026). China figure: Wu Dan Li (2024), Zhejiang University School of Management, `http://www.som.zju.edu.cn/2024/1029/c63655a2981194/page.htm` (accessed January 6, 2025).

[2] National Retail Federation (2025), "2025 Retail Returns Landscape," `https://nrf.com/research/2025-retail-returns-landscape` (accessed July 28, 2026).

[3] Eightx (2026), "Average eCommerce Return Rate by Category," `https://eightx.co/blog/average-ecommerce-return-rate` (accessed June 24, 2026); McKinsey & Company (2020), "Returning to order: Improving returns management for apparel companies," `https://www.mckinsey.com/industries/retail/our-insights/returning-to-order-improving-returns-management-for-apparel-companies` (accessed June 24, 2026).

when the product is kept, and bears net return-handling costs when it is returned.

Our model generates direct behavioral effects and an indirect effect through the equilibrium price response. At a given product price, RI raises willingness to purchase more broadly because every buyer receives coverage without paying a separate premium, but it also lowers the return cost of every purchaser. Under CI, the premium limits coverage and participation, so uninsured purchasers face stronger return discipline, but consumers who expect to return are especially likely to buy insurance. Equilibrium prices need not follow the direct demand effect because three forces interact. The seller-paid RI premium creates cost pass-through, the number of purchases ultimately kept affects the profitability of raising the price, and the price itself changes the total volume of returns through who purchases and whether purchases are returned. One important force is *surplus extraction*: when return rates are low, a price increase becomes more profitable. This force need not imply a higher observed price because pass-through and return mitigation may work in the opposite direction.

To assess these effects jointly, we consider the welfare consequences along both ex-ante and ex-post margins. *Ex-ante efficiency* concerns whether purchases create positive expected social surplus. Broader participation under RI can correct for monopoly underprovision when marginal buyers are socially valuable, but it can also create overprovision by attracting frequent returners whose purchases impose more cost than value. *Ex-post efficiency* concerns whether a purchased product is returned only when returns create social value after accounting for shipping, hassle, handling, and recovery value. CI directly strengthens return discipline among uninsured buyers, although its aggregate ex-post advantage also depends on equilibrium prices and insurance take-up. Return policy must therefore preserve valuable participation while discouraging both inefficient purchases and excessive returns. RI tends to perform better when markups are large and marginal buyers remain socially valuable despite return risk, while CI tends to perform better when markups are thin, return-handling costs are high, and screening marginal purchases and returns is more valuable.

Despite these differences, both RI and CI share design limitations on both the ex-ante and ex-post efficiency margins. In addition to aggravating ex-post over-return among insured buyers, subsidizing returns can attract consumers who are especially likely to return, worsening the composition of purchases. Worse, under CI, standardized insurance premiums are set separately from the seller's product price and need not reflect product-level margins, return-handling costs, or recovery values.

The third research question therefore considers a counterfactual price–refund menu that we call *return-penalty discount (RPD)*. Instead of subsidizing return shipping while keeping the product price and refund rule unchanged, RPD preserves a full-price option with full refund rights and adds a discounted option with a stricter refund. The platform sets the return penalty, while the seller chooses the list price and discount using product-level information on margins and return costs. Consumers who expect to keep the product may select the discount, improving purchaser composition and return discipline. Nevertheless, RPD is not necessarily more efficient. Stricter refund terms can strengthen surplus extraction and exclude socially valuable buyers. If the penalty is too severe, they can also deter returns that would be socially efficient. We evaluate both full forfeiture of the refund and partial forfeiture such as restocking fees. Under monopoly, preserving valuable participation, discouraging inefficient purchases, and correcting return behavior remain in tension, so which force dominates is an empirical question.

We study these tradeoffs with transaction data from a mid-sized women's clothing retailer on Taobao that repeatedly switched between RI and CI during 2017–2023. Within-retailer switches provide policy variation that is not confounded by retailer identity. For products sold under both policies, RI is associated with lower prices and higher return rates. Under CI, insured transactions return much more often than uninsured ones, consistent with selection into coverage and a behavioral response to reimbursement. Price comparisons vary with product composition and aggregation, which motivates a framework in which prices, purchases, insurance, and returns adjust jointly.

We estimate a structural model of these decisions under monopoly pricing for six apparel categories. The estimates recover category-specific purchase values, match uncertainty, and return responses, which we combine with seller costs and recovery values to evaluate equilibrium welfare. The model reproduces the main qualitative differences between the policies. It underpredicts purchases and widens the conditional RI–CI return gap relative to the data, so we use it primarily for within-category policy rankings. As a benchmark, we make returns socially efficient while retaining monopoly pricing. Even then, about 25%–30% of first-best welfare remains unrealized because prices exclude efficient purchases. Correcting for returns alone does not eliminate the participation distortion.

The RI–CI comparison shows how the value of participation relative to return losses determines rankings. For dresses, where markups are high and scrap recovery constitutes a large share of the retail price, RI's broader participation includes socially valuable marginal buyers excluded under CI and outweighs its extra return distortion, raising welfare by about 2–3 percentage points of first-best. In other categories, where markups are thinner and scrap recovery covers only a modest share of price, return losses absorb more of the surplus from a purchase that is kept. In these categories, CI excludes some efficient purchases but improves composition and return discipline enough to outperform RI by about 15–30 percentage points of first-best welfare. The starkest contrast is for tops: roughly 80% of first-best welfare is lost under RI through severe ex-post distortion and poor ex-ante screening, while CI attains about 50 percentage points more of first-best welfare.

Full-forfeiture RPD shows why stronger return discipline is not enough by itself. In most categories, purchases concentrate on the discounted branch, full-price demand is negligible, and return probability on the discount branch is near zero. The policy therefore replaces over-return under RSI with under-return. Some consumers keep products that would be socially efficient to return. Even so, it substantially reduces ex-post distortion. It also strengthens surplus extraction and raises the purchase cutoff, excluding marginal consumers, even though the average price paid is lower. The reduction in return distortion dominates this participation loss in every category except dresses: outside the dress category, full-forfeiture RPD outperforms RI by more than 15 percentage points of first-best welfare and slightly outperforms CI by up to about 7 percentage points, whereas for dresses it falls short of both RSI policies by roughly 17–20 percentage points because many excluded buyers would have created positive social surplus.

Partial-forfeiture RPD softens both failures of full forfeiture. Restocking fees of 10%–15% of the discounted price preserve much of the return discipline while reducing under-return and restoring participation. These fees outperform both RSI policies outside the dress category, attaining about 75–78% of first-best welfare. For dresses, partial forfeiture restores much of what full forfeiture loses—raising welfare from about

57% of first-best under full forfeiture to about 69–70%—but remains below RI (about 76%). For tops, the thin-margin category, full forfeiture slightly outperforms partial-forfeiture RPD in this range (by about 1–2 percentage points of first-best welfare). Partial-forfeiture RPD's advantage outside the dress category persists when recovery varies, but RI's baseline advantage for dresses disappears when recovery is low. Resampling largely supports these patterns, though welfare distributions overlap for some intermediate categories. The evidence favors category-specific penalties. When tailoring is infeasible, moderate partial-forfeiture RPD is a practical default.

1.1 Literature Review

We organize the related literature to follow our analysis: welfare foundations, return shipping insurance, alternative policy design, extensions we hold fixed, and empirical evidence.

Canonical theory by Che (1996) identifies a basic trade-off. Generous refunds improve risk sharing for experience-good buyers, but they also let sellers screen toward high-valuation keepers and charge higher prices, so some consumers who would have kept at a lower price return instead. In vertically differentiated markets, firms further meter return propensity through refund terms, pairing strict refunds with low prices and generous refunds with high prices to extract surplus from consumers who learn match value only after purchase (Inderst and Tirosh, 2015). Su (2009) extends the analysis to supply chains, where consumer returns can distort incentives under common contracts and where full versus partial refunds have different coordination implications.

Further work explains why equilibrium return policies often exceed the socially efficient level. Courty and Hao (2000) show that a more generous refund reduces the information rent paid to high-valuation consumers, who earn rents only when they keep the product. Inderst and Ottaviani (2013) study ex-ante monopoly contracting when the seller has an informational advantage and gives pre-purchase advice before the buyer knows true match value. Credible communication then requires a refund policy that is more generous than socially optimal. This bonding mechanism flips the usual monopoly underprovision result and generates distortions on both margins: ex-ante overprovision and ex-post over-return. We extend this ex-ante and ex-post efficiency tradeoff to RI and CI and clarify when each dominates in social surplus.

Theory on return shipping insurance compares RI and CI. Li et al. (2021) show that RI complements partial refunds and tends to dominate CI when premiums are small, holding return probability fixed so insurance mainly shifts who purchases. Li et al. (2023) incorporate anticipated regret and informational heterogeneity to study profit-maximizing RSI policies. Under RI, automatic coverage raises prices through cost passthrough and demand expansion. Under CI, informed buyers skip optional coverage and uninformed buyers buy when return shipping is costly relative to a cheap premium. The separate purchase lowers willingness to pay and leads the retailer to discount the base product. Self-selection by information separates the two groups, sustains demand from both, and can yield a win-win-win for the retailer, insurer, and consumers. We instead let reimbursed shipping lower each buyer's private return cost, separating adverse selection from ex-post over-return without fixed return types or regret frictions.

Beyond automatic RI and optional CI, Najafi et al. (2024) analyze a forfeiture menu closely related to our return-penalty discount (RPD). The menu raises profit only when return-cost savings from screening outweigh the markdown, and fails when screening is too weak or the markdown mainly cannibalizes margin on likely keepers. We use an estimated model to evaluate how strict return discipline should be when the objective is social surplus rather than profit alone.

We abstract from three extensions held fixed in the baseline. Related work maps return costs into consumer search: prepurchase search can make returns complement active search or substitute for costly pre-search, and in price-directed sequential search higher return costs intensify duopoly price competition and raise consumer surplus (Jerath and Ren, 2024; Janssen and Williams, 2024; Petrikaitė, 2018). Under asymmetric information, generous return protection is more costly for low-quality sellers to mimic (Moorthy and Srinivasan, 1995; Zhang et al., 2022). Competition in return policy can favor high-return sellers under money-back guarantees and intensify price rivalry when return rights are mandated (McWilliams, 2012; Chesnokova, 2007). We isolate match friction, seller pricing, and reimbursed returns.

The empirical literature documents how return leniency shifts demand, spending, and realized returns. Anderson et al. (2009) structurally estimate joint purchase and return decisions and show that the return option raises demand with category-specific effects. A meta-analysis of 21 studies finds that leniency raises purchases more than returns (Janakiraman et al., 2016). Sahoo et al. (2018) link review environments to returns using theory-motivated transaction-level regressions. Return fees, free-return adoption, free-shipping promotions, and customer-level returns have been tied to spending, orders, and profitability in field and quasi-experimental work (Bower and Maxham, 2012; Patel et al., 2021; Shehu et al., 2020; Petersen and Kumar, 2009, 2015). Browsing, deliberation, and customer segments also predict return likelihood and profitability (Ibragimov et al., 2024; Bechwati and Siegal, 2005; Abbey et al., 2018). These studies establish behavioral and profit effects. Evaluating social surplus requires equilibrium objects that shift jointly when policy changes: participation, prices, return behavior, and seller-side costs. Few studies isolate return-policy changes in settings clean enough to study social welfare empirically, as return data are scarce and store and product assortment differences often confound policy comparisons.

The only such study we can identify is the recent working paper of Newberry and Zhou (2024), who are the first to estimate a structural demand and supply model with the option to return. Using Tmall tablet sellers, they show that leniency mainly provides insurance rather than signaling and can lower consumer surplus once sellers reprice. In their paper, return policy is summarized by a scalar leniency index that shifts a common return friction for all consumers. Our within-store RI/CI switches identify a more general policy that can differ across consumers and that shifts both keep and return utilities. Moreover, their focus is consumer surplus and firm profit rather than the social-surplus tradeoff between screening and ex-post return behavior. Relative to their single-category setting, comparing apparel subcategories also contrasts the inefficiency-loss channels across products.

Our contribution relative to prior work is threefold. First, we estimate a unified welfare framework for alternative return policies, using within-store switches between RI and CI that hold retailer identity fixed while allowing prices and return decisions to adjust. Second, we decompose ex-ante and ex-post inefficiency losses to clarify where distortions arise and why the welfare ranking of RI and CI varies with category-level margins and return frictions. Third, we propose return-penalty discount (RPD) as a platform policy option that can improve social welfare relative to existing schemes.

The remainder of the paper proceeds as follows. Section 2 describes Taobao RSI institutions and repeated within-store switches between RI and CI. Section 3 presents the data and stylized facts on prices and return rates. Section 4 sets up the structural model, and Section 5 characterizes equilibrium under RI, CI, and return-penalty discount (RPD). Sections 6 and 7 discuss identification, estimation, and calibration. Section 8 reports counterfactual welfare simulations by category. Section 9 concludes.

2 Institutional Background

2.1 E-Commerce in China

China has the world's largest e-commerce market. In 2023, online retail sales reached approximately 15.4 trillion yuan, accounting for 27.6% of total retail sales[4]. Clothing and textiles, the apparel category studied here, account for the largest share of online retail, at 22%, followed by daily necessities (14.5%) and household appliances (10.6%). Taobao, operated by Alibaba Group, is the dominant platform, with over 645 million monthly active users as of September 2024[5].

2.2 Return Shipping Insurance

Before 2010, return shipping cost allocation on Taobao was ambiguous: consumers typically bore costs for preference-based returns, while sellers covered costs arising from product defects. Disputes were frequent and costly to resolve. Taobao and Huatai Insurance introduced Return Shipping Insurance (RSI) that year to standardize return-shipping reimbursement and lower these costs. The RSI market reached approximately 8 billion yuan in 2022, up 66% from 4.96 billion yuan in 2019[6]. ZhongAn, China's largest online insurer, reported issuing around 15 billion RSI policies in 2019, representing over 50% of the internet non-auto insurance market[7].

RSI takes two forms. Under *Retailers' Return Shipping Insurance (RI)*, the seller purchases insurance on behalf of all buyers. Under *Consumers' Return Shipping Insurance (CI)*, individual consumers purchase coverage at checkout. RI dominates, accounting for over 90% of all RSI policies. Taobao platform policies reinforce this concentration through preferential visibility, traffic allocation, and access to promotional events for RI-enrolled products. The insurance market on Taobao is centralized: the platform partners with a small set of insurers (Ping An, CPIC, ZhongAn Online) and assigns providers algorithmically. Neither consumers nor sellers choose their insurer, so coverage terms are uniform across providers.

[4] Ministry of Commerce (2024), *2023 China Online Retail Market Development Report*, `https://cif.mofcom.gov.cn/cif/html/upload/20240313102933492_2023%E5%B9%B4%E4%B8%AD%E5%9B%BD%E7%BD%91%E7%BB%9C%E9%9B%B6%E5%94%AE%E5%B8%82%E5%9C%BA%E5%8F%91%E5%B1%95%E6%8A%A5%E5%91%8A.pdf` (accessed July 3, 2026).

[5] iiMedia Research (2024), Statista, monthly active users of leading shopping apps in China, September 2024, `https://www.statista.com/statistics/1044676/china-leading-shopping-apps-monthly-active-users/` (accessed July 3, 2026).

[6] Shanghai Securities News (2022), Sina Finance, `https://finance.sina.com.cn/roll/2022-09-27/doc-imqmmtha8965373.shtml` (accessed July 3, 2026).

[7] ZhongAn Online P&C Insurance Co. Ltd., "Introduction to Return Shipping Insurance," `https://en.airc.org.tw/re/seminar/2009161495602.pdf` (accessed July 3, 2026).

2.3 RSI Contract Structure

Coverage applies on a per-order basis. When consumers purchase multiple items from the same store, a single insurance record covers the entire order. On approved returns, buyers receive a full refund of the product price. RSI separately reimburses return shipping up to the coverage limit. Coverage activates upon shipment confirmation and extends for up to 90 days (15 days for clothing and certain categories).[8] The coverage amount equals the delivery fee for a 1 kg package, which depends on the distance between seller and buyer, ranging from 5 to 25 RMB.

Under RI, sellers make enrollment decisions at the store level, applying uniformly to all products. Insurers set the premium from the seller's historical return rate over the past three months and the category-average return rate, typically ranging from 1 to 5 RMB.[9] CI premiums typically exceed these RI premiums because adverse selection into the insurance market concentrates higher-return-risk buyers among those who purchase coverage.

Automatic RI and optional CI do not coexist within the same store: consumers are offered a CI add-on at checkout *if and only if* the seller has not enrolled the store in RI. When the seller enrolls in RI, coverage is automatic for all buyers and a separate CI purchase is not offered for that store. The CI premium is consumer-specific, determined mainly by the buyer's province of residence and return history, independent of the product. Buyers with higher historical return rates face higher CI premiums, consistent with selection by return propensity. Store-level RI enrollment and mutual exclusivity between the two policies generate within-store time-series variation in return policy. Section 3 describes the transaction data and repeated within-store switches that exploit this variation.

3 Data and Stylized Facts

Our dataset comprises transaction records from a single mid-sized women's clothing retailer on Taobao, spanning from 2017 to 2023. Each transaction lies in one RSI policy. CI is observed only when the store is not on RI. Over this period, total revenue is approximately 62 million yuan.

The data include sales, returns, and insurance at the transaction level, with consumer information limited to addresses. Insurance premium and coverage are observed for nearly all RI transactions, excluding international orders and buyers ineligible because of high historical return rates. Under CI, coverage is recorded only when the consumer buys insurance; for non-purchasers we impute it from same-province addresses. Premiums are not recorded, so we impute the premium from coverage using the 2025 premium–coverage schedule (Appendix A). The cleaned analysis sample contains 362,071 transactions (238,906 under RI and 123,165 under CI).

We code a return as a binary indicator equal to one when the buyer receives a strictly positive refund and zero otherwise, excluding partial refunds for minor product defects. We keep single-item, single-quantity orders with confirmed shipment and no premium shipping, and drop non-product listings and promotion days on which the store temporarily switched RSI policy. Restricting to one item per order rules out

[8] The 90-day coverage window refers to insurance reimbursement eligibility, not to the return policy itself; these are separate.

[9] We treat premiums as exogenous and do not model feedback from seller pricing into future premiums.

complementarities between purchase and return decisions across products, while restricting to quantity one rules out bracketing, in which buyers order multiple sizes or colors of the same listing and return all but one.

Figure 1 plots 30-day moving averages of the daily return rate from 2017 through 2023, with blue and red shading marking RI and CI periods. These shaded intervals show the store's repeated switches between RSI policies over the six years. The series drifts up over the sample and exceeds 40% toward the end of 2023. The sharp rise at the end of the window plausibly coincides with rapid growth in *live-stream* commerce on Taobao, where purchases are typically more impulsive and hence more return-prone.[10]

Figure 1: 30-day moving average of the daily return rate (%), 2017–2023, with RI and CI periods shaded.

Structural estimation focuses on six categories: dress, tops, pants, sweater, two-piece, and outerwear. Aggregation and market construction are in Appendix A. Table 1 reports prices, premiums, coverage, and return rates by policy, pooling across those categories.

Average list prices are higher under CI than under RI (about 92 versus about 87 RMB). Traditional premium pass-through would predict the opposite: the seller-paid RI premium is a visible cost that could raise list prices under RI. We do not observe higher RI prices, which can suggest that other pricing forces are at work. One such force could be surplus extraction. Under CI, uninsured buyers face higher private return costs and return less often, which can strengthen the seller's incentive to set higher prices.

Average return rates are higher under RI than under CI, with pooled means of about 27% versus about 21%. Under CI, insurance is purchased on 2.91% of transactions. Return rates are much higher among insured than uninsured transactions (46.3% versus 20.4%). The gap is consistent with ex-post distortion from lower private return costs among insured buyers and adverse selection into coverage, where buyers with higher return propensities are more likely to insure.

Pooled comparisons may be confounded by product selection. Figure 2 restricts to 1,140 products that sell under both RI and CI and have more than five transactions per product. Panel (a) plots the normalized

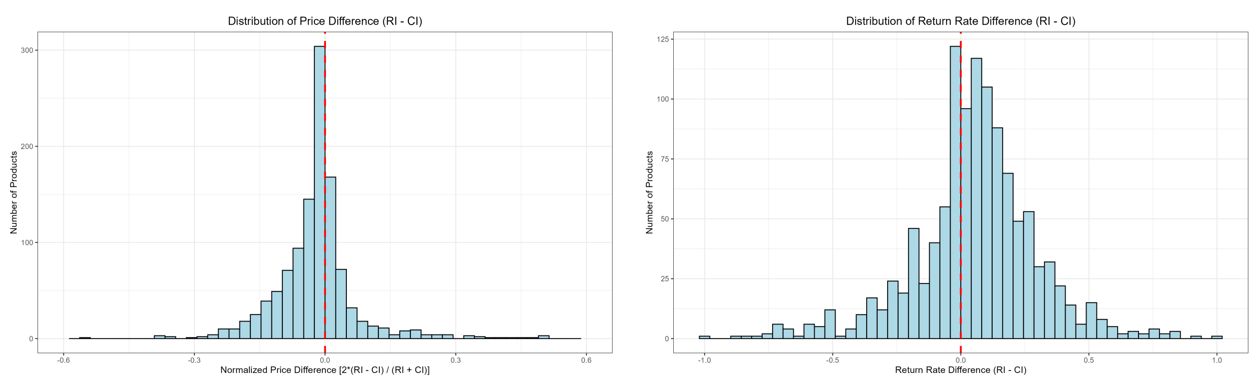
[10] Removing a pooled annual cycle (day-of-year harmonics 1–4) prior to the same 30-day moving average yields a broadly similar pattern, including the pronounced late-sample increase, and still shows no clear ranking of return rates between RI and CI periods.

Table 1: Summary Statistics

Variable	Unit	N	Mean	Sd	Min	Median	Max
CI							
Price	RMB	123,165	91.56	33.22	18.05	84.30	244.70
Premium	RMB	123,165	5.01	0.22	4.60	5.10	7.10
Coverage	RMB	123,165	10.62	1.46	8.00	11.00	25.00
Return Rate	%	123,165	21.14	40.83	0.00	0.00	100.00
RI							
Price	RMB	238,906	87.39	49.14	18.00	79.00	407.81
Premium	RMB	238,906	2.67	0.39	1.40	2.50	3.30
Coverage	RMB	238,906	10.59	1.45	8.00	11.00	25.00
Return Rate	%	238,906	27.31	44.55	0.00	0.00	100.00

Notes: Return-rate medians are zero because the outcome is binary at the transaction level.

price difference $2(\bar{p}^{\mathrm{RI}} - \bar{p}^{\mathrm{CI}})/(\bar{p}^{\mathrm{RI}} + \bar{p}^{\mathrm{CI}})$, where $\bar{p}$ is the product-level mean transaction price under each policy. As in the pooled sample, a larger share of products still show higher prices under CI than under RI. Panel (b) reports $\bar{r}^{\mathrm{RI}} - \bar{r}^{\mathrm{CI}}$, where $\bar{r}$ is the product-level mean return rate under each policy. Most mass lies to the right of zero, indicating higher return rates under RI.



(a) Normalized price difference. (b) Return-rate difference.

Figure 2: Within-product price and return-rate differences (RI − CI).

Taken together, the time series documents repeated within-store policy switches. Pooled and within-product comparisons show that a larger share of products have higher prices under CI than under RI, and that return rates are higher under RI once products are held fixed. The next section develops the structural model used to estimate RSI effects.

4 Model

We consider a retail shopping model with Return Shipping Insurance (RSI) in which a monopolistic seller offers a single apparel category in period t. Market size N_t varies across categories and seasons (Appendix A). Categories are separable: each of N_t potential buyers decides whether to purchase and, conditional on purchase, whether to return the product, with no cross-category spillovers. The return-shipping insurance policy (RI or CI) is exogenous.

The seller sets retail price p_t and produces each unit at marginal cost $\theta_t^c = X_t \alpha^c + \zeta_t^c$. The store-level shock ζ_t^c is common across products and reflects shared inputs such as marketing, customer acquisition and service, labor, utilities, and rent. If a product is returned, the seller pays a handling fee, repossesses the unit, and recovers scrap value. Net return handling cost h_t equals gross handling minus scrap recovery; because resale value scales with p_t, h_t can be negative when recovery exceeds the fee. We calibrate h_t in Section 7.

The game proceeds in three stages:

1. **Pricing stage**: The seller sets p_t, taking the prevailing RSI policy as given.

2. **Purchase stage**: Under RI, consumers decide whether to buy with automatic coverage. Under CI, each consumer chooses among no purchase, purchase without insurance, and purchase with optional return-shipping insurance.

3. **Return stage**: Ex-post preference shocks are realized and consumers decide whether to keep or return; returned units impose net handling cost h_t on the seller.

Consumers differ along two dimensions: ex-ante valuation and return-shipping coverage. Ex-ante value for consumer i at time t is

$$\nu_{it} = \mu_t + \xi_t + \omega \epsilon_{it}, \tag{1}$$

where $\mu_t = X_t \beta^\mu$ is mean valuation from observables X_t; ξ_t is a product-and-time-specific demand shock unobserved by the econometrician but orthogonal to X_t; ω is the standard deviation of idiosyncratic ex-ante tastes within the category and is time-invariant within the category[11]; and $\epsilon_{it} \sim \mathcal{N}(0,1)$. Hence $\nu_{it} \sim \mathcal{N}(\mu_t + \xi_t, \omega^2)$; let F_ν denote the distribution of ν_{it}. Coverage k_i is the insurer's return-shipping reimbursement upon a return. It depends on the consumer's location and is drawn from a nonparametric distribution $\Psi(\cdot)$, independent of ν_{it}.[12] All other shocks have parametric distributions.

Premiums are set by insurers and are exogenous to the seller's price choice in period t.[13] Under RI, the seller pays insurer premium I_t^r per purchase. Under CI, consumers who buy coverage pay return-shipping premium $I^c(k_i)$ at checkout, indexed to location-specific coverage k_i. Returning the product imposes non-monetary hassle κ (repackaging, shipment coordination, and refund delay), which RSI does not

[11] We allow ω to differ across categories. In estimation (Section 7), $\omega_j = C_j' \beta^\omega$ for a category-indicator vector C_j; only the non-time-varying category component enters this index. Period-to-period shifts in demand are absorbed by μ_t (via observables) and by ξ_t. Individual-level buyer characteristics are not observed in our transaction data, so we do not estimate random coefficients that link preferences to demographics or similar covariates. Instead, ω measures cross-sectional dispersion of tastes around $\mu_t + \xi_t$ within the category.

[12] While k_i varies by location, which may correlate with demographic factors like income, we assume the customer base is balanced conditional on k_i.

[13] We do not endogenize insurer pricing: insurers pool return risk across many sellers while our data cover one retailer, and several major insurers set standardized premiums rather than a single monopolist, so we treat premiums as given each period. Premium menus and eligibility (including RI ineligibility for some high-return buyers) are likewise taken as platform or insurer rules. We do not model feedback from current returns into next-period premiums; CI take-up still identifies selection on return propensity given the observed menu.

reimburse. Under either RSI scheme, consumers who return receive a full refund of the product price; RSI also reimburses monetary return shipping up to k_i.[14]

After purchase, ex-post shocks $\varepsilon_{it}^{\text{keep}}$ and $\varepsilon_{it}^{\text{return}}$ are realized, capturing fit with the consumer's preferences for style, quality, and size when keeping or returning.[15] The ex-post shocks follow a Type I extreme value distribution with location $-\gamma/\sigma_t$ and scale $1/\sigma_t$, where γ is Euler's constant, ensuring zero mean. We specify ex-post precision as $\sigma_t \equiv X_t^{\sigma}\beta^{\sigma}$; higher σ_t indicates lower uncertainty about product fit.[16] Let BC_{it} equal 1 if consumer i purchases CI coverage and RSI_{it} equal 1 if $BC_{it} = 1$ or the seller provides RI. Conditional on purchase, the consumer chooses between keeping and returning. Realized utility on each branch is mean utility plus the corresponding shock, with mean utilities

$$\overline{u}_{it}^{\text{keep}} = \nu_{it} - p_t - I^c(k_i)\, BC_{it}, \tag{2}$$

$$\overline{u}_{it}^{\text{return}} = -(\kappa + k_i) + k_i\, RSI_{it} - I^c(k_i)\, BC_{it}. \tag{3}$$

These primitives pin down consumer utilities and seller costs. The next section characterizes equilibrium purchase, insurance, and return decisions, retailer pricing, and return-policy design.

5 Equilibrium Analysis

We focus on equilibrium within a single category and period, suppressing period subscripts. Superscript l denotes the RSI scheme (r for RI, c for CI). We analyze consumer behavior under each RSI scheme and then compare the mechanisms that matter for pricing and welfare.

5.1 Consumer Behavior under RI

Expected consumer surplus from purchasing under scheme l is

$$CS_i^l \equiv \mathbb{E}\{\max(\overline{u}_i^{\text{keep}} + \varepsilon_i^{\text{keep}},\ \overline{u}_i^{\text{return}} + \varepsilon_i^{\text{return}})\}, \tag{4}$$

with the expectation taken over ex-post shocks. Under RI, automatic coverage covers every purchaser. The mean keep and return utilities are therefore $\overline{u}_i^{\text{keep}} = \nu_i - p$ and $\overline{u}_i^{\text{return}} = -\kappa$. With Type I extreme value shocks, expected consumer surplus has a closed form:

$$CS_i^r = \frac{1}{\sigma} \ln(e^{\sigma(\nu_i - p)} + e^{-\sigma\kappa}). \tag{5}$$

[14] Given that our data is restricted to single-product orders where items typically weigh less than one kilogram, k_i fully covers typical return-shipping expenses when reimbursed.

[15] This formulation with separate shocks for keeping and returning is a modeling choice. An equivalent approach is to assume a single ex-post shock for keeping with a logistic distribution, setting $\varepsilon_{it}^{\text{return}} = 0$. The two formulations yield identical choice probabilities when the difference $\varepsilon_{it}^{\text{keep}} - \varepsilon_{it}^{\text{return}}$ follows a logistic distribution.

[16] The cumulative distribution function is $F(x) = \exp(-\exp(-\sigma x + \gamma))$, yielding a variance of $\pi^2/(6\sigma^2)$. We estimate σ_t in Section 7.

This expression is the discrete-choice inclusive value. A smaller σ raises expected consumer surplus ($\partial CS_i^r/\partial\sigma < 0$) because greater shock dispersion increases the option value of the ex-post keep-or-return decision.[17] With the outside option normalized to zero, consumers purchase when $CS_i^r \geq 0$, which defines the cutoff $\hat{v}^r \equiv p + \frac{1}{\sigma}\ln(1 - e^{-\sigma\kappa})$.

Conditional on purchase, consumer i keeps the product if $\overline{u}_i^{\text{keep}} + \varepsilon_i^{\text{keep}} \geq \overline{u}_i^{\text{return}} + \varepsilon_i^{\text{return}}$. The return probability is

$$q_i^r \equiv \frac{e^{-\sigma\kappa}}{e^{\sigma(v_i - p)} + e^{-\sigma\kappa}}. \tag{6}$$

This is the standard logit choice probability, with σ scaling mean indirect utilities. When σ is small, ex-post uncertainty is high and the keep-or-return decision is driven primarily by shock realizations, so q_i^r approaches 50%. As $\sigma \to \infty$, mean utilities dominate and the return decision becomes deterministic.

5.2 Consumer Behavior under CI

Under CI, optional insurance creates endogenous segmentation between insured and uninsured purchasers. Consumers simultaneously choose among not buying, purchasing the product alone, or purchasing the product with return-shipping insurance at premium $I^c(k_i)$. Expected consumer surplus for the two purchase options is:

$$CS_i^{bc} = \frac{1}{\sigma}\ln(e^{\sigma(v_i - p)} + e^{-\sigma\kappa}) - I^c(k_i) \tag{7}$$

$$CS_i^n = \frac{1}{\sigma}\ln(e^{\sigma(v_i - p)} + e^{-\sigma(\kappa + k_i)}) \tag{8}$$

with and without insurance, respectively.

The insurance value $CS_i^{bc} + I^c(k_i) - CS_i^n$ equals the integral of the return probability over unreimbursed monetary return costs $x \in [0, k_i]$,

$$CS_i^{bc} + I^c(k_i) - CS_i^n = \int_0^{k_i} \frac{e^{-\sigma(\kappa + x)}}{e^{\sigma(v_i - p)} + e^{-\sigma(\kappa + x)}}\, dx. \tag{9}$$

Because each integrand term is itself a return probability, insurance value is larger when returns are more likely and therefore decreases with v_i. Consumer i purchases insurance when this value exceeds the premium, yielding the insurance-indifference cutoff $\tilde{v}^{bc}$:

$$v_i \leq \tilde{v}^{bc} \equiv p - \kappa + \frac{1}{\sigma}\ln\left(\frac{e^{-\sigma I^c(k_i)} - e^{-\sigma k_i}}{1 - e^{-\sigma I^c(k_i)}}\right). \tag{10}$$

The cutoff $\tilde{v}^{bc}$ increases with coverage and price (through a higher return probability) and decreases with premiums and hassle cost.

Define purchase thresholds $\hat{v}^{bc}$ and $\hat{v}^n$ as the valuations at which expected consumer surplus equals zero

[17] Differentiating yields $\frac{\partial CS_i^r}{\partial\sigma} = \frac{1}{\sigma}\left[(v_i - p)(1 - q_i^r) - \kappa q_i^r - CS_i^r\right]$. The bracketed term is the probability-weighted average of mean utilities minus the inclusive value. The inclusive value exceeds this average because it includes the expected ex-post shock conditional on the optimal choice, which is positive. Hence the derivative is negative.

with and without insurance, respectively:

$$\hat{v}^{bc} \equiv p + \frac{1}{\sigma} \ln \left(e^{\sigma I^c(k_i)} - e^{-\sigma\kappa} \right), \tag{11}$$

$$\hat{v}^{n} \equiv p + \frac{1}{\sigma} \ln \left(1 - e^{-\sigma(\kappa + k_i)} \right). \tag{12}$$

Together, $\hat{v}^{bc}$, $\hat{v}^n$, and $\tilde{v}^{bc}$ characterize consumer i's optimal choice. The consumer buys only the product if $v_i \geq \max\{\tilde{v}^{bc}, \hat{v}^n\}$, buys the product with insurance if $\hat{v}^{bc} \leq v_i < \tilde{v}^{bc}$, and otherwise does not buy. An insured segment exists if and only if $\tilde{v}^{bc} \geq \hat{v}^{bc}$, equivalent to

$$e^{-\sigma\kappa}(1 - e^{-\sigma k_i}) \geq (e^{\sigma I^c(k_i)} - 1). \tag{13}$$

This holds when coverage k_i is large relative to the premium $I^c(k_i)$ and hassle cost is not so high that returns are rare regardless of coverage. When (13) fails ($\tilde{v}^{bc} < \hat{v}^{bc}$), no purchaser buys insurance because consumers who would value coverage at the margin obtain nonpositive expected consumer surplus from purchasing with insurance, so every buyer is uninsured.

Return probabilities differ by insurance status. Insured consumers ($\hat{v}^{bc} \leq v_i < \tilde{v}^{bc}$) return with probability q_i^r, as under RI, since the CI premium is sunk at purchase. Uninsured consumers return with probability

$$q_i^n \equiv \frac{e^{-\sigma(\kappa + k_i)}}{e^{\sigma(v_i - p)} + e^{-\sigma(\kappa + k_i)}}. \tag{14}$$

They bear the full monetary cost k_i and therefore return less often than insured consumers.

5.3 Retailer's Decisions

The seller prices against gross demand and returns. Let D_i^l denote the purchase indicator implied by the cutoffs above, and let q_i^l denote the corresponding return probability under scheme $l \in \{r, c\}$ conditional on purchase: q_i^r is given by (6), and q_i^c equals q_i^r for insured purchasers and q_i^n in (14) for uninsured purchasers.

Aggregating over F_v and coverage distribution Ψ, gross demand and aggregate returns are

$$D^l \equiv N\, \mathbb{E}_{v_i, k_i}\left[D_i^l\right], \tag{15}$$

$$R^l \equiv N\, \mathbb{E}_{v_i, k_i}\left[D_i^l\, q_i^l\right]. \tag{16}$$

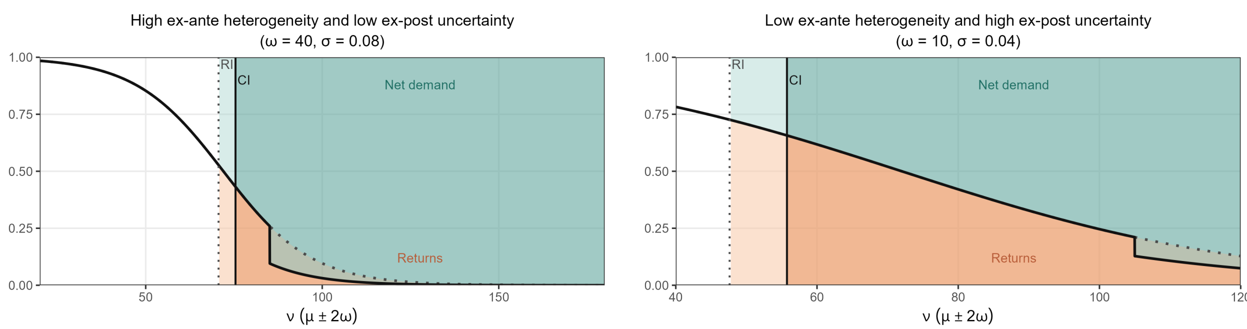
Net demand, $D^l - R^l$, is the number of units ultimately kept. Let *RI* denote an indicator equal to one under RI and zero under CI. The seller's aggregate profit is

$$\begin{aligned} \pi^l(p) = {} & (p - \theta^c - I^r RI)(D^l - R^l) \\ & - (h + \theta^c + I^r RI)R^l. \end{aligned} \tag{17}$$

The first term is the margin earned on kept units, while the second is the loss on returned units. At the same price, RI generates higher gross demand than CI because consumers face no explicit premium, but it also attracts more lower-valuation buyers and distorts return incentives for all purchasers, increasing returns. The equilibrium price therefore maximizes (17) against net demand $D^l - R^l$ and return cost, not gross demand alone. Three channels shape pricing: cost pass-through, surplus extraction through net demand, and return mitigation; these forces need not move prices in the same direction.

Cost Pass-through. Under RI, the seller pays the insurance premium I^r, which enters as a marginal cost, so a higher premium raises prices through pass-through. Under CI, the premium does not enter profit directly and affects profit only through demand and returns. Incidence among purchasers also differs. Under CI, consumers pay premiums directly, so insurance costs fall on lower-valuation buyers who return more often. Under RI, premiums are passed through to the product price and borne by all purchasers, so high-valuation keepers effectively subsidize insurance for lower-valuation frequent returners.

Surplus Extraction. Kept sales enter profit as $(p - \theta^c - I^r RI)(D^l - R^l)$, so stronger net demand supports higher equilibrium prices, which we call *surplus extraction*. At a common price, RI expands gross demand (no premium) but also raises the return rate (automatic coverage), so net demand need not rise. Figure 3 shows the comparison. In the left panel, the horizontal axis is ex-ante value v over $\mu \pm 2\omega$. The downward-sloping curves are purchaser return probabilities $q(v)$, which separate purchases into the orange return region and the teal net-demand region. The vertical lines are purchase cutoffs, with consumers buying to the right of each cutoff. Dotted lines are RI and solid lines are CI. The CI cutoff lies farther right because gross demand is lower at the same price. Under CI, $q(v)$ also drops sharply at the insurance cutoff, where private return costs jump.



(a) Large ω, large σ ($\omega = 40$, $\sigma = 0.08$).

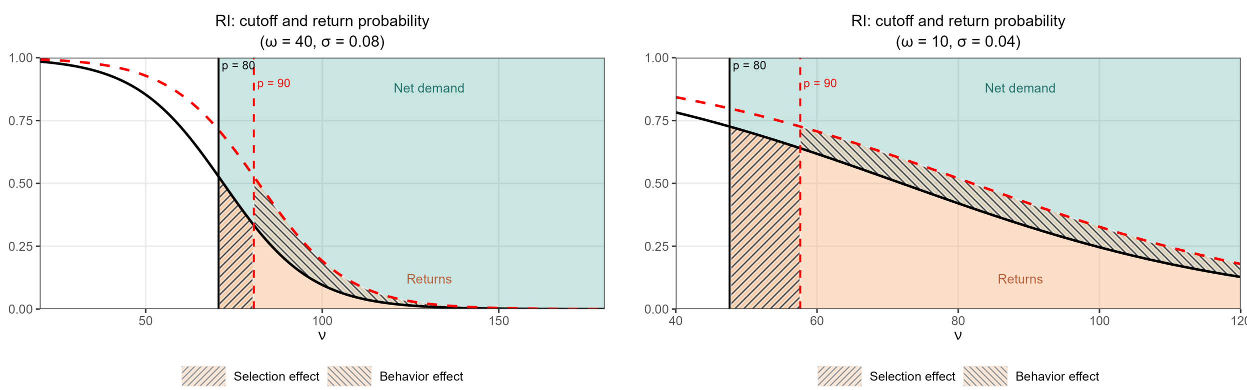
(b) Small ω, small σ ($\omega = 10$, $\sigma = 0.04$).

Figure 3: Comparison of returns and net demand under RI and CI. The horizontal axis covers $\mu \pm 2\omega$. Both panels use $\mu = 100$ and a common price.

Whether net demand is higher under RI or under CI is ambiguous and depends on the degree of ex-ante heterogeneity ω and of ex-post uncertainty (inversely related to σ). When ex-ante heterogeneity is large and ex-post uncertainty is small (large ω and large σ), as in Figure 3 (left), gross demand is relatively inelastic and return probability is sensitive to reimbursement. Low ex-post uncertainty means mean keep-or-return payoffs dominate idiosyncratic shocks, so coverage strongly shifts $q(v)$. CI then generates higher net demand because uninsured consumers have substantially lower return rates and the reduction in gross demand from requiring consumers to pay the premium is small. When instead ex-ante heterogeneity is small and ex-post uncertainty is large (small ω and small σ), as in Figure 3 (right), return probabilities respond less to coverage and the premium shifts the purchase cutoff more relative to the concentrated value distribution, so RI's broader participation can dominate and CI need not raise net demand.

An important pricing implication is that when return probabilities are highly sensitive to coverage, a smaller but better-composed purchaser pool under CI can raise net demand and give the seller an incentive to charge a higher price. The surplus extraction force should not be equated mechanically with a higher equilibrium price, because the observed price also reflects pass-through and return mitigation. Section 5.4.3 formalizes extensive-margin surplus extraction as foregone efficient sales.

Return Mitigation. A higher price affects aggregate returns through two opposing forces. It screens out lower-valuation buyers, who return more often, which tends to lower gross returns (a selection effect). Among remaining purchasers, however, a higher price raises individual return probabilities, because keeping requires a larger positive ex-post shock (a behavior effect). Because these forces pull in opposite directions, the net effect is again ambiguous. Figure 4 illustrates both channels under RI for the same (ω, σ) cases as Figure 3 (CI is similar). Solid curves and cutoffs are the baseline price; dashed red curves and cutoffs are the purchase cutoff and return probability under a higher price. Striped regions mark the selection effect (between cutoffs) and the behavior effect (the upward shift in $q(v)$ among remaining purchasers).



(a) Large ω, large σ ($\omega = 40$, $\sigma = 0.08$).

(b) Small ω, small σ ($\omega = 10$, $\sigma = 0.04$).

Figure 4: Selection and behavior effects of a price increase under RI. Parameters match Figure 3.

Which force dominates depends on the same primitives ω and σ. When ex-ante heterogeneity is large and ex-post uncertainty is small (large ω and large σ), as in Figure 4 (left), gross demand is relatively inelastic and $q(v)$ responds strongly to price. The seller prefers a lower price to curb returns, since cutting price reduces each buyer's return probability without much increase in sales. When instead ex-ante heterogeneity is small and ex-post uncertainty is large (small ω and small σ), as in Figure 4 (right), purchase volume responds strongly to price and $q(v)$ responds less. The seller prefers a higher price to curb returns, since raising price mainly discourages purchases, with little change in each individual's return probability. When returns are costly, the incentive to mitigate returns is stronger under RI. Whether a stronger return-mitigation incentive raises or lowers the RI price depends on these primitives. Prices tend to be higher under RI when ω and σ are small, and lower under RI in the opposite case.

Summary. The three channels imply that RI need not have higher equilibrium prices than under CI even though the seller pays the premium. When the premium is low, the pass-through channel is weak, and surplus extraction and return mitigation can reverse the ranking depending on ω and ex-post uncertainty. In general, RI prices tend to be higher when ω and σ are small because RI can expand net demand, and return mitigation mainly works through screening. Thus, the price comparison is an equilibrium outcome rather than a direct measure of insurance incidence.

5.4 Return Policy Design

The preceding subsections characterize equilibrium under RI and CI. We next define welfare benchmarks, characterize monopoly pricing with efficient returns as a second-best benchmark, compare RI and CI, and extend the analysis to return-penalty discount (RPD). Counterfactual simulations apply this framework at estimated parameters.

5.4.1 First Best and Welfare Decomposition

Ex-post efficiency concerns the return decision conditional on purchase. A return is ex-post efficient if and only if

$$v_i + \varepsilon_i^{\text{keep}} \leq -(\kappa + k_i + h) + \varepsilon_i^{\text{return}}. \tag{18}$$

The left-hand side is the social value of keeping (valuation plus ex-post keep shock). The right-hand side is the social value of returning (negative of hassle cost κ, shipping cost k_i, and handling cost h, plus ex-post return shock). Production cost θ^c does not enter either side because the comparison is conditional on purchase and θ^c is therefore sunk. The socially optimal return probability implied by (18) is

$$q_i^*(v_i, k_i; \sigma, \kappa, h) \equiv \frac{e^{-\sigma(\kappa+k_i+h)}}{e^{\sigma v_i} + e^{-\sigma(\kappa+k_i+h)}}. \tag{19}$$

Compared with private return probability q_i^l, q_i^* internalizes h and evaluates the margin at v_i because price is a transfer. Under RSI, consumers instead use $v_i - p$ and do not internalize the seller's loss on return (h and p), so $q_i^l > q_i^*$, creating excessive returns.

Ex-ante efficiency concerns the purchase decision. Expected seller profit from consumer i is

$$\pi_i^l \equiv (1 - q_i^l)(p - \theta^c - I^r RI) - q_i^l(h + \theta^c + I^r RI), \tag{20}$$

and insurer expected profit from consumer i is

$$\tilde{\pi}_i^l \equiv I^r \cdot RI + I^c(k_i) \cdot BC_i - k_i \cdot RSI_i \cdot q_i^l. \tag{21}$$

We define the *individual social surplus* from allocating the product to consumer i as the sum of consumer surplus, seller surplus, and insurer surplus:

$$S_i^l \equiv CS_i^l + \pi_i^l + \tilde{\pi}_i^l. \tag{22}$$

A transaction is ex-ante efficient only if $S_i^l \geq 0$, but S_i^l is evaluated at private return behavior q_i^l, so the purchase benchmark embeds ex-post behavior and differs across return policies. Ex-ante and ex-post efficiency are therefore entangled. To compare efficiency losses on a common basis across policies, we hold ex-post behavior at q_i^* when evaluating ex-ante efficiency and decompose efficiency losses relative to first best.

We aggregate welfare by taking expectations over potential consumers and omitting market size N, so welfare here should be interpreted as welfare per potential consumer. With the outside option normalized to zero, non-purchasers contribute zero. Expected consumer surplus and welfare under scheme l are

$$CS^l \equiv \mathbb{E}_{\nu_i,k_i}\left[CS_i^l \cdot D_i^l\right], \tag{23}$$

$$W^l \equiv \mathbb{E}_{\nu_i,k_i}\left[S_i^l \cdot D_i^l\right], \tag{24}$$

where under CI the purchase indicators and surplus terms are branch-specific on insured and uninsured margins.

To quantify how return policy affects welfare, we compare welfare W^l to first best and split the gap into ex-post and ex-ante components. The split uses an intermediate allocation that keeps observed purchase decisions D_i^l fixed but replaces q_i^l with q_i^*. Substituting q_i^* into the individual social-surplus expression yields the closed form

$$S_i^*(\nu_i) \equiv -\theta^c + \frac{1}{\sigma}\ln\left(e^{\sigma\nu_i} + e^{-\sigma(\kappa+k_i+h)}\right). \tag{25}$$

Let $\hat{\nu}^*(k_i)$ denote the first-best purchase cutoff solving $S_i^*(\nu_i) = 0$, and set $D_i^* \equiv \mathbf{1}\big(\nu_i \geq \hat{\nu}^*(k_i)\big)$. Define $W^{OR} \equiv \mathbb{E}_{\nu_i,k_i}[S_i^*(\nu_i) \cdot D_i^l]$ as welfare with optimal returns at observed purchases and $W^{FB} \equiv \mathbb{E}_{\nu_i,k_i}[S_i^*(\nu_i) \cdot D_i^*]$ as first-best welfare. The gap to first best then satisfies

$$W^{FB} - W^l = \underbrace{\left(W^{OR} - W^l\right)}_{\Delta^{ex\text{-}post}} + \underbrace{\left(W^{FB} - W^{OR}\right)}_{\Delta^{ex\text{-}ante}}. \tag{26}$$

The first term, $\Delta^{ex\text{-}post}$, measures surplus lost when return probabilities differ from q_i^* among purchasers under scheme l, holding D_i^l fixed to isolate ex-post inefficiency. The second term, $\Delta^{ex\text{-}ante} = \mathbb{E}_{\nu_i,k_i}\left[S_i^*(\nu_i)(D_i^* - D_i^l)\right] = W^{FB} - W^{OR}$, is the remaining gap after every purchaser's return probability is set to q_i^*.

Because D_i^* maximizes expected surplus under $S_i^*(\nu_i)$, $\Delta^{ex\text{-}ante} = W^{FB} - W^{OR}$ is always nonnegative. It can arise from efficient trades left inactive ($D_i^* = 1$, $D_i^l = 0$) or from admitting buyers with $S_i^*(\nu_i) < 0$ ($D_i^* = 0$, $D_i^l = 1$). The latter case rarely arises and does not appear in our counterfactuals: the option value of returns makes $S_i^*(\theta^c) > 0$, so $\hat{\nu}^* < \theta^c$, and under monopoly pricing the actual purchase cutoff $\hat{\nu}^l$ is typically above $\hat{\nu}^*$. Our counterfactuals therefore always underprovide relative to first best for $\Delta^{ex\text{-}ante}$, reflecting consumers who would purchase at first best but do not under scheme l.

In what follows, we use $\Delta^{ex\text{-}ante}$ only for this first-best decomposition and assess the welfare effect of participation by individual surplus S_i^l with return probability q_i^l. We call this the extensive margin to separate it from the ex-ante margin under q_i^*. Overprovision arises when the actual purchase cutoff lies below the efficient cutoff solving $S_i^l(v_i) = 0$, so some admitted buyers have negative social surplus, while underprovision arises when the purchase cutoff lies above that efficient cutoff, so some buyers with $S_i^l \geq 0$ remain excluded.

5.4.2 Second Best under Monopoly Pricing

In our model, marginal cost pricing ($p = \theta^c$) and a return refund of $-h$ implement the first best. On return, the buyer receives scrap value minus the handling fee borne by the seller rather than recovering p, so the buyer fully internalizes the seller's loss on return in addition to the hassle cost $\kappa + k_i$.[18] Under monopoly, return policy can improve return discipline and participation, but once returns are efficient the remaining distortion is monopoly pricing above θ^c, which stems from market structure.

To quantify how large the residual welfare loss from market structure is (a loss that return policy design itself cannot correct), we use an *optimal refund* benchmark (Online Appendix B.2). With a return refund of $-h$, return probabilities coincide with the first-best rule q_i^*, so $\Delta^{ex\text{-}post} = 0$. The seller chooses p to maximize $(p - \theta^c)$ times demand under this refund, because each sale earns $p - \theta^c$ whether the unit is kept or returned. Among single-price policies with this uniform refund, the resulting allocation is second best: the remaining gap to the first best is entirely ex-ante distortion from monopoly pricing. Neither RI nor CI attains this benchmark, because both refund p on return. RI can nonetheless admit buyers whom the benchmark excludes because lenient returns can weaken surplus extraction, though only at the cost of greater over-return.

This benchmark is a welfare diagnostic rather than a direct policy prescription. Net handling cost h combines logistics, inspection, and resale value that vary across sellers and categories and are imperfectly observed, so mandating refund $-h$ listing by listing is impractical. The same benchmark refund is also ruled out where distance-selling law requires full product-price reimbursement on no-reason returns, as under China's seven-day no-reason return rules and the EU Consumer Rights Directive.[19] Product-refund forfeiture beyond these statutory floors raises consumer-protection and reputation concerns. Restocking fees and partial refunds are tolerated when disclosed at checkout and tied to processing costs, but opaque or punitive forfeiture invites complaints and regulatory scrutiny. We therefore evaluate deployable alternatives, including return-penalty discount (RPD) in Section 5.4.4.

5.4.3 Assessment of RI and CI

Relative to the *optimal refund* benchmark, we assess RI and CI first on the ex-post margin and then on the extensive margin. On the extensive margin we judge over- and underprovision by S_i^l, which can run in either direction even when $\Delta^{ex\text{-}ante} > 0$.

[18] Expected seller profit from consumer i is then zero on every path, so $S_i^* = CS_i^*$ and efficient purchase and return decisions maximize consumer surplus as well as social surplus.

[19] China's *Law on Protection of Consumer Rights and Interests* (Art. 25) and *Interim Measures for Seven-Day No-Reason Returns on Online Purchases* (`https://www.gov.cn/zhengce/zhengceku/2020-11/03/content_5557118.htm`). EU Directive 2011/83/EU, Art. 13 (`https://eur-lex.europa.eu/eli/dir/2011/83/oj`). Exceptions and prior buyer confirmation narrow these rights.

On the ex-post margin, both RSI schemes refund p on return and are more lenient than the partial-refund benchmark. CI generally outperforms RI on this margin. On return the seller bears $-(p+h)$ and, when shipping is reimbursed, the insurer bears $-k_i$. The externality gap is therefore $-(p+h+k_i)$ under RI and for insured CI purchasers, but only $-(p+h)$ for uninsured purchasers who bear $-k_i$ themselves. Holding price fixed, RI therefore imposes the larger gap on every purchaser, whereas under CI only insured buyers face $-(p+h+k_i)$. Price is endogenous, however, and can be higher under CI. Provided the equilibrium price under CI does not exceed that under RI by too much and insurance take-up is not too large, the efficiency gain from stronger return discipline for uninsured consumers dominates, so CI outperforms RI on the ex-post margin.

On the extensive margin, overprovision can arise when the seller chooses price without internalizing the externality on the insurer: he pays a fixed premium per order regardless of return probability and thus does not bear the marginal expected claims cost of attracting high-return buyers. Equilibrium can then expand participation to include marginal purchasers even when they contribute negative social surplus. This admits too many consumers with high return propensity who are socially unprofitable once reimbursement costs are accounted for.

RSI is also likely to soften the underprovision in the partial-refund benchmark for two reasons. First, holding price fixed, a full product refund together with return-shipping reimbursement expands participation, in contrast to the partial refund of $-h$ and no reimbursement under the benchmark. Second, lenient returns can give rise to a lower price because of weaker surplus-extraction incentives. In general, RI may mitigate underprovision more than CI because surplus-extraction incentives are weaker and coverage is provided at no separate charge. The welfare ranking across the two RSI schemes and the benchmark therefore depends on the profit margin. With large markups, RI's broader participation among buyers with positive marginal social surplus can dominate its added return distortion. With thin markups, overprovision and ex-post distortion under RI can dominate instead, making RI worse than the benchmark.

To quantify surplus extraction on the extensive margin, we relate it to the socially valuable purchases each RSI policy leaves unrealized, measured by *foregone efficient sales*: the shortfall of actual purchases relative to the planner's extensive margin where marginal social surplus at purchase is zero. In a textbook single-product monopoly without returns, this shortfall coincides with the usual underproduction relative to the efficient quantity, and a higher profit margin is associated with more sales prevented. Across return schemes, however, posted prices and profit margins poorly measure surplus extraction. RI and CI attach different return benefits and insurance costs for buyers and shift costs differently for sellers, so from the consumer's perspective they are effectively different products rather than the same good sold at different markups. Comparing foregone efficient sales is thus a better way to compare surplus extraction across policies.

Section 8.3.3 visualizes this trade-off between the extensive and ex-post margins at the estimates. We next assess return-penalty discount (RPD).

5.4.4 Assessment of RPD

Under return-penalty discount (RPD), each consumer chooses either a discounted price with no or partial product refund on return or a full price with full refund rights while bearing the private return cost, as in common "final sale" or restocking-fee menus.[20] RPD preserves a full-refund option while adding a second branch with stricter refund terms. On the discount branch, a share of the discounted price is forfeited on return. Equilibrium and welfare analysis appear in Online Appendix B.1.

Relative to RSI, RPD is a natural alternative for correcting excessive returns, especially when markups are thin and social return costs are high. RSI expands low-end participation by subsidizing returns and thereby worsens return composition and over-return. RPD does not expand participation through return subsidies; forfeiture on the discount branch instead attenuates over-return among buyers who select that branch.

RPD also coordinates screening in the hands of the party with product-level information. Under CI, the insurer sets standardized premiums while the seller chooses only product price, so price is a blunt screen for return-intensive demand and optional coverage can admit socially inefficient purchases that price alone cannot undo. Even when baseline return risk is low, a premium set mainly from return probability can be too low if handling costs are high. Cheap coverage can then induce enough additional returns to impose a large negative externality on the seller and on society. Under RPD, the seller jointly chooses list price and discount under platform-set refund terms, allowing the menu to attract likely keepers while limiting reimbursed-return incentives. This can improve welfare relative to CI, but the same control may strengthen surplus extraction and raise foregone efficient sales. Which force dominates is an empirical question.

We take full forfeiture on the discount branch as the leading case because it delivers the strictest return discipline and is straightforward to implement. Ex ante, stricter refund terms can exclude socially valuable marginal buyers. Ex post, no product refund narrows the return subsidy to an externality gap of h alone; when recovery exceeds handling ($h < 0$), full forfeiture can overshoot into under-return. Partial-forfeiture RPD, which mitigates this distortion, is evaluated separately in the counterfactual simulations.

6 Identification

The objects to be identified are the coverage distribution $\Psi(\cdot)$, mean ex-ante value and taste dispersion captured by μ_t and ω_j, and ex-post precision σ_t. We normalize the non-monetary return hassle cost to $\kappa = 5$ RMB because, as emphasized by Anderson et al. (2009), high ex-post uncertainty and low hassle costs affect purchase and return behavior in similar directions. Both raise purchases and returns. In our setting, insurance purchase under CI is also tied to return risk, so σ_t and κ would be difficult to separately pin down from purchase, return, and insurance moments alone.[21]

[20] On Amazon Seller Central, merchant-fulfilled sellers may charge a restocking fee of up to 20% of item price when the buyer changes their mind about a purchase and returns the item in its original condition outside the return window (Seller Central Help, `https://sellercentral.amazon.com/help/hub/reference/external/G201725780`).

[21] We do not normalize κ to zero because eliminating return costs entirely would imply that, under RI, the purchase probability is always one.

Under RI, fully reimbursed shipping makes the purchase threshold independent of k_i, so $\Psi(\cdot)$ is non-parametrically identified from the empirical coverage distribution and purchase does not select on coverage. CI purchase, return, and insurance moments further restrict the same demand primitives through endogenous insurance selection under CI.

The purchase probability under RI conditional on price is $\mathbb{E}[D^r_{it}(k_i, p_t)] = \mathbb{E}[\mathbf{1}(\nu_{it} \geq \hat{\nu}^r_t)]$, where $\hat{\nu}^r_t = p_t + \frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})$. With $\nu_{it} = \mu_t + \xi_t + \omega_j\epsilon_{it}$, define $w^r_{it} \equiv \nu_{it} - \frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})$ and mean willingness to pay $\bar{w}^r_t \equiv \mathbb{E}[w^r_{it}]$, so $\nu_{it} \geq \hat{\nu}^r_t$ is equivalent to $w^r_{it} \geq p_t$. The purchase probability therefore equals

$$\mathbb{E}[D^r_{it}(k_i, p_t)] = 1 - \Phi\left(\frac{p_t - \bar{w}^r_t}{\omega_j}\right).$$

Variation in p_t across category–months traces out the purchase schedule; its slope identifies ω_j and its location identifies $\bar{w}^r_t$, given a valid price instrument. Mean willingness to pay decomposes into intrinsic value and return option value:

$$\bar{w}^r_t \equiv \underbrace{\mu_t + \xi_t}_{\text{Intrinsic value}} + \underbrace{\left[-\frac{1}{\sigma_t}\ln(1 - e^{-\sigma_t\kappa})\right]}_{\text{Return option value}}.$$

The return option value increases when ex-post uncertainty is higher (lower σ_t) and when hassle cost κ is lower.

For a fixed purchase rate, a high intrinsic component of $\bar{w}^r_t$ implies fewer returns, whereas a large return option value (lower σ_t) implies more returns. Re-expressing expected returns under RI in terms of w^r_{it} gives

$$\mathbb{E}_{w^r_{it}}\left[\frac{e^{-\sigma_t\kappa}}{(1 - e^{-\sigma_t\kappa})e^{\sigma_t(w^r_{it} - p_t)} + e^{-\sigma_t\kappa}}\,\mathbf{1}\big(w^r_{it} \geq p_t\big)\right]. \tag{27}$$

For a fixed purchase probability, expected returns in (27) are decreasing in σ_t. Joint purchase and return variation therefore identifies σ_t, and given σ_t and ω_j, we recover intrinsic value $\mu_t + \xi_t$. Variation in X_t (with $\mu_t = X_t\beta^\mu$) then separates μ_t from ξ_t.

Given calibrated net handling cost h_t, the pricing first-order condition maps observed prices into marginal production cost θ^c_t. Inverting that condition at observed prices and estimated demand responses yields implied θ^c_t in estimation below. Because each market aggregates products within a category–month, X_t shifts with assortment size, in-season and newly published shares, and principal components of product characteristics; variation in these aggregates across periods identifies α^c and the store-level cost shock ζ^c_t through $\theta^c_t = X_t\alpha^c + \zeta^c_t$.

7 Estimation

We estimate the model by GMM, pooling moment conditions across categories j. The estimation sample contains 6 apparel categories and 313 category × month markets after smoothing, resampling, and the 5%

purchase-probability screen in Appendix A.[22] Taste dispersion ω_j is fixed within a category over time. Market size is defined at the category $\times$ season level as twice the maximum monthly sales within that category–season, and monthly purchase probabilities equal smoothed sales divided by that size. As is standard when the outside option is unobserved, absolute market size is not separately identified from mean utility. The twice-peak rule is therefore a scale normalization that keeps purchase probabilities interior and seasonally comparable. The subsections below report return-handling calibration, demand estimation, supply-side cost recovery, selected parameter estimates, and fit on endogenous moments.

7.1 Return-handling calibration

Net return handling cost is h_t = gross handling − scrap value. We set gross handling to 15 RMB per return, in line with industry reports that each clothing return without revenue costs sellers about 15 yuan in sunk marketing, packaging, and shipping[23]. For scrap value, we calibrate a three-way split of returned inventory using industry surveys. About one-half of returned apparel units are resold at full retail price (e.g., 48%). Condition grading assigns roughly 20–30% of units to discount or outlet channels and 20–30% to secondary liquidation; we combine these into a 40% markdown tail. Roughly 10% of returns are written off[24]. Units in the markdown tail are resold below list; we assume they recover 50–70% of p_t, i.e. markdowns of 30–50% off list price. Weighting yields average scrap value $0.5\,p_t + 0.4 \times (0.5–0.7)\,p_t \approx 0.75\,p_t$, so $h_t = 15 - 0.75\,p_t$ (RMB). We also consider scrap shares of $0.65\,p_t$ and $0.85\,p_t$; Section 8.6.1 reports the implied welfare rankings.

7.2 Demand Side

In order to estimate the demand side parameters, we follow the nested fixed-point logic of Berry et al. (1995). The nonlinear parameters $\Theta = (\beta^{\omega}, \beta^{\sigma})$ determine category-level dispersion in ex-ante valuations, $\omega_j = C_j'\beta^{\omega}$ for a category-indicator vector C_j, and the ex-post precision index, $\sigma_t = X_t^{\sigma}\beta^{\sigma}$. Mean utility is $\mu_t = X_t\beta^{\mu}$, where X_t contains category-month demand shifters such as assortment size, in-season share, newly published share, category indicators, and product characteristics such as style, length, and material texture. As in the identification argument, we normalize $\kappa = 5$ RMB and fix $\Psi(\cdot)$ at the empirical RI coverage distribution. The index X_t^{σ} is more parsimonious than X_t because σ_t is identified from joint purchase and return moments and is less sharply pinned down than mean utility. It includes category indicators, which

[22] Very low purchase rates leave mean utility poorly pinned down in the purchase-probability inversion—large negative values are nearly observationally equivalent—so we drop those cells.

[23] China Venture News and Zhou Jiabao (2025), reprinting *Times Finance*; cites Yinman founder Fang Jianhua on per-return seller costs, `https://www.chinaventure.com.cn/news/116-20250102-384596.html` (accessed May 19, 2026).

[24] Eightx (2026), "Average eCommerce Return Rate by Category," `https://eightx.co/blog/average-ecommerce-return-rate` (accessed May 19, 2026); SELVANE (2024), summarizing Optoro return grading, `https://www.selvane.co/blogs/knowledge/the-hidden-cost-of-fast-fashion-returns-what-happens-to-garments-after-you-send-them-back` (accessed May 19, 2026); McKinsey & Company (2020), "Returning to order: Improving returns management for apparel companies," `https://www.mckinsey.com/industries/retail/our-insights/returning-to-order-improving-returns-management-for-apparel-companies` (accessed May 19, 2026). SELVANE (2024) reports that 25–30% of units are never resold; we follow the lower landfill estimate in McKinsey & Company (2020) for pure write-offs and fold low-recovery liquidation into the markdown tail.

absorb persistent differences in ex-post fit uncertainty across apparel types, and newly published share, which allows σ_t to vary when the assortment is heavily refreshed. Assortment size and in-season share enter μ_t and implied marginal cost but not σ_t.

For each candidate Θ, we recover the mean-utility index $\delta_t(\Theta) = \mu_t + \xi_t$ by inverting purchase probabilities. When only one RSI policy is observed in a market, the inversion matches that policy's sales-weighted, coverage-adjusted purchase rate. When RI and CI coexist, a single δ_t minimizes the sum of squared discrepancies between model-implied and observed purchase rates under RI and under CI. We estimate $\beta^\mu(\Theta)$ by IV regression of $\delta_t(\Theta)$ on X_t using instruments Z_t, which include X_t and the average price of other categories in the same period, and define $\xi_t(\Theta) = \delta_t(\Theta) - X_t\beta^\mu(\Theta)$. The other-category price is a Hausman-style cost instrument. Shared cost factors such as labor, logistics, and materials move prices across categories together. After aggregation to coarse categories, demand shocks are less correlated across categories than costs. Direct monthly cost shifters such as provincial wages and electricity bills are unavailable at our frequency, so we rely on this proxy. The first-stage partial F of this instrument for own-category price is 100.07 ($p = 0.000$; $N = 313$).

Given each candidate Θ, we form two sets of moments. Orthogonality conditions $Z_t\xi_t(\Theta)$ impose $E[Z_t\xi_t] = 0$. Squared prediction errors $m_t(\Theta)^2$ match sample return and insurance rates to their model-implied counterparts at Θ, where $m_t(\Theta)$ is the Ψ-mass-weighted gap between each sample and implied rate.[25] We choose $\hat{\Theta}$ to minimize $\bar{g}_n(\Theta)' W \bar{g}_n(\Theta)$, updating W iteratively until convergence, and set $\hat{\beta}^\mu = \beta^\mu(\hat{\Theta})$.

7.3 Supply Side

Given $\hat{\Theta}$, we invert the pricing first-order condition at observed scheme-specific prices using the estimated demand system. When only one RSI policy is observed, we recover θ_t^c by inverting that condition at the policy's observed price. When both policies are observed, we invert each policy's condition to obtain $\theta_{t,\mathrm{RI}}^c$ and $\theta_{t,\mathrm{CI}}^c$ and set θ_t^c to their weighted average, with weights $(dD_t^r/dp_t)^2$ and $(dD_t^c/dp_t)^2$. Because each first-order condition is linear in θ_t^c with slope dD_t^l/dp_t, this average minimizes the sum of squared pricing residuals across the two branches. This yields 309 implied cost observations.[26] We regress θ_t^c on X_t and time fixed effects without an intercept to obtain $\hat{\alpha}^c$ and $\hat{\zeta}_t^c$ from $\theta_t^c = X_t\alpha^c + \zeta_t^c$.

7.4 Parameter Estimates

Table 2 reports selected estimates of $\hat{\Theta}$ and $\hat{\alpha}^c$, with implied $\omega_j = C_j'\hat{\beta}^\omega$ and $\sigma_t = X_t^\sigma\hat{\beta}^\sigma$. Assortment size is the number of listed products in the category–month. Newly published share is the fraction of SKUs first published in the transaction year. In-season share is the fraction whose season tag matches the calendar month. Higher σ_t means lower ex-post fit uncertainty.

We assess sampling uncertainty with a transaction-level bootstrap of $B = 500$ replicates. Each draw resamples transactions with replacement and re-estimates the model; GMM moment conditions are recentered

[25] Identification of $(\beta^\omega, \beta^\sigma)$ relies on cross-market variation in purchase and return (equation (27)). Signed prediction errors can cancel—for example, a large overprediction in one market offset by underprediction in another—so we enter $m_t(\Theta)^2$ rather than $m_t(\Theta)$. At Θ_0 and under correct specification, each $m_t(\Theta_0)$ is zero.

[26] All 313 category–month markets with observed prices enter demand estimation and moment-fit comparisons. Supply inversion drops four cells in which $|\partial D^l/\partial p|$ falls below 10^{-5} at $\hat{\Theta}$, where the inversion is numerically unstable.

Table 2: Selected parameter estimates at $\hat{\Theta}$ and $\hat{\alpha}^c$.

Parameter	Estimate	SE	95% CI
β^{ω}			
ω_j (dress)	59.217	50.610	[58.965 , 250.506]
ω_j (two-piece)	35.730	6.202	[25.911 , 47.897]
ω_j (outerwear)	35.679	15.850	[19.197 , 83.563]
ω_j (pants)	31.732	23.857	[28.211 , 140.728]
ω_j (sweaters)	21.710	10.912	[16.534 , 74.759]
ω_j (tops)	8.553	4.337	[0.099 , 17.665]
β^{σ}			
σ_t: newly published share	0.076	0.010	[0.033 , 0.084]
σ_t: dress	0.003	0.000	[0.002 , 0.003]
σ_t: two-piece	0.061	0.009	[0.046 , 0.080]
σ_t: outerwear	0.048	0.010	[0.031 , 0.072]
σ_t: pants	0.050	0.007	[0.020 , 0.056]
σ_t: sweaters	0.052	0.009	[0.017 , 0.068]
σ_t: tops	0.129	0.043	[0.085 , 0.269]
β^{μ}			
μ_t: Assortment size	-0.125	0.038	[-0.213 , -0.070]
μ_t: Newly published share	148.054	36.079	[146.660 , 327.353]
μ_t: In-season share	2.346	10.220	[-46.314 , 8.125]
θ^c			
θ_t^c: Assortment size	-0.097	0.016	[-0.135 , -0.069]
θ_t^c: Newly published share	0.099	15.580	[-64.999 , 2.317]
θ_t^c: In-season share	13.429	2.184	[12.230 , 22.398]

at their values evaluated on the original sample at $\hat{\Theta}$. The table reports bootstrap standard errors and percentile 95% confidence intervals, defined by the 2.5th and 97.5th quantiles of the bootstrap distribution. Histograms appear in Online Appendix D.1; most bootstrap distributions are approximately symmetric, but we report percentile intervals throughout because some parameters are skewed.

Ex-ante dispersion is widest for dresses (59.217) and narrowest for tops (8.553). The σ_t category intercepts rank in the opposite order: precision is lowest for dresses (0.003) and highest for tops (0.129), so the dress category has the greatest ex-post fit uncertainty. Mean utility rises with newly published share (148.054). Because the regressor is a share on $[0, 1]$, a one-percentage-point increase (0.01 on the share scale) raises μ_t by about 1.5 RMB. The estimate is large in part because most transactions in our sample occur in a SKU's first listing year, so high newly published share marks category-months whose assortment is concentrated in first-year styles and therefore higher μ_t. Assortment size enters mean utility slightly negatively (-0.125), which may reflect some degree of substitution within the category.

In-season share enters implied marginal cost positively (13.429). Category-months with more off-season or carryover listings tend to have lower prices and lower inverted θ_t^c. Newly published share enters implied marginal cost only slightly (0.099) and is insignificant.

7.5 Model Fit

Table 3 compares model-implied and sample values of prices, purchase probabilities, conditional return probabilities, and conditional CI insurance purchase probabilities at the category × month level.[27] Each entry is an unweighted mean or standard deviation across the 313 estimation markets.

Purchase probabilities need not match exactly even though purchase enters the δ_t inversion. Where RI and CI are both observed, a single mean-utility index must fit both policies, so the inversion minimizes a weighted sum of squared purchase discrepancies rather than matching either policy on its own. Given $\hat{\Theta}$, the table reports list prices, conditional return and insurance rates, and their dispersion.

Table 3: Model vs. data at $\hat{\Theta}$ (category × month cell means and SDs).

	Mean		SD	
	mod.	data	mod.	data
Prices				
RI price (RMB)	107.42	99.89	55.02	39.68
CI price (RMB)	98.33	95.37	49.43	33.41
Purchase probabilities				
RI purchase probability (%)	18.99	26.20	20.45	15.51
CI purchase probability (%)	15.59	23.04	18.39	12.83
Return and insurance probabilities				
RI return probability (%)	37.51	29.33	16.88	15.29
CI return probability (%)	16.54	22.90	15.74	12.97
CI insurance purchase probability (%)	7.26	2.61	20.18	7.77

Notes: Return and insurance purchase probabilities are conditional on purchase; probabilities in percentage points.

The model reproduces the main qualitative patterns in the table. Mean RI list prices, purchase rates, and conditional return rates exceed their CI counterparts in both the model and the data, and simulated price levels are of the right order of magnitude.[28] Purchase probabilities are underpredicted by about seven percentage points in both policies. Conditional return rates miss in opposite directions: the model overpredicts RI returns and underpredicts CI returns. This wider conditional gap may partly reflect an underestimate of ex-post uncertainty. When ex-post uncertainty is low, return probabilities are more sensitive to return-shipping coverage, so the model's RI–CI return gap can exceed that in the data and overstate the welfare cost of RI relative to CI.

Estimation targets three coverage-weighted unconditional moments (RI return, CI return, and CI insurance purchase). At $\hat{\Theta}$, unconditional return rates average about 7% under RI and 3% under CI in the model, compared with about 8% and 5% in the data. The implied RI–CI gap is 4.5 percentage points in the model

[27] We report conditional rates for interpretability. Estimation targets unconditional rates per consumer exposure. Conditional moments would reweight discrepancies toward buyers, giving similar weight to a percentage-point miss even when the purchasing population differs across markets, and would divide by purchase rates in a way that is unstable when purchase is small.

[28] In Table 1, tops account for about 31% of listings and constitute the only estimation category in which CI means exceed RI means, with RI means higher in the other five. That composition partly explains why pooled CI prices exceed RI prices there but mean RI prices exceed CI prices here.

and 2.4 percentage points in the sample, compared with 21 and 6.4 percentage points for the conditional gaps. Return fit is therefore less poor than the conditional comparison suggests, although purchase remains underpredicted and CI unconditional returns remain too low. The table also shows higher conditional CI insurance take-up in the model than in the data (7.3% vs 2.6%). Unconditional insurance rates implied by these purchase and conditional rates are only about 1% in the model and 0.6% in the data, so the insurance moment contributes a small level discrepancy even though the conditional gap looks wide. Level fit remains imperfect overall. Even so, we use the estimates primarily to compare *relative* welfare across policies within each category.

8 Counterfactual Simulations

We apply the welfare framework at the estimated parameters and report category-specific policy rankings from equilibrium primitives and welfare decompositions. Section 8.6 reports robustness checks on welfare rankings and decompositions.

8.1 Simulation Design

For the welfare counterfactuals, we construct one *representative* snapshot per category. Covariates, premium-related inputs, demand shocks, and production-cost shocks are averaged across periods using consumer-count weights, and total market size is the sum of those counts. We then solve for equilibrium prices under each policy at the estimated parameters and evaluate purchase probabilities, return rates, and welfare by aggregating over ex-ante valuations and the coverage distribution. Prices and quantities in the welfare tables are therefore model-implied equilibria for this representative environment, not realized outcomes from a single calendar month.

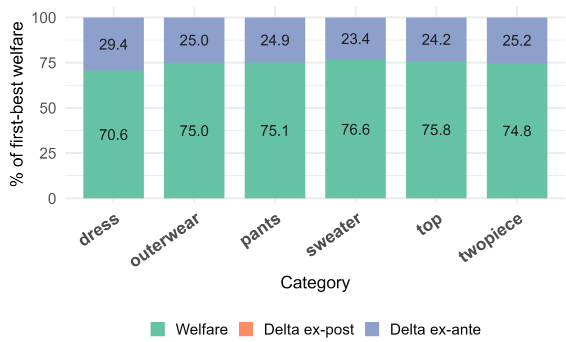
An alternative is to recompute equilibrium and welfare period by period and then average those welfare numbers. We prefer the representative snapshot because it targets long-run average performance. Period-level primitives can be noisy—thin months, for example, leave mean utility sparsely identified—and welfare is highly nonlinear in those primitives, so extreme draws can dominate a period-by-period average. Averaging primitives first, then evaluating welfare once, dampens that noise while preserving the consumer-weighted composition of the category.

8.2 Second-Best Numerics

Before comparing return policies, we quantify how much of the welfare gap to first best remains once returns alone are made efficient, isolating the distortion from market structure. Figure 5 reports the optimal refund benchmark from Section 5.4.2. Panel (a) decomposes welfare as shares of first best. By construction $\Delta^{ex\text{-}post} = 0$. Welfare remains below first best as monopoly pricing screens out efficient purchases. The entire shortfall is ex-ante distortion, accounting for about 30% of first-best welfare for dresses and about 25% in the other categories.

Panel (b) reports the purchase cutoff solving $CS_i(\nu_i) = 0$, the efficient cutoff solving $S_i(\nu_i) = 0$ under efficient return behavior, and marginal social surplus at the purchase cutoff. Marginal social surplus is strictly positive in every category, so the ex-ante shortfall in Panel (a) is monopoly underprovision on the extensive margin. In valuation space that underprovision is the purchase–efficient cutoff gap, the foregone-efficient-sales measure of surplus extraction from Section 5.4.3. The gap is largest for dresses (about 40) and smallest for tops (about 7). These cutoffs are the benchmark against which RI, CI, and RPD are compared below.

Figure 5: Efficient returns do not eliminate monopoly underprovision.



(a) Welfare decomposition.

Category	Purch. cutoff	Eff. cutoff	Marg. SS
dress	56.65	16.13	21.47
outerwear	113.81	86.24	26.25
pants	91.98	65.55	24.38
sweater	110.64	88.93	20.85
top	48.61	41.36	7.24
twopiece	102.04	76.87	24.21

(b) Purchase cutoff, efficient cutoff, and marginal social surplus.

8.3 RI vs. CI

We first compare the two RSI policies using equilibrium primitives, welfare decompositions, and purchase cutoffs. The tables and figures also include full-forfeiture RPD for comparison. We interpret RPD in the next subsection.

8.3.1 Equilibrium primitives

Table 4 reports equilibrium primitives for RI and CI by category. Columns include prices, production costs, calibrated net return-handling costs, ex-post shock dispersion from σ_t, and model-implied purchase and return probabilities.

Table 4: Category-level equilibrium primitives

						Purchase probability		Return probability	
Category	Production cost	Return handling	Ex-post SD	RI price	CI price	RI	CI	RI	CI
dress	42.06	-48.90	28.95	71.67	68.19	0.11	0.07	0.54	0.34
outerwear	87.73	-76.51	15.77	121.42	115.55	0.27	0.24	0.32	0.11
pants	67.96	-59.55	17.08	100.34	94.27	0.33	0.28	0.37	0.13
sweater	89.99	-72.56	16.64	121.85	113.35	0.43	0.40	0.41	0.15
top	41.39	-25.48	7.58	55.48	49.05	0.17	0.29	0.28	0.04
twopiece	77.95	-65.99	14.65	109.64	103.55	0.21	0.19	0.31	0.10

Notes: Monetary values are in RMB. Return probabilities are conditional on purchase.

Simulated purchase probability is higher under RI than under CI in every category except tops. This pattern is consistent with the estimation sample, where purchase probability is higher under RI than under CI at the category–month level on average. Across all categories, the return probability conditional on purchase is lower under CI than under RI.

Category-level variation begins with gross margins, which dictate the social surplus generated when a product is kept. The dress category exhibits the highest markup, $(p - \theta^c)/p$, at approximately 38%–41% across pricing policies, whereas other categories hover between 25% and 32% (e.g., 25% for tops). Economically, a higher markup reflects a wider wedge between ex-ante consumer valuations and production costs. This wedge represents the ex-ante consumption surplus realized upon a kept purchase, implying that potential social gains from expanding market participation are exceptionally high in high-markup segments.

When a consumer returns the product, the welfare outcome depends on the net return-handling cost (h_t), which captures salvage value net of shipping and processing frictions. Although a high return probability typically generates ex-post distortions, this effect is moderated by the efficiency of the return channel. In the dress category, the net salvage recovery of −48.90 RMB preserves 68%–72% of the retail price, heavily insulating the system from return frictions. For tops, where margins are lower, the net recovery of −25.48 RMB covers only 46%–52% of the price, leaving the system highly vulnerable to return logistics.

Ultimately, the return probability weighs these two competing welfare forces: a higher probability shifts the welfare weight from ex-ante consumption surplus to the ex-post salvage recovery process. The key question is whether a wide gross margin and a high net salvage recovery rate can together dominate the welfare-reducing effects of a high return probability. We turn to the quantitative welfare decomposition in the next subsection to address this question.

8.3.2 Welfare decomposition

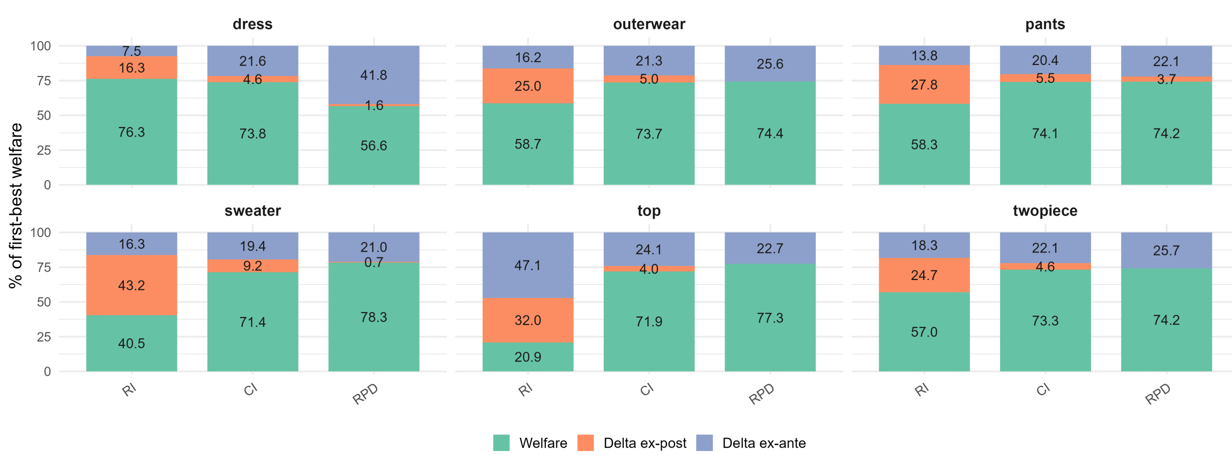


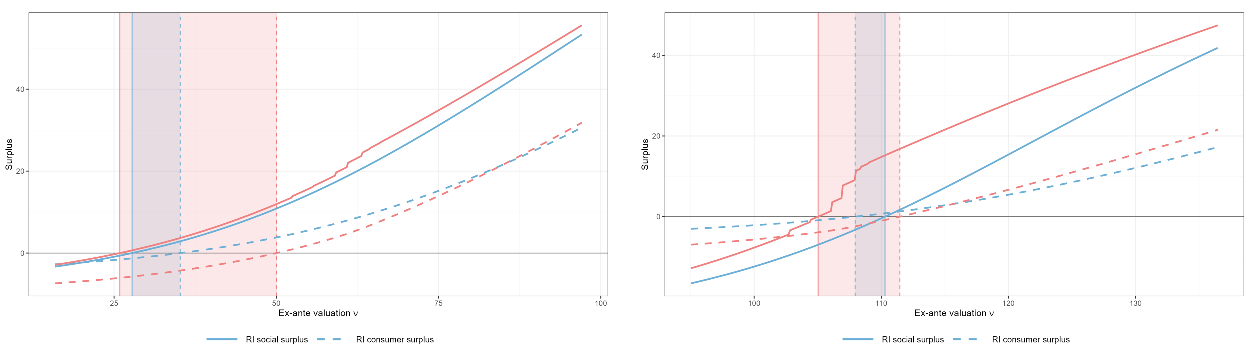
Figure 6: Welfare decomposition by category (RI, CI, full-forfeiture RPD on the discount branch), normalized as a share of first-best welfare.

Figure 6 shows how these primitives aggregate into welfare. Each stacked bar expresses welfare and the shortfalls from ex-post and ex-ante distortions as percentages of first-best welfare. Ex-post distortion is larger under RI than under CI in every category, consistent with the lower return probabilities under optional insurance.

For dresses, RI attains a larger overall share of first-best welfare than CI. Automatic coverage attracts additional buyers who remain socially valuable on net, so the gain in ex-ante efficiency dominates the incremental ex-post distortion from returns.[29] Outside the dress category, the ranking reverses: optional insurance attains a higher share of first-best welfare. Markups are lower, and return costs account for a larger share of surplus per kept sale. Curbing returns and restricting marginal entry improve ex-ante and ex-post efficiency together, and the improvement is large enough for it to outperform automatic coverage even though equilibrium purchase volume is higher under the latter. RI is especially harmful in thin-markup categories such as tops, where CI outperforms it on both margins. Roughly 80% of the first-best benchmark for tops is dissipated under RI in the decomposition, driven by poor ex-ante screening and ex-post over-return.

8.3.3 Extensive–ex-post margin trade-off

Figure 7 plots individual social surplus $S_i^l(v)$ (solid) and consumer surplus $CS_i^l(v)$ (dashed) under equilibrium RI and CI prices for dresses and outerwear. Curves and cutoffs average across coverage types using empirical coverage masses. Solid vertical lines mark efficient cutoffs where $S_i^l = 0$, and dashed vertical lines mark purchase cutoffs where $CS_i^l = 0$. Purchasers lie to the right of the purchase cutoff. Realized social surplus under scheme l integrates $S_i^l(v)$ over that set, weighted by the density of v. The signed purchase–efficient gap is the purchase cutoff minus the efficient cutoff (the shaded band). If the gap is positive, the scheme underprovides and excludes socially valuable buyers. If it is negative, the scheme overprovides and admits buyers with negative social surplus.



(a) Dress.

(b) Outerwear.

Figure 7: Extensive-margin cutoffs under RI and CI. Curves and cutoffs average across coverage types using empirical coverage masses; CI social surplus jumps at insurance cutoffs.

For dresses, both policies underprovide, but CI's gap is much larger (about 24 versus 8 under RI). In the uninsured region, CI's solid curve lies only modestly above RI's, so the ex-post margin gain from having

[29] Ex-post uncertainty need not imply a higher marginal *social* value of additional participation: greater dispersion in ex-post valuations raises private option value but also tends to increase return probabilities, and once return-handling and refund costs are accounted for, the social surplus contributed by marginal entrants need not rise mechanically with uncertainty.

buyers bear return shipping is small. This could be partly due to large ex-post uncertainty for dresses, under which return behavior responds little to coverage. Against that small gain, RI's smaller underprovision on the extensive margin dominates: the wide band of socially valuable buyers between the two purchase cutoffs raises welfare relative to CI. That RI–CI difference in purchase cutoffs is wide because dress buyers return often. When return probabilities are high, the CI premium weighs heavily in expected consumer surplus relative to price. Free coverage under RI therefore raises willingness to pay at the margin much more than optional coverage under CI.

For outerwear, the provision ranking between RI and CI flips. CI still underprovides, whereas RI overprovides, generating negative marginal social surplus. The margin trade-off reverses relative to dresses. In the uninsured region, CI's solid curve lies well above RI's, so the ex-post margin gain from having buyers bear return shipping is large. On the other hand, RI's broader participation adds little on the extensive margin: the RI–CI difference in purchase cutoffs is narrow, and part of the buyers RI admits lie in the shaded overprovision band. Return discipline therefore dominates.

Table 5 reports the same cutoffs and marginal social surplus for all categories and adds RPD next to RI and CI, matching the objects in Figure 5(b). The remaining categories follow outerwear: CI underprovides and RI overprovides. Relative to the second-best gaps in that figure (about 7 to 40), RSI softens underprovision on the extensive margin, and RI softens more than CI and can overshoot. The RI gap falls to about 8 for dresses and turns negative outside dresses (about -2 to -6), while CI gaps stay positive outside dresses (about 4 to 6).

Table 5: Purchase cutoffs, efficient cutoffs, and marginal social surplus by category

	Purchase cutoff			Efficient cutoff			Marginal social surplus		
Category	RI	CI	RPD	RI	CI	RPD	RI	CI	RPD
dress	35.19	50.02	66.71	27.45	25.56	35.08	2.87	11.96	26.48
outerwear	107.94	111.44	114.17	110.28	105.70	87.71	-3.27	16.76	26.45
pants	84.87	89.28	90.30	87.31	83.02	76.97	-2.93	15.48	16.19
sweater	107.07	108.66	109.49	113.17	104.26	94.94	-8.87	11.02	15.76
top	52.16	48.60	48.36	53.64	43.91	41.38	-3.33	5.93	6.98
twopiece	97.80	100.14	102.32	100.01	94.23	77.93	-3.32	15.94	24.37

Notes: Entries average across coverage types using empirical coverage masses. Efficient cutoffs use each scheme's equilibrium return behavior.

The CI gap exceeds RI's for two reasons. First, when marginal entrants remain uninsured, stronger ex-post discipline raises their social surplus relative to RI and can lower the efficient cutoff. Second, the purchase cutoff itself is higher than under RI. Both enlarge foregone efficient sales. CI reduces return distortions but can leave more socially valuable purchases unrealized. Dresses versus the other categories illustrate the trade-off between surplus extraction and return control.[30]

[30]When marginal entrants would adopt CI coverage instead, ex-post behavior coincides with RI and the efficient cutoff is unchanged. The higher purchase cutoff nevertheless still enlarges the gap.

These rankings use each scheme's private return behavior q_i^l. The second-best efficient cutoff under q_i^* remains below every q_i^l-efficient cutoff, so the first-best residual $\Delta^{ex\text{-}ante}$ is still underprovision even where RI overprovides on the extensive margin for some categories (Section 5.4.1).

8.4 Full-forfeiture RPD

We report full-forfeiture RPD, the leading case from Section 5.4.4. Table 6 reports the list price, discount d, demand-weighted average paid price, and segment-level purchases and returns.

Depending on equilibrium prices and the wedge between the purchase cutoff and the discount–full-price segmentation cutoff, buyers may only fall on the discount branch. In most categories full-price demand is negligible, purchases concentrate on the *discount* branch, and return probability there is close to zero. In contrast to RSI, ex-post distortion takes the form of under-return rather than over-return. With no product refund, the externality on the seller when a consumer returns narrows to $-h$ alone, which is positive in our setting when scrap value recovery exceeds handling. Ex-post distortion is nevertheless smaller under full-forfeiture RPD than under either RSI policy (Figure 6).

Table 6: RPD counterfactual: prices and segment-level purchase probabilities, returns, and return probabilities by category

				Discount option			Full price		
Category	Price	Discount	Effective price	Purchase prob.	Return	Return prob.	Purchase prob.	Return	Return prob.
dress	105.00	37.70	67.30	0.04	0.00	0.01	0.00	0.00	
outerwear	202.10	87.93	114.17	0.22	0.00	0.00	0.00	0.00	
pants	95.30	2.40	93.79	0.17	0.00	0.00	0.10	0.02	0.24
sweater	114.56	4.90	109.77	0.38	0.00	0.00	0.01	0.00	0.34
top	49.06	0.69	48.36	0.30	0.00	0.00	0.00	0.00	
twopiece	181.15	78.83	102.32	0.18	0.00	0.00	0.00	0.00	

Participation, on the other hand, is weaker because surplus extraction is stronger. The RPD purchase cutoff is about 67, against about 50 under CI and 35 under RI, even though RPD's effective price (about 67) lies slightly below the CI price (about 68). Forfeiting the product payment on return removes the implicit full-refund subsidy, so the purchase cutoff can rise above the CI cutoff even when the demand-weighted price paid is lower.

Relative to the RI and CI purchase–efficient gaps above, full-forfeiture RPD underprovides by more: for dresses the purchase–efficient gap rises to about 32 and marginal social surplus to about 26 (Table 5). Outside dresses the same ordering generally holds—RI often overprovides, CI underprovides, and RPD underprovides by more.

This stronger screening still raises welfare relative to RI and CI in every category except dresses. In the dress category the extensive margin is especially valuable. Marginal entrants often have relatively low valuations and high return propensity, yet many such purchases remain socially worth encouraging because kept sales generate high surplus. RSI preserves a channel for those buyers. The discount branch under full-forfeiture RPD largely shuts it down, so the participation loss outweighs the ex-post gain.

8.5 Partial forfeiture and restocking fees

Full forfeiture is extreme. Softening the return penalty on the discount branch to a partial forfeiture $\phi \in (0, 1)$ restores some participation while retaining most of the return discipline. A natural retail implementation is a restocking fee: on return the buyer pays $\phi(p - d)$ and recovers share $1 - \phi$ of the discounted payment. Full-forfeiture RPD is the extreme case where $\phi = 1$. We focus on restocking fees in the 5–20% range, which are common in retail practice on eligible returns.[31]

Figure 8 varies ϕ on the discount branch.

Figure 8: Partial-forfeiture RPD on the discount branch: welfare decomposition by category for $\phi \in \{0.05, 0.10, 0.15, 0.20\}$, normalized within category and ϕ.

Lower ϕ reduces under-return and raises participation relative to full forfeiture while retaining most of the return discipline from refund forfeiture on that branch. At $\phi = 0.10$–0.15, welfare exceeds both RSI policies in every category other than dresses and matches or exceeds the optimal refund benchmark, attaining about 75–78% of first-best compared with roughly 75–77% under that benchmark. For dresses, welfare remains below RI (about 76% of first-best) but at low ϕ approaches the optimal-refund benchmark (about 71% at $\phi = 0.05$–0.10) and lies well above full-forfeiture RPD (about 57%).

The pattern favors category-specific fees over a single platform rule. In the high-markup dress category, a low fee (about 5–10% of $p - d$) limits the participation loss from full-forfeiture RPD, but RI still ranks first. For tops, where margins are thin and curbing ex-post return distortion and inefficient purchases on the extensive margin matters most, full-forfeiture RPD dominates both RSI policies and partial-forfeiture RPD in the 5–20% band. Elsewhere, welfare peaks in the 5–20% band common in retail practice, so moderate partial-forfeiture RPD on the discount branch outperforms both RSI policies and full-forfeiture RPD.

[31] On Amazon Seller Central, merchant-fulfilled sellers may charge a restocking fee of up to 20% of item price when the buyer changes their mind about a purchase and returns the item in its original condition outside the return window (Seller Central Help, `https://sellercentral.amazon.com/help/hub/reference/external/G201725780`).

8.6 Robustness check

We next assess sensitivity to the scrap-value calibration and sampling uncertainty in welfare rankings and decompositions.

8.6.1 Scrap-value sensitivity

The baseline decomposition in the previous discussion uses the scrap calibration in Section 7.1 (about 75% of price on average). Because $h_t = 15 -$ scrap value, alternative recovery rates shift the return externality and can reorder counterfactuals even when demand primitives are fixed. The Online Appendix (C) recomputes welfare at scrap shares of $0.65\,p_t$ and $0.85\,p_t$.

When recovery is low (65%), returns are costlier, ex-post distortion under RI rises, and CI ranks above RI in every category, including dresses. The baseline RI advantage for dresses therefore does not survive. When recovery is high (85%), RI still beats CI only for dresses, while the ranking between CI and full-forfeiture RPD splits by category. CI is ahead for dresses, outerwear, pants, and two-piece. Full-forfeiture RPD is ahead for sweaters and tops. Full-forfeiture rankings are therefore not invariant to scrap-recovery assumptions. The qualitative lesson is nevertheless stable. Reimbursement-heavy policies fare worse when recovery is weak and better when recovery is strong.

At 65% and 85% scrap, partial-forfeiture RPD largely confirms the baseline pattern from the previous discussion. Outside the dress category, $\phi \approx 0.05$–0.20 still beats both RSI policies at both scrap shares. For dresses, partial-forfeiture RPD at 85% scrap remains below both RSI policies even at low ϕ, and RI ranks first, as at baseline. At 65% scrap it beats RI but remains close to CI.

8.6.2 Inference on welfare ranking and decomposition

Online Appendix D (Appendix D) reports bootstrap inference for welfare levels and decompositions by policy within each category. For most objects, the bootstrap histograms are approximately symmetric and close to normal around the full-sample estimate, which is expected under the recentered bootstrap. Table 7 reports pairwise bootstrap win rates for actual welfare.

Welfare rankings are largely robust across categories. RI continues to rank above the other policies for dresses, while partial-forfeiture RPD ranks highest outside the dress category when implemented at the welfare-maximizing restocking fee over $\phi \in \{0.05, 0.10, 0.15, 0.20\}$. In some categories, such as two-piece and outerwear, the welfare distributions under CI, full-forfeiture RPD, and partial-forfeiture RPD are difficult to distinguish by eye. Visual comparisons should therefore not be over-interpreted. The decomposition terms are nevertheless informative and align with the mechanisms discussed above. Relative to CI, RI exhibits lower ex-ante distortion but higher ex-post distortion in all categories except tops, where it performs worse on both margins. Partial-forfeiture RPD substantially reduces ex-ante distortion relative to full forfeiture while delivering similar ex-post distortion, or only slightly higher.

Overall, the bootstrap evidence corroborates the welfare rankings and decomposition patterns discussed in the previous subsection.

9 Conclusion

Counterfactuals at the estimated parameters confirm that welfare-maximizing return policy is category-specific. We first report how welfare divides across agents, then discuss policy implications, and close with limitations and directions for future work.

9.1 Welfare across agents

The counterfactual analysis above reports the welfare decomposition of efficiency losses relative to first best. A complementary decomposition asks how welfare divides among consumers, the seller, and the insurer. Figure 9 plots expected consumer surplus, seller profit, and insurer profit by category under each return policy, normalized as shares of first-best welfare within category and scheme. For partial-forfeiture RPD we use the welfare-maximizing restocking fee from Section 8.5. Consumer surplus and seller profit stack above zero; insurer profit is weakly negative whenever reimbursement flows through the insurer and is shown below zero. The three components sum to welfare.

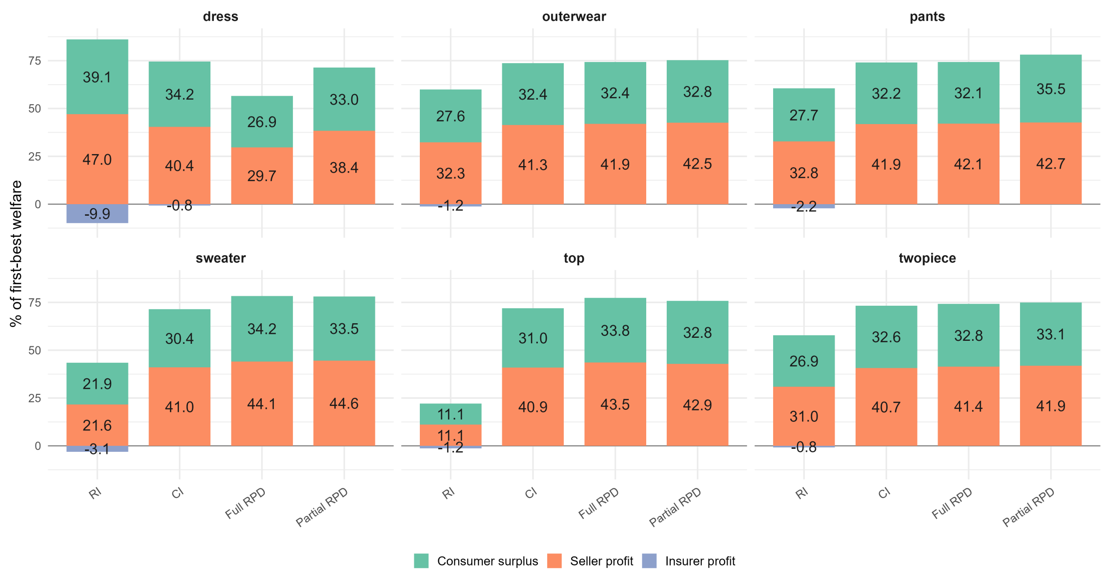


Figure 9: Welfare across agents by category and return policy.

Within each category, rankings of welfare, consumer surplus, and seller profit across policies are broadly aligned. Insurer profit is near zero under RPD and under CI outside the dress category, where coverage take-up is zero. At our exogenous premiums, expected claims exceed premium revenue on the insured margin, so insurer profit is negative under RI in every category and under CI only for dresses, where high return probabilities induce insurance take-up.[32]

[32]Industry accounts report early losses on Taobao return-shipping insurance and later underwriting profits after rates were adjusted to claims. Huatai reportedly lost about 14 million RMB on the product in 2012 (Wang Liangdi, 2017, Peking University Financial Law Center, `https://www.finlaw.pku.edu.cn/flyxjr/gk_hljryfl_20181025180041616718/2017_jrfy_20181029112500112638/zdsbq3y/37finlaw239868.htm`, accessed July 23, 2026). ZhongAn's other-insurance line (mainly return-shipping insurance) lost money in 2014–2017 and earned about 70 million RMB in underwriting profit in 2019 (Tu Yinghao,

Welfare and consumer surplus are broadly similar under RPD and CI on average across categories. The policies nevertheless allocate consumer surplus across buyers very differently. Under RPD menus, aggressive surplus extraction prices out many low-valuation consumers who would otherwise purchase, leaving those buyers with zero surplus. Under CI, the equilibrium product price is usually lower, and low-valuation buyers retain the option to purchase return-shipping insurance, which can raise their consumer surplus relative to RPD.

9.2 Policy implications

Platforms typically cannot condition return rules on product-level markups or handling costs. When category-by-category tailoring is infeasible, partial-forfeiture RPD is a practical default. In our counterfactuals it outperforms both RSI policies in most categories and nearly matches RI for dresses. When platforms have additional information on margins or return frictions, welfare can be improved further by imposing a stricter penalty on returns where those costs are high and relaxing it where kept-sale surplus is large.

Against that benchmark, current practice often pushes the opposite way. Requiring sellers to adopt RI is extremely costly in thin-margin categories such as tops, where our counterfactuals already rank RI lowest on welfare. Pinduoduo may require RI even on very cheap products priced as low as five yuan, and Taobao can bar sellers from Singles' Day promotions unless they offer RI. Platforms should avoid mandating automatic insurance in such settings.

The cost of lenient reimbursement rises further when return-handling costs are large. Both the extensive-margin loss from inefficient purchases and the wedge between social and private return costs grow with return-handling costs. *Refund-without-return* policies on e-commerce platforms are such a case.[33] The buyer keeps the good while the seller bears essentially the full disposition cost, so handling costs can exceed our baseline calibration. Full-forfeiture RPD is then likely to gain further against both partial-forfeiture RPD and RSI.

Price regulation could in principle address both margins at once, narrowing the wedge between private and social return costs and alleviating monopoly underprovision, but we do not analyze it. Sellers within a category differ in margins and costs, and competitive markets normally let firms set prices; platform-imposed caps would require granular cost data and invite fairness concerns that the platform favors some sellers. Regulating refund forfeiture or conditioning RSI on observable category traits is less intrusive.

9.3 Limitations and future research

We treat the RSI policy as given when recovering demand and inverting seller pricing. Endogenizing enrollment would leave demand estimation largely unchanged conditional on the observed scheme: purchase, insurance, and return behavior still identify the same primitives. The supply side would change. Optimality of the chosen scheme becomes an additional constraint on costs, so the mapping from observed prices to

National Business Daily, June 17, 2020, `https://www.jwview.com/jingwei/html/m/06-17/327298.shtml`, accessed July 23, 2026). Negative insurer profit in our counterfactuals therefore mainly reflects fixed observed premiums under changing return behavior, not permanent steady-state losses.

[33] In Chinese e-commerce this practice is known as *jin tuikuan* ("refund only"): money back without sending goods back.

production cost is no longer a single first-order condition under an exogenous policy. Recovered costs would have to rationalize not only within-scheme pricing but also the seller's preference for RI versus CI—and, near indifference, would be pinned by equalized profits across schemes. We deliberately avoid that layer. Policy choice can respond to return rates, and returns respond to the policy, so a fully endogenous model requires dynamics or belief updating about future premiums and claims. We leave that reverse-causality loop aside to keep identification transparent: observed prices under each scheme already discipline demand and monopoly pricing in our static setup.

Several extensions would change the policy ranking. First, under oligopoly, buyers screened out by a high price or a strict refund can divert to other products or sellers. That leakage mitigates the social cost of screening out valuable transactions relative to our single-seller benchmark and therefore tilts welfare toward stricter return terms. Second, we study platform-regulated refund menus (RPD). Delegating refund choice to sellers adds a screening instrument—menus of price and refund contracts in the spirit of sequential screening—which can expand participation for some types while strengthening surplus extraction, with an ambiguous net welfare effect. Third, our counterfactuals are risk-neutral; risk aversion would raise the value of automatic coverage on the margin. Fourth, we take insurer premiums as exogenous. Endogenizing them would reintroduce double marginalization between the insurer and the seller, with an ambiguous effect on the welfare ranking across categories.[34] Fifth, on a two-sided platform, return policy also shifts cross-network externalities between buyers and sellers; characterizing the platform-optimal refund then requires modeling both sides jointly, which we leave for future work.

[34] Insurer and retailer markups push the retail price above the vertically integrated benchmark. The wedge is weaker than in a fixed-cost insurance model because expected claims scale with the return probability, which itself rises with price. Remaining pass-through still widens the private–social gap in return costs. Ex ante, higher prices are costly when marginal purchases have positive social surplus (as for dresses) but can help in thin-margin categories where RSI already overprovides. Under CI, a higher premium also reduces coverage take-up and thereby aggregate returns.

A Data Construction

This appendix describes the construction of our analysis sample from raw transaction records. The process involves three main stages: (i) filtering the sample, (ii) classifying RSI schemes and imputing missing values, and (iii) aggregation, filtering and smoothing.

A.1 Sample Filtering

We restrict the sample to transactions from 2017 through 2023. To ensure clean identification of purchase and return behavior, we apply several exclusion criteria. First, we drop orders containing multiple products or quantities, as these introduce complications such as product complementarity and reduced marginal return costs. Second, we exclude orders lacking a logistic ID, which typically indicates cancellation before shipment. Third, we remove transactions with non-zero shipping fees (reflecting premium delivery requests), partial refunds for minor product defects, and obvious non-products such as shipping-fee-only or promotional giveaway listings.

Additionally, we drop transactions occurring on days when the seller temporarily switches RSI schemes for promotional events. The platform requires sellers to offer RI to participate in promotions (e.g., Singles' Day), so a seller may switch from CI to RI, or vice versa, for a single day within a month. Removing these promotion days ensures that the sample reflects the store's usual RSI scheme rather than event-driven deviations.

A.2 RSI Scheme Classification and CI Imputation

RSI scheme classification.
Missing insurance fields are ambiguous: under RI they may reflect ineligibility, under CI a decision not to buy coverage. We assign schemes using complete records (both premium and coverage $\Rightarrow$ RI; coverage only $\Rightarrow$ CI), propagate the day-level majority label to other rows that day, treat long runs of consecutive missing records as CI, and drop remaining ambiguous cases and confirmed ineligibility under RI (1.8% of RI transactions).

CI coverage and premium imputation.
Since both premium and coverage under CI depend on consumer location, we model the premium as a linear function of coverage, $I^c(k_i) = ak_i + b$, with coefficients calibrated from 2025 Taobao data. We assume agents do not internalize the impact of current return behavior on future premiums. For transactions classified under CI in our sample, we impute coverage from the nearest same-province observation by date and impute premiums from this linear fit because premiums are not recorded for 2017–2023.[35]

[35] We treat the 2025 premium–coverage slope as stable; it implies premiums of roughly 60% of coverage, consistent with older public documentation (e.g. `https://zhuanlan.zhihu.com/p/270357719`).

A.3 Aggregation, Filtering and Smoothing

Table 1 pools the filtered sample by RSI policy. List prices are computed in two steps: we first take the mean transaction price within each category and calendar month, then pool those category–month means by policy, weighting each cell by its transaction count. Premiums, coverage, and return rates pool transactions directly by policy.

The logic proceeds in order. First, we *aggregate* transactions to coarse categories and construct category–month measures of *sales* (transaction counts) and *prices*: within each category and calendar month we sum sales and take the mean transaction price. Second, we *screen* categories on average monthly sales, keeping only those with at least fifty transactions per month on average; this step leaves the dress, tops, pants, sweater, two-piece, and outerwear categories, which map to products j in the model. Third, we *smooth* category-level monthly sales and prices with a five-month centered moving average, and we *resample* transactions so that the number of transactions in each category–month matches the corresponding smoothed sales. Fourth, we define *market size* at the category $\times$ *season* level (spring through winter), because apparel demand is highly seasonal. For each category–season, market size is twice the maximum monthly sales within that category–season; we then form monthly purchase probabilities as smoothed sales relative to that market size. Fifth, we *filter out* category–months whose implied purchase probability is below 5%, so very thin cells do not drive estimation.

When the store switches return policy within a period, we compute the share of transaction time spent under each scheme and scale the observed scheme-specific transactions to represent the full period under that scheme.

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Online Appendix

B Return-Policy Extensions

B.1 Return-Penalty Discount: Equilibrium

Section 5.4.4 introduces return-penalty discount (RPD). Buyers choose a *discount* branch with list price $p - d$ and forfeiture share $\phi \in [0, 1]$ of $p - d$ on return, or a *full-price* branch with list price p and a full product refund. On both branches the buyer bears $\kappa + k_i$ when returning; on the discount branch the buyer recovers $(1 - \phi)(p - d)$.

Expected surplus on the two branches is

$$CS_i^f = \frac{1}{\sigma} \ln\left(e^{\sigma(v_i - p)} + e^{-\sigma(\kappa + k_i)}\right), \tag{28}$$

$$CS_i^d = d - p + \frac{1}{\sigma} \ln\left(e^{\sigma v_i} + e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}\right). \tag{29}$$

The consumer prefers the discount branch when $CS_i^d \geq CS_i^f$, which defines

$$\tilde{v}^{RPD} = -(\kappa + k_i) + \frac{1}{\sigma} \ln\left(\frac{e^{\sigma(p-d)} - e^{\sigma(1-\phi)(p-d)}}{1 - e^{-\sigma d}}\right). \tag{30}$$

Purchase thresholds are

$$\hat{v}^d = \frac{1}{\sigma} \ln\left(e^{\sigma(p-d)} - e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}\right), \tag{31}$$

$$\hat{v}^f \equiv p + \frac{1}{\sigma} \ln\left(1 - e^{-\sigma(\kappa+k_i)}\right). \tag{32}$$

The consumer purchases when $\max\{CS_i^d, CS_i^f\} \geq 0$. Both branches are active when $\tilde{v}^{RPD} \geq \hat{v}^f$, or equivalently (when $1 - e^{-\sigma d} > 0$)

$$\frac{e^{\sigma(p-d)} - e^{\sigma(1-\phi)(p-d)}}{1 - e^{-\sigma d}} \geq e^{\sigma p}\left(e^{\sigma(\kappa+k_i)} - 1\right). \tag{33}$$

Conditional on purchase,

$$q_i^d = \frac{e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}}{e^{\sigma v_i} + e^{\sigma[(1-\phi)(p-d)-(\kappa+k_i)]}}, \tag{34}$$

$$q_i^f = \frac{e^{-\sigma(\kappa+k_i)}}{e^{\sigma(v_i-p)} + e^{-\sigma(\kappa+k_i)}}. \tag{35}$$

With branch indicators D_i^d, D_i^f and aggregates D^d, R^d, D^f, R^f, the seller chooses (p, d) to maximize $\pi^{RPD} = \pi^d + \pi^f$, where $\pi^d = (p - d - \theta^c)D^d - [(1 - \phi)(p - d) + h]R^d$ and $\pi^f = (p - \theta^c)D^f - (p + h)R^f$.

Individual social surplus on each branch is

$$S_i^d \equiv CS_i^d + (p - d - \theta^c) - q_i^d\big[(1-\phi)(p-d) + h\big], \tag{36}$$

$$S_i^f \equiv CS_i^f + (1 - q_i^f)(p - \theta^c) - q_i^f(\theta^c + h). \tag{37}$$

Define $CS^{RPD} \equiv \mathbb{E}_{\nu_i,k_i}[D_i^d\, CS_i^d + D_i^f\, CS_i^f]$ and $W^{RPD} \equiv \mathbb{E}_{\nu_i,k_i}[D_i^d\, S_i^d + D_i^f\, S_i^f]$. Counterfactuals through Section 8.4 set $\phi = 1$; Section 8.5 varies ϕ when full forfeiture overshoots on the return margin.

B.2 Optimal Refund Policy

Section 5.4.2 defines the benchmark. With list price p and return refund $-h$,

$$CS_i^{\text{rs}} = -p + \frac{1}{\sigma}\ln\left(e^{\sigma\nu_i} + e^{-\sigma(h+\kappa+k_i)}\right), \tag{38}$$

$$\hat{\nu}^{\text{rs}} = \frac{1}{\sigma}\ln\left(e^{\sigma p} - e^{-\sigma(h+\kappa+k_i)}\right), \tag{39}$$

where $\hat{\nu}^{\text{rs}}$ solves $CS_i^{\text{rs}} = 0$ when $e^{\sigma p} > e^{-\sigma(h+\kappa+k_i)}$. The consumer purchases when $\nu_i \geq \hat{\nu}^{\text{rs}}$ and returns with $q_i^{\text{rs}} = q_i^*$ from (19). Aggregates are

$$D^{\text{rs}} \equiv \mathbb{E}_{\nu_i,k_i}\big[\mathbf{1}(\nu_i \geq \hat{\nu}^{\text{rs}})\big], \tag{40}$$

$$R^{\text{rs}} \equiv \mathbb{E}_{\nu_i,k_i}\big[\mathbf{1}(\nu_i \geq \hat{\nu}^{\text{rs}})\, q_i^{\text{rs}}\big], \tag{41}$$

and the seller chooses p to maximize $\pi^{\text{rs}}(p) = (p - \theta^c)D^{\text{rs}}$, with $S_i^{\text{rs}} = CS_i^{\text{rs}} + (p - \theta^c)$.

C Scrap-Value Sensitivity

Section 8.6.1 of the main text varies scrap value to $0.65\, p_t$ and $0.85\, p_t$. For each calibration, we repeat the normalized policy decomposition in Figure 6 and the partial-RPD grid in Figure 8.

C.1 Scrap value $0.65\ p_t$

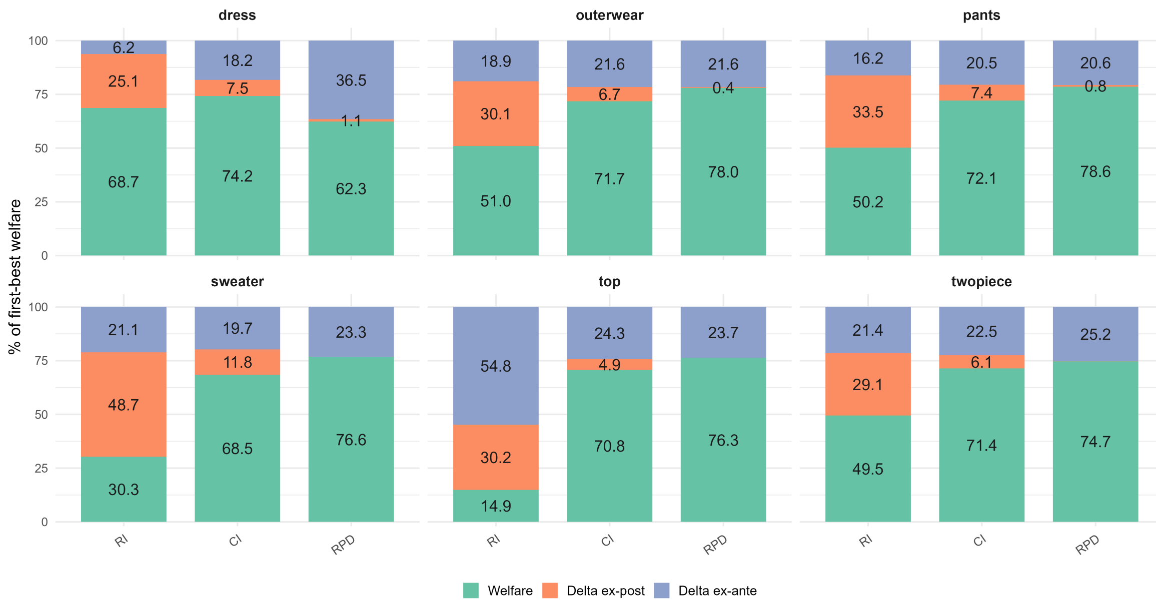


Figure 10: Welfare decomposition (RI, CI, RPD) as a share of first best. Scrap $0.65\ p_t$.

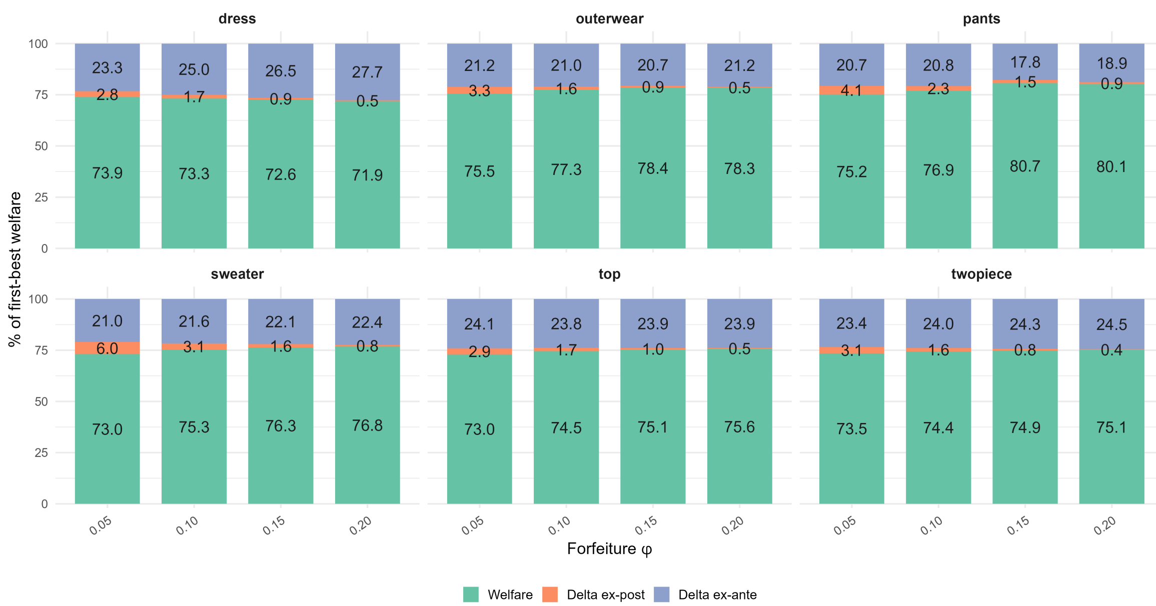


Figure 11: Partial-forfeiture RPD by ϕ and category. Scrap $0.65\ p_t$.

C.2 Scrap value $0.85\ p_t$

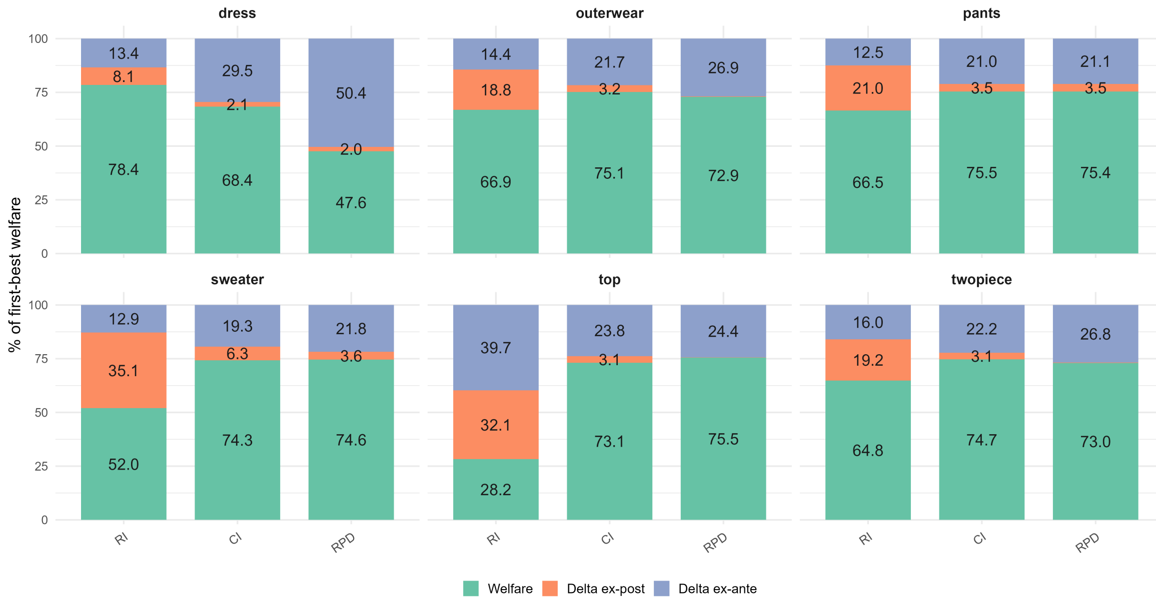


Figure 12: Welfare decomposition (RI, CI, RPD) as a share of first best. Scrap $0.85\ p_t$.

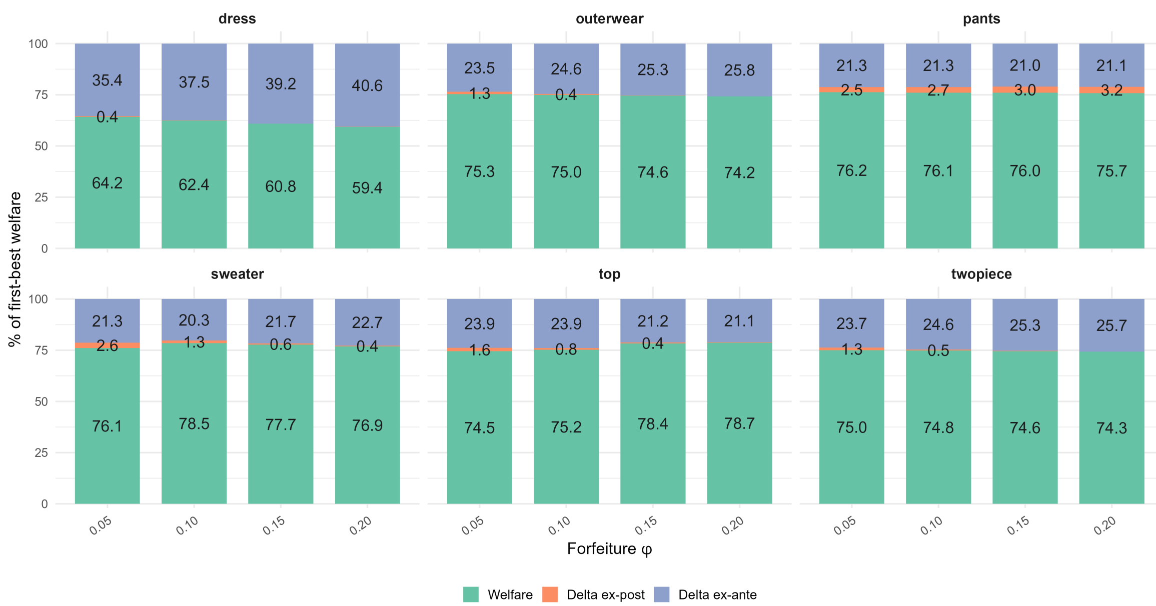


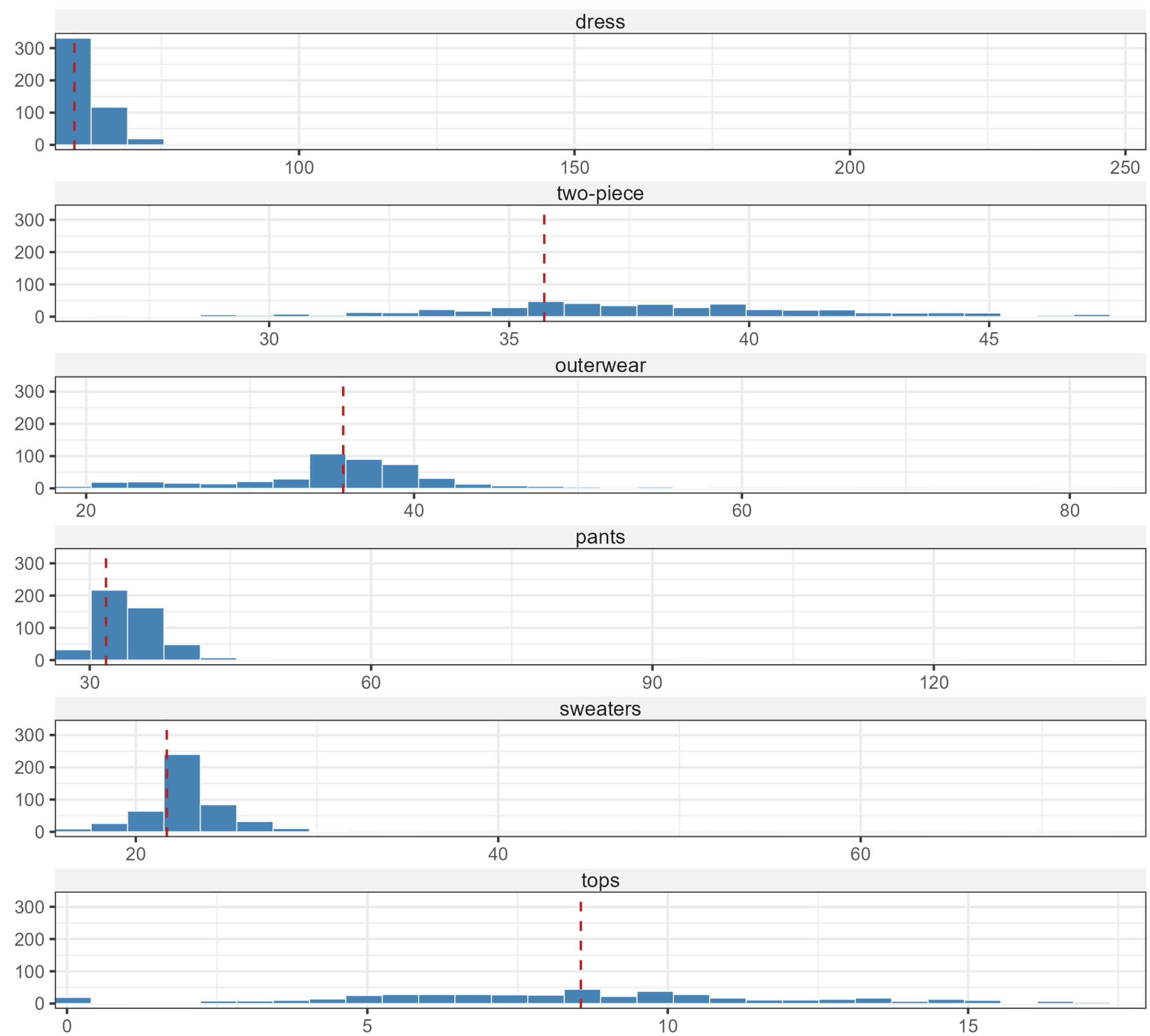
Figure 13: Partial-forfeiture RPD by ϕ and category. Scrap $0.85\ p_t$.

D Bootstrap Inference

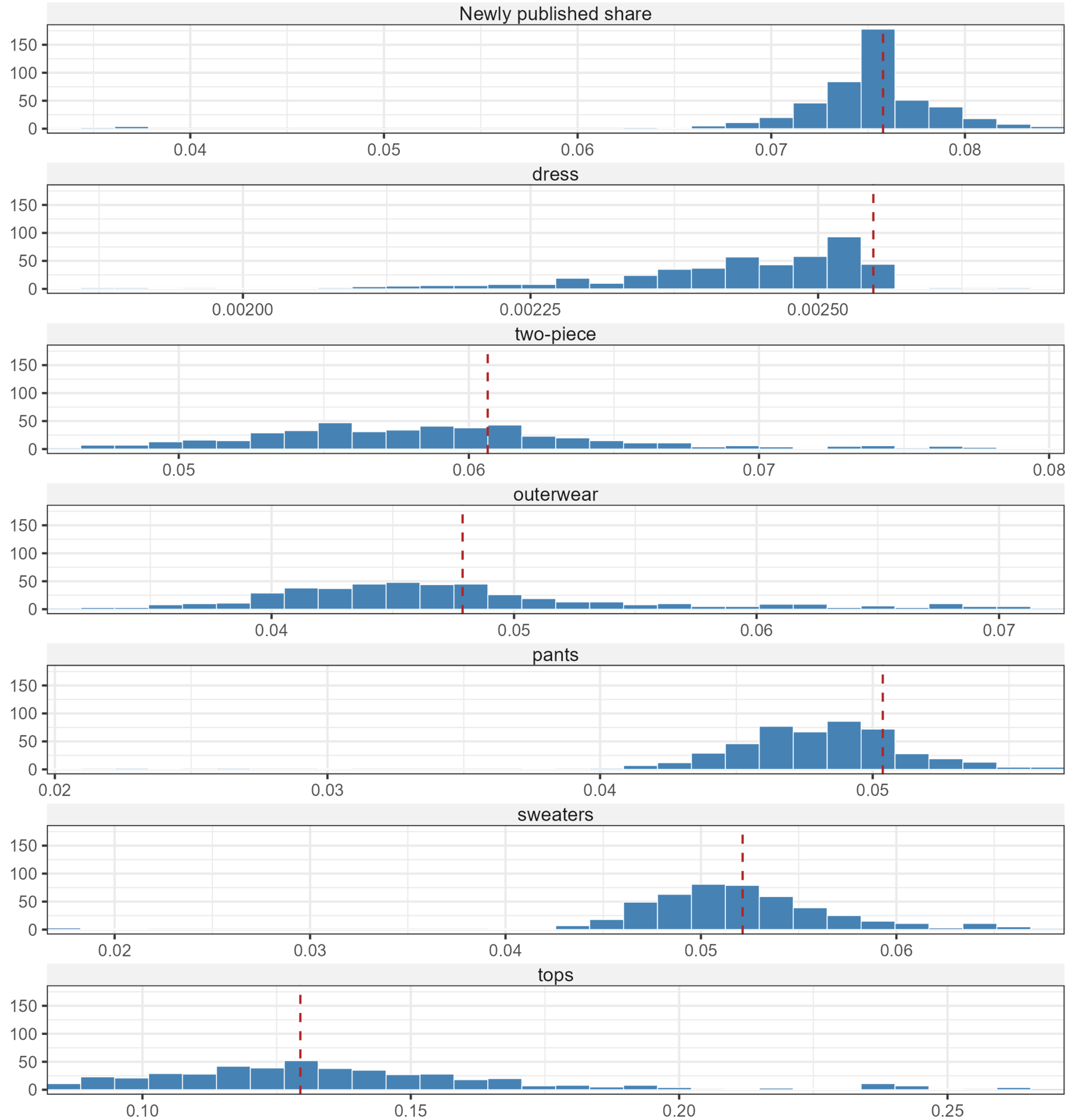
Each panel plots bootstrap replicates in 30 bins. For readability, the horizontal axis spans the 2.5th–97.5th percentile range of the draws displayed in that panel. Parameter panels apply this range separately to each coefficient. Welfare panels use a common range pooled across return policies within the category and scale outcomes relative to first-best welfare. Bootstrap standard errors and confidence intervals in the main text use all replicates. The dashed vertical line marks the full-sample point estimate for the object in that panel.

D.1 Point estimates

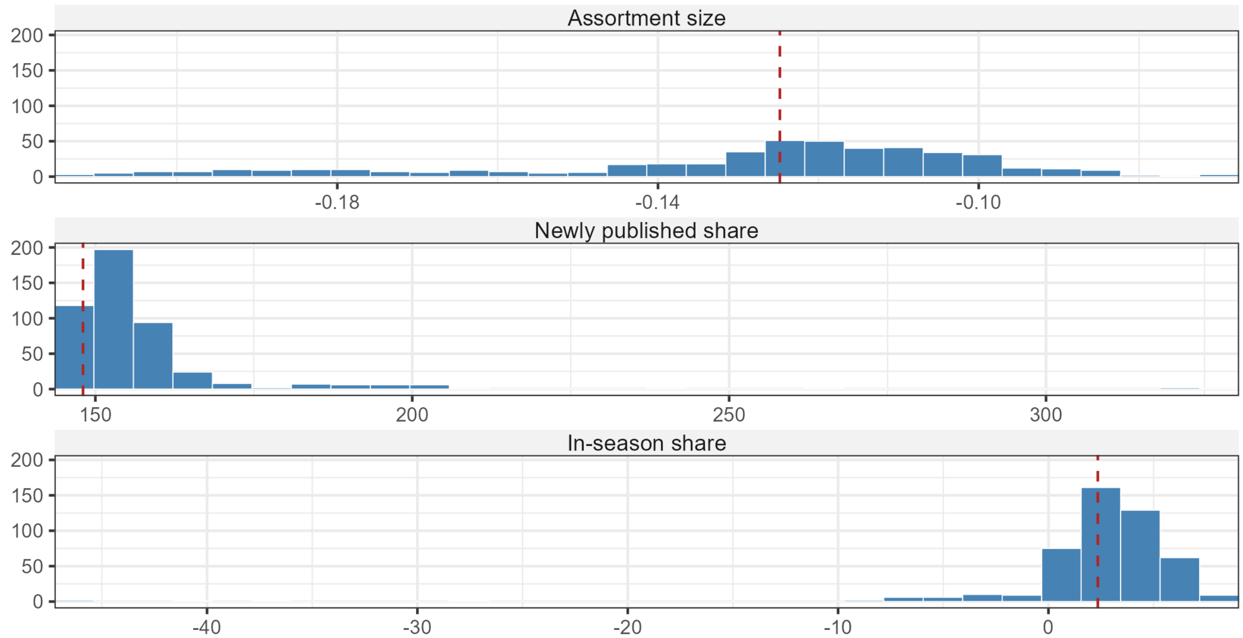
Ex-ante uncertainty (β^{ω})



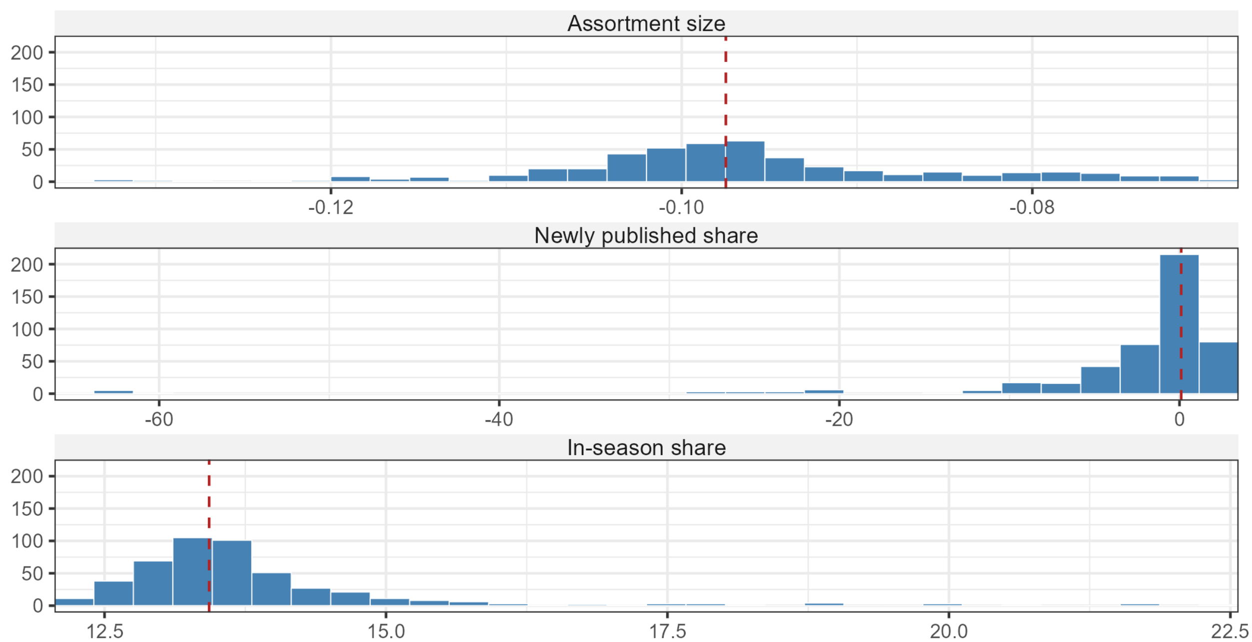
Ex-post uncertainty (β^{σ})



Mean utility (β^{μ})

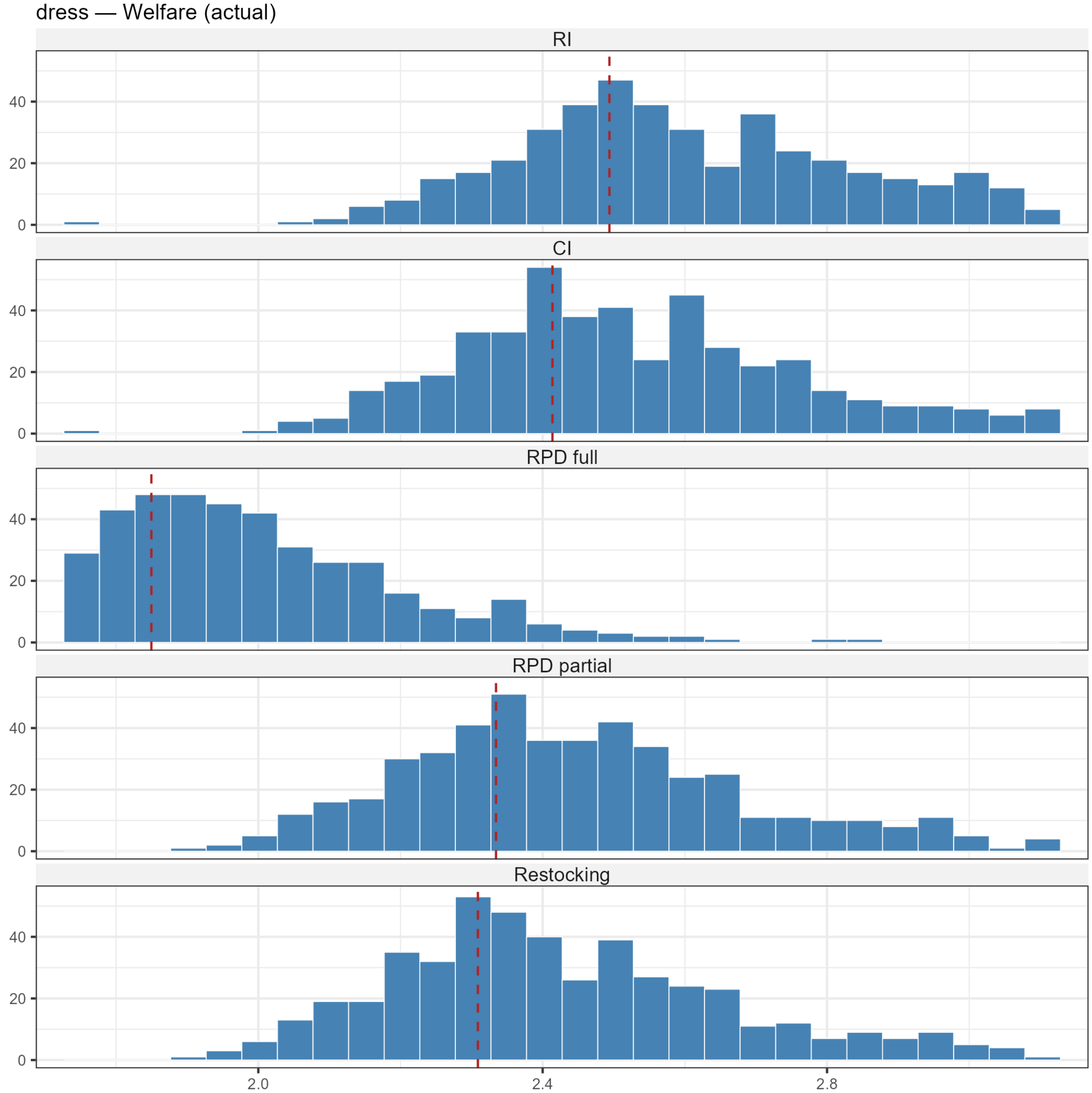


Production cost (θ^{c})

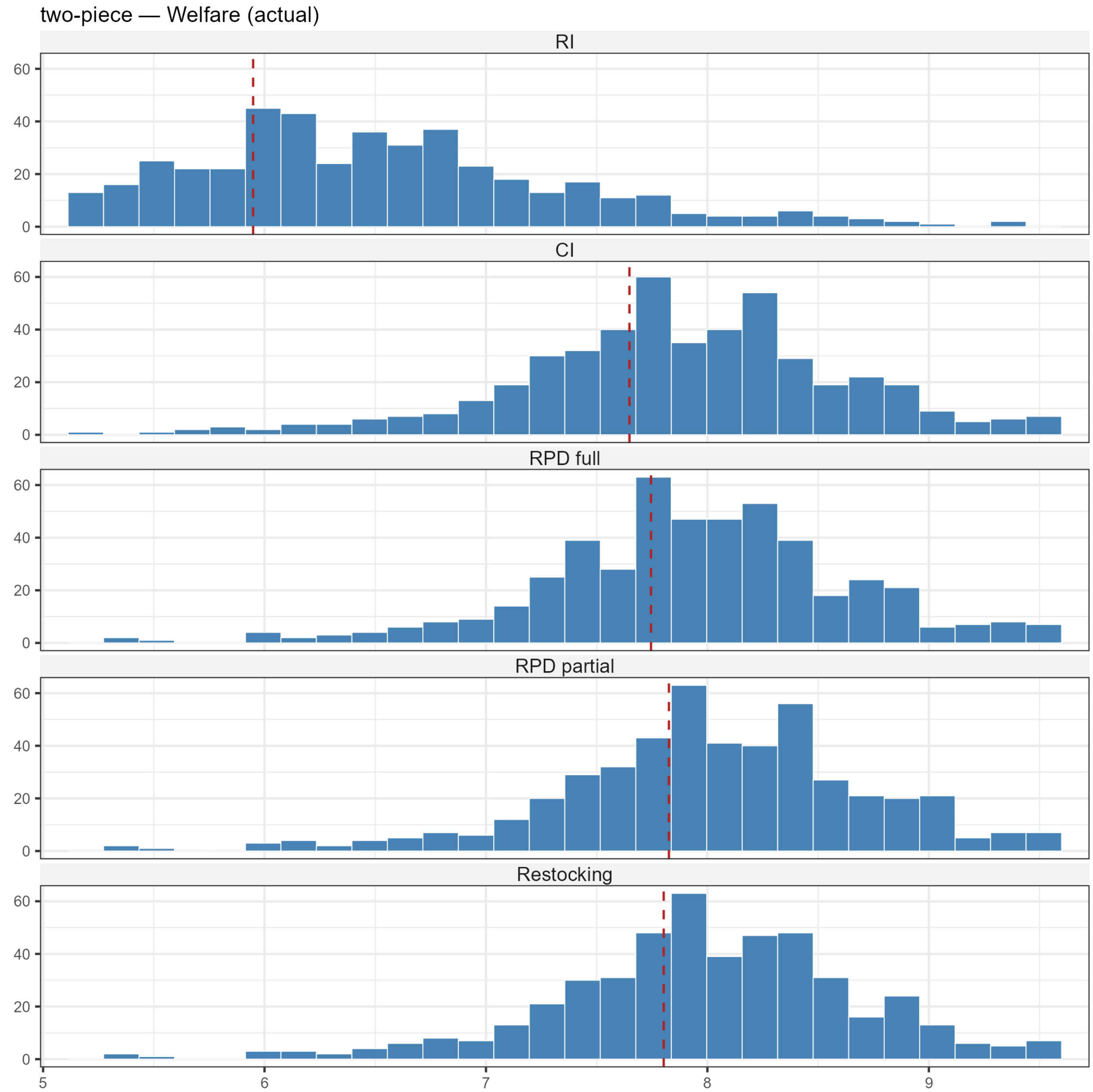


D.2 Actual welfare

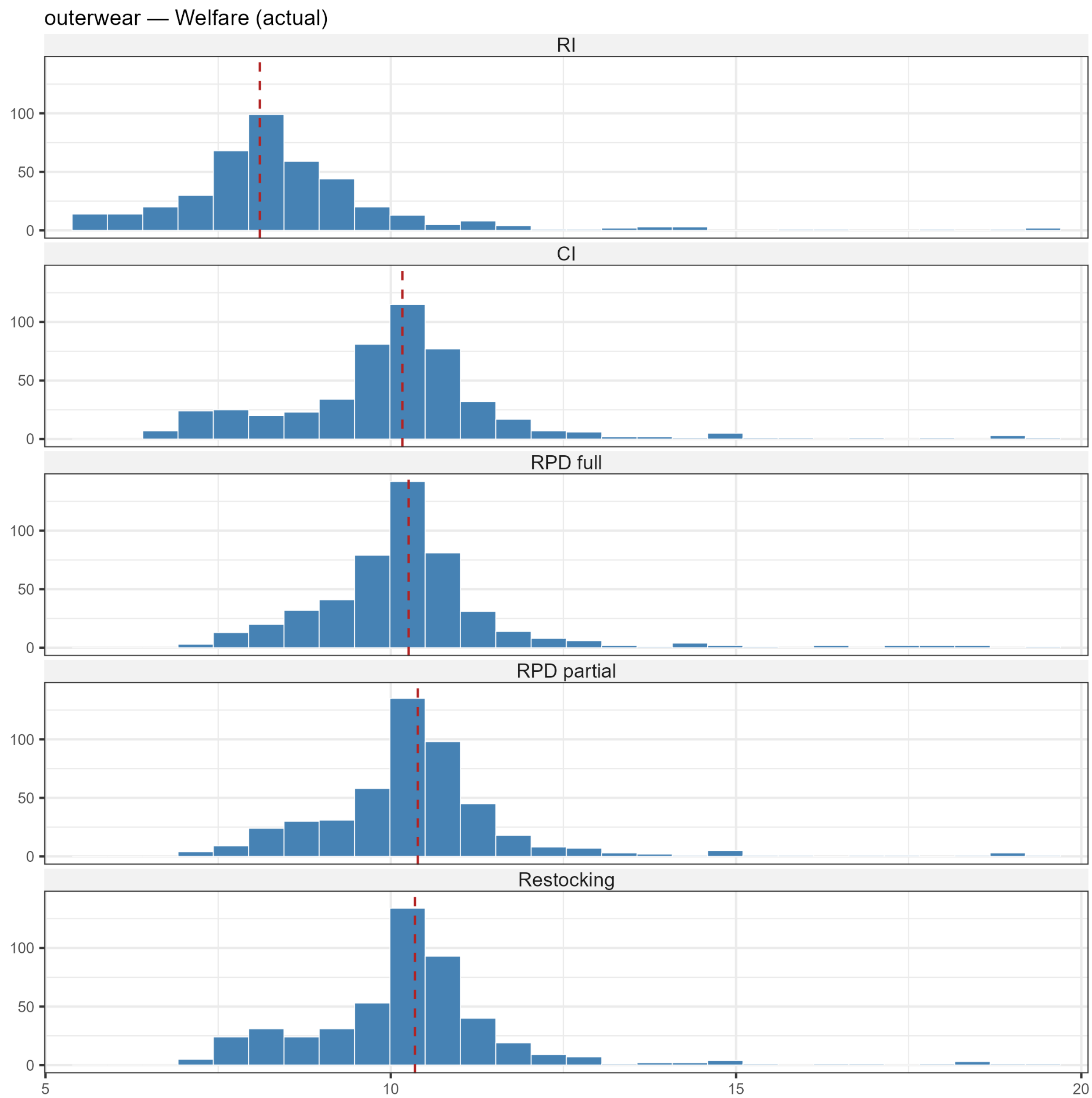
Dress



Two-piece

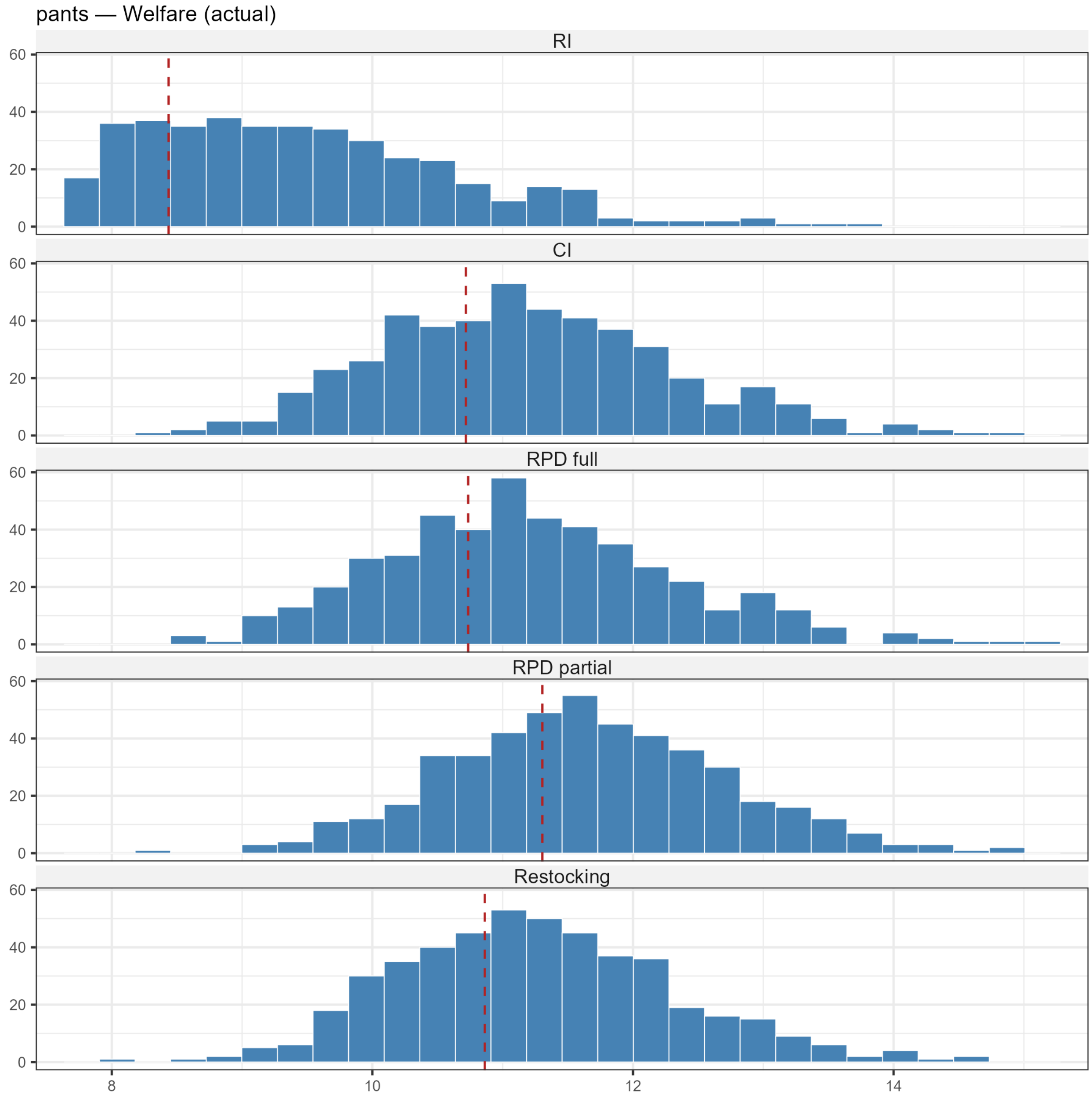

two-piece — Welfare (actual)
RI
CI
RPD full
RPD partial
Restocking
0
20
40
60
5
6
7
8
9

Outerwear

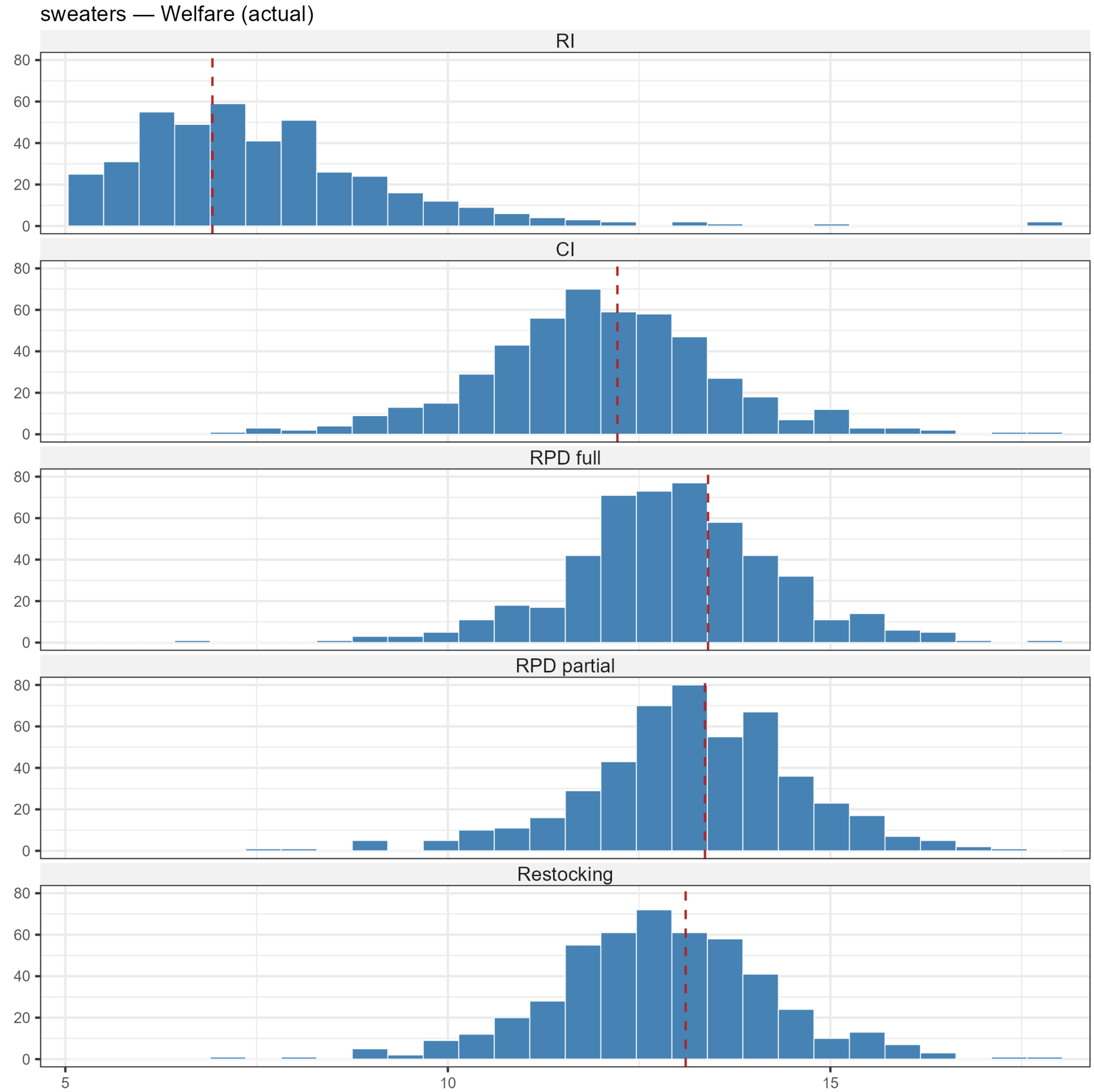


outerwear — Welfare (actual)
RI
CI
RPD full
RPD partial
Restocking
100
50
0
5
10
15
20

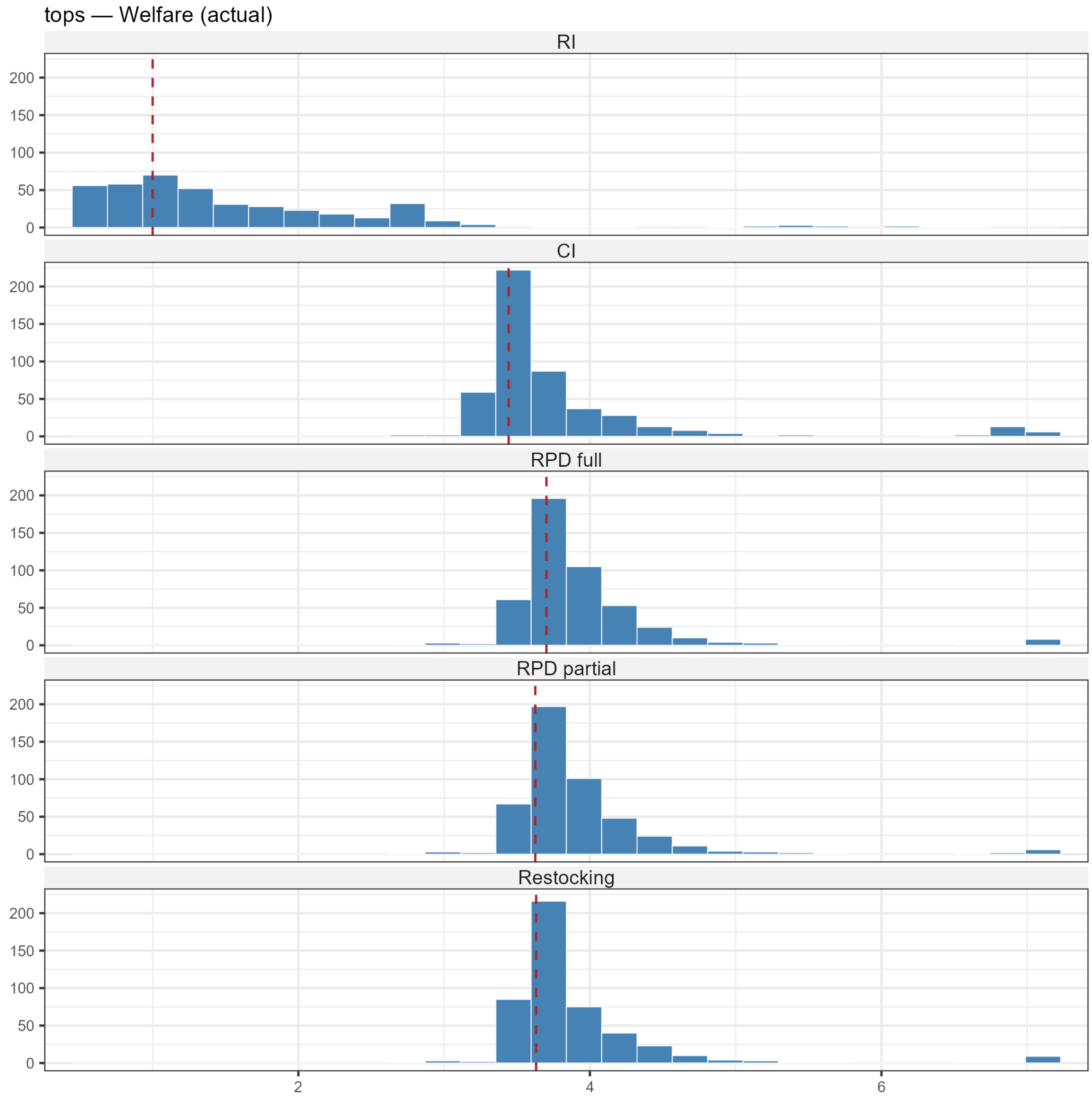
Pants


pants — Welfare (actual)
RI
CI
RPD full
RPD partial
Restocking
60
40
20
0
8
10
12
14

Sweaters


sweaters — Welfare (actual)
RI
CI
RPD full
RPD partial
Restocking
80
60
40
20
0
5
10
15

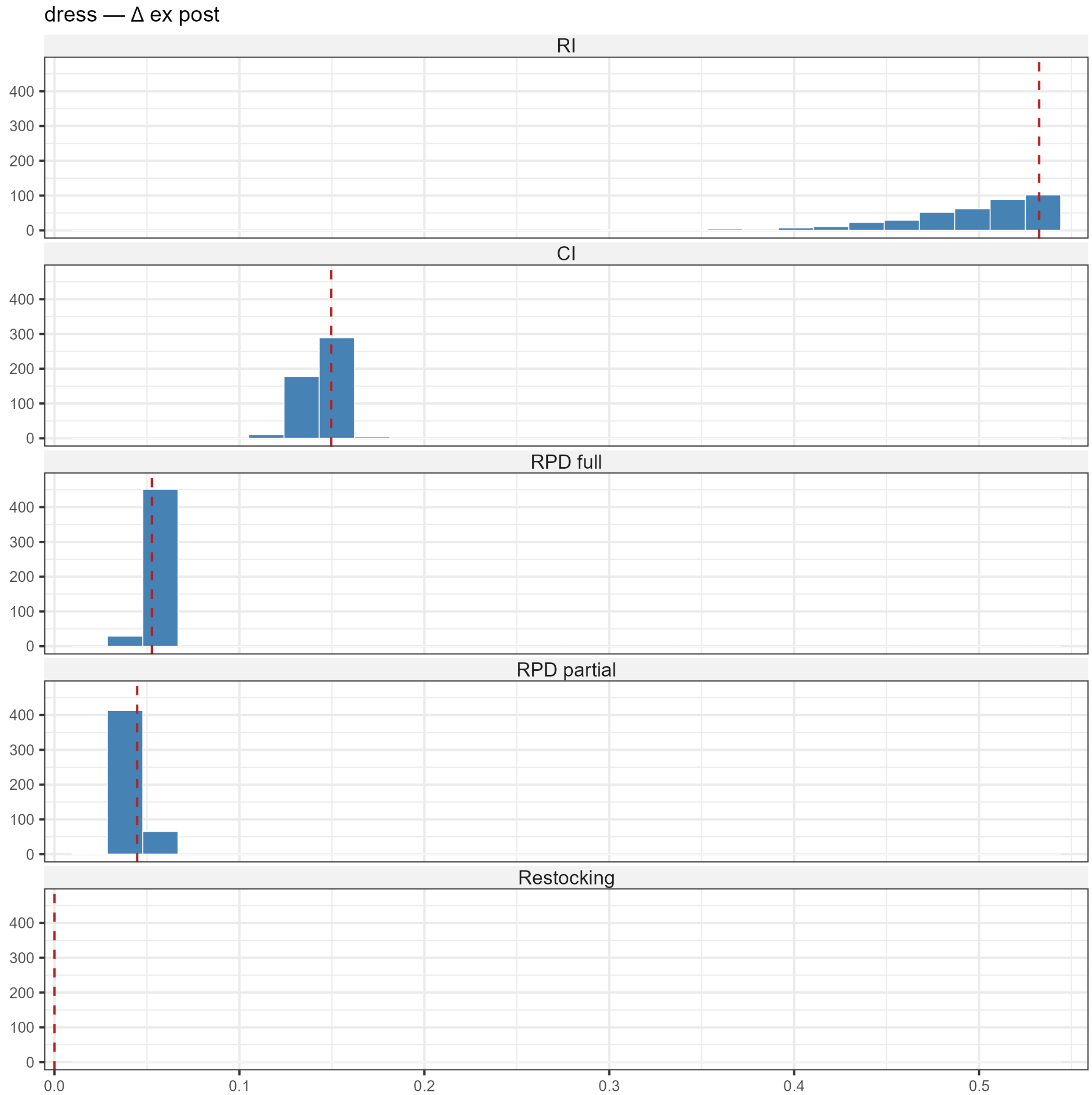
Tops



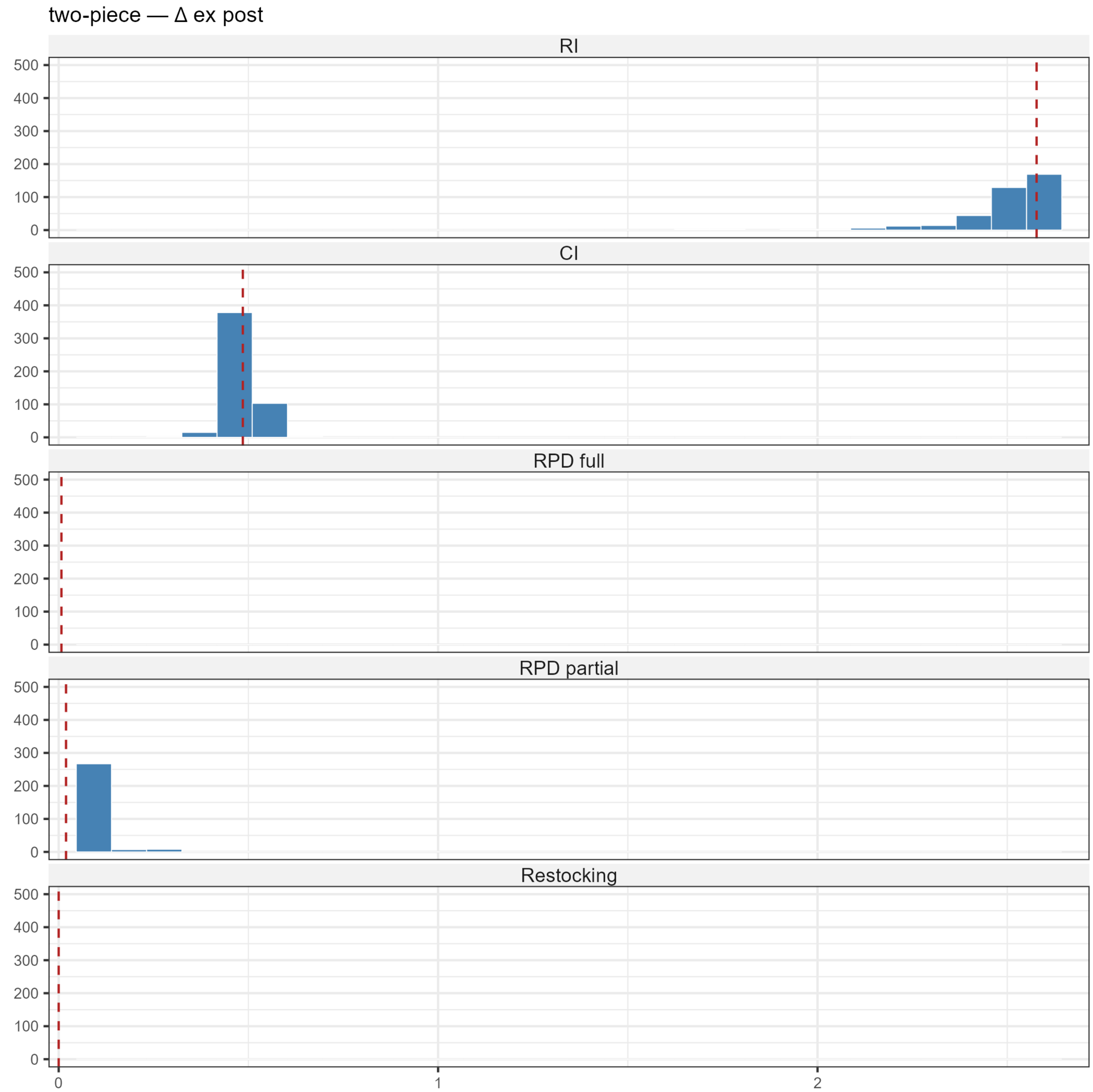
tops — Welfare (actual)
RI
CI
RPD full
RPD partial
Restocking
0
50
100
150
200
2
4
6

D.3 Δ ex-post

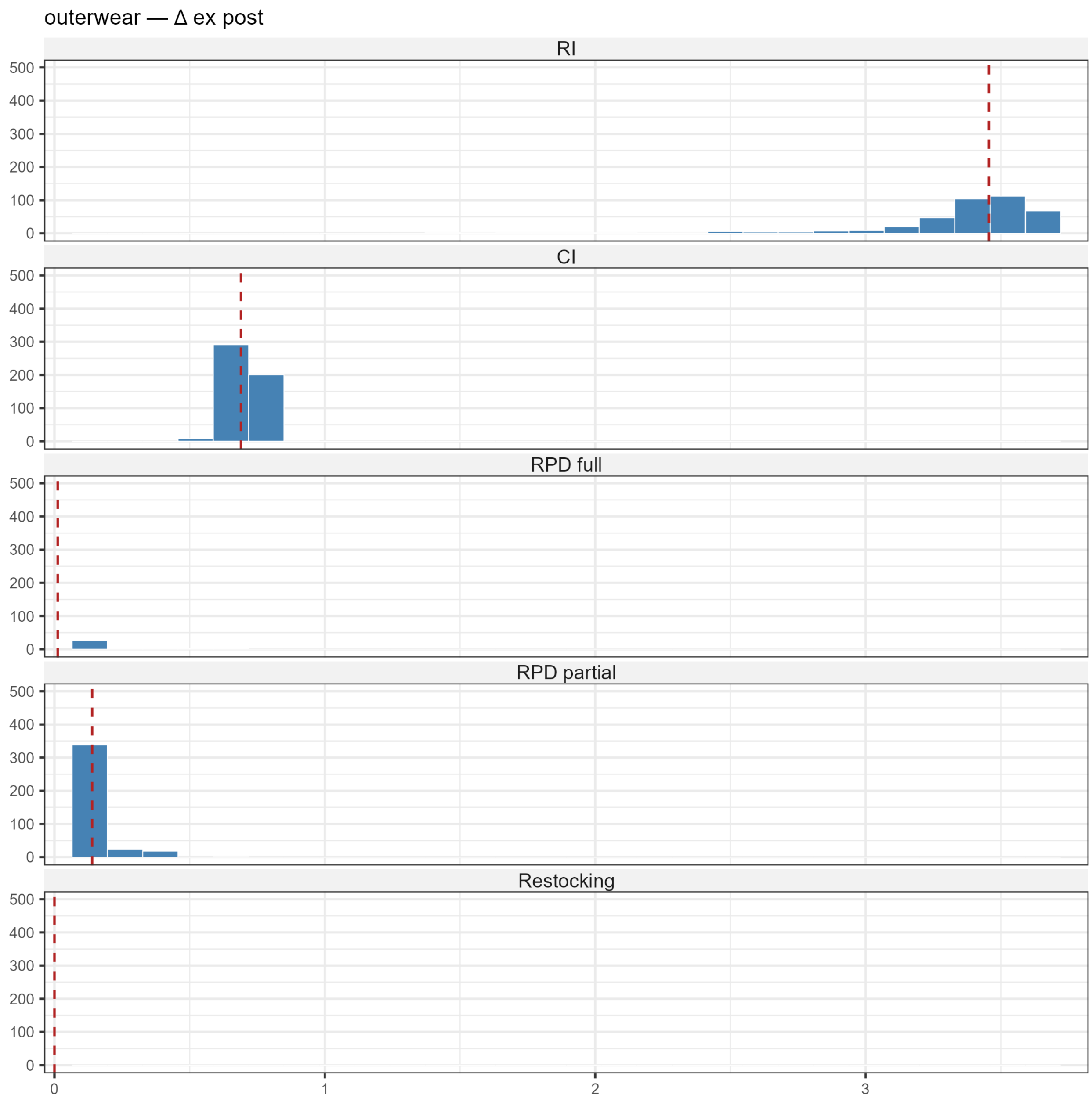
Dress



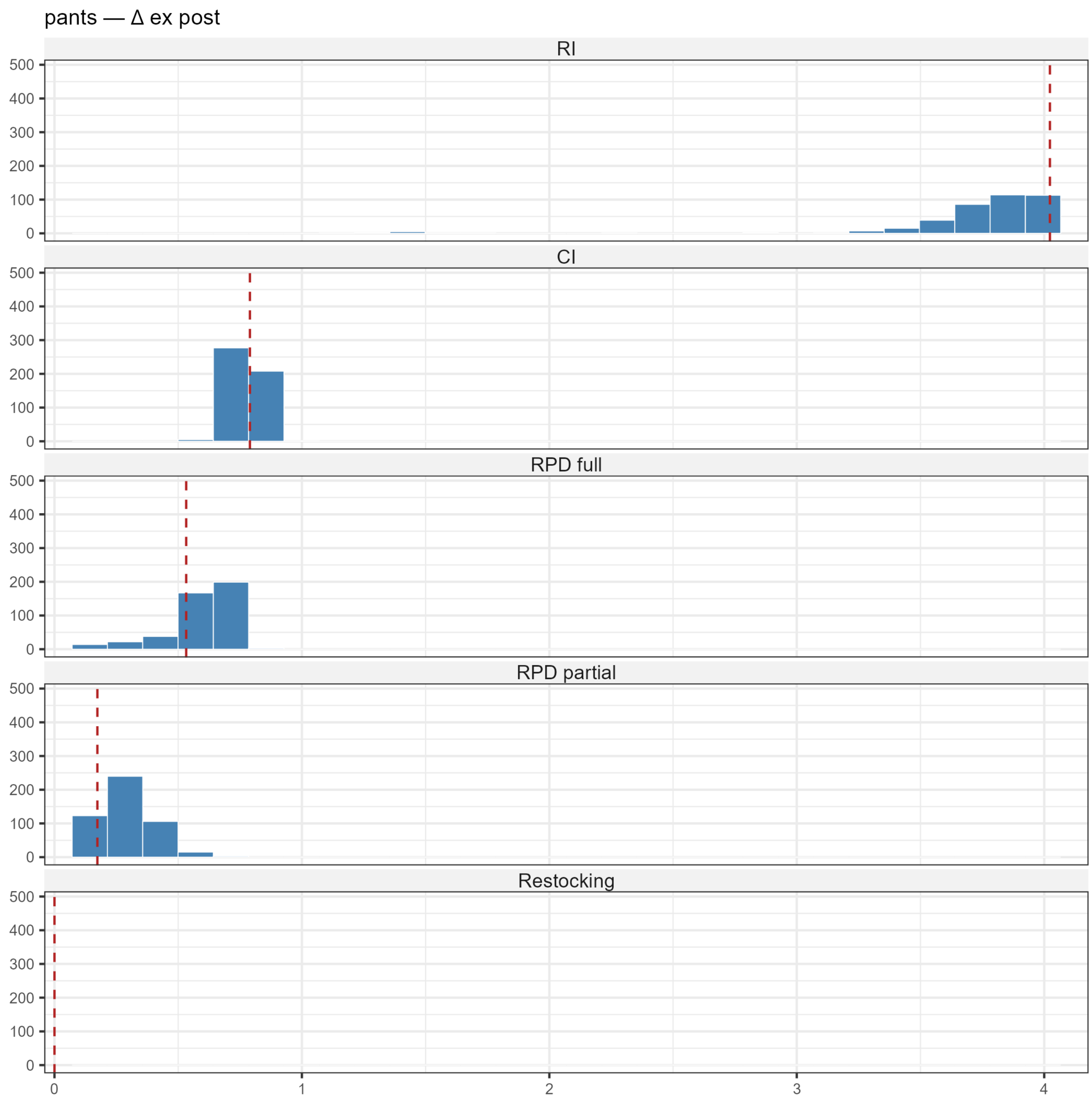
Two-piece


two-piece — Δ ex post
RI
CI
RPD full
RPD partial
Restocking
500
400
300
200
100
0
0
1
2

Outerwear

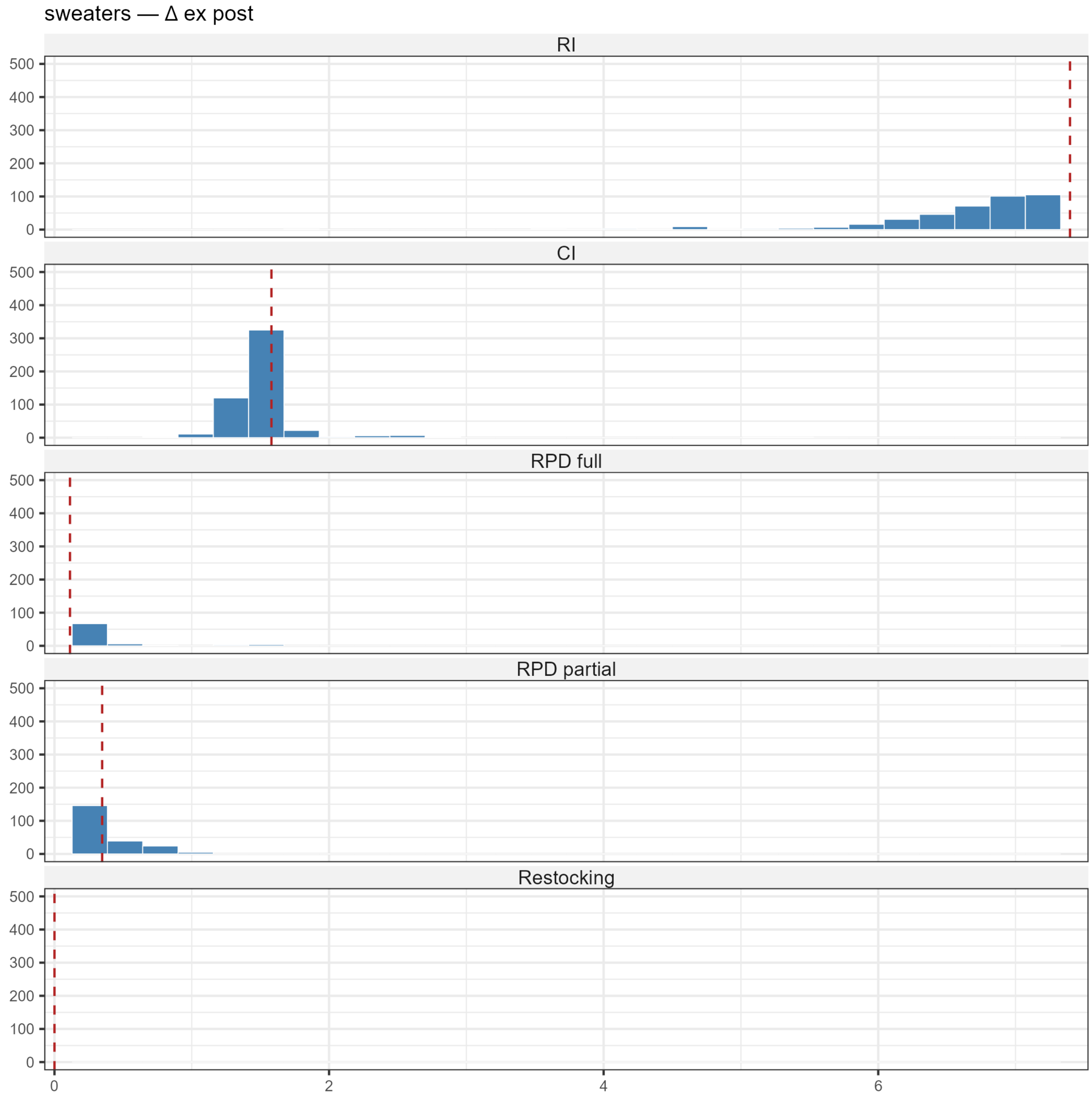

outerwear — Δ ex post
RI
CI
RPD full
RPD partial
Restocking
500
400
300
200
100
0
0
1
2
3

Pants

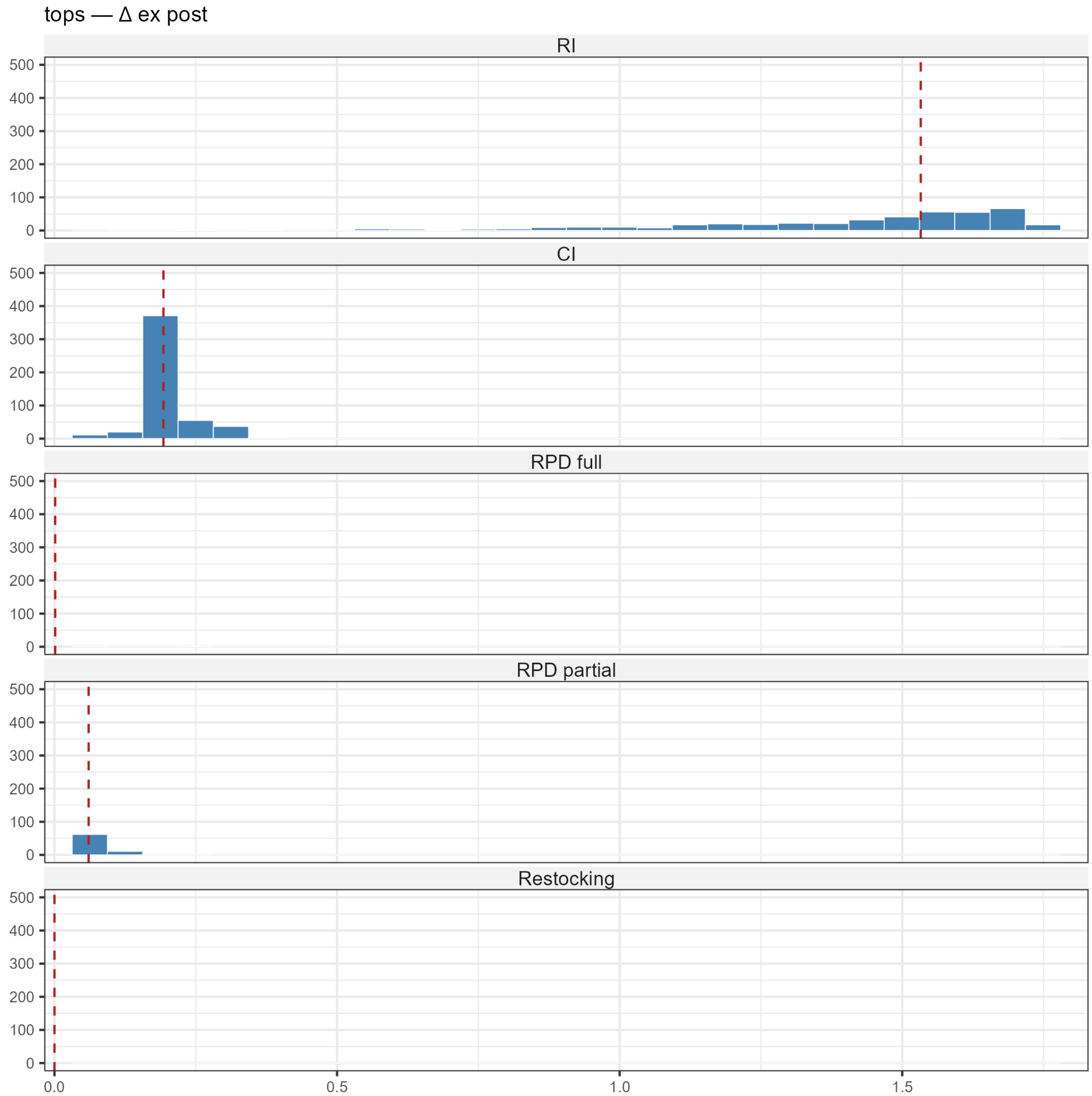


pants — Δ ex post
RI
CI
RPD full
RPD partial
Restocking
500
400
300
200
100
0
0
1
2
3
4

Sweaters

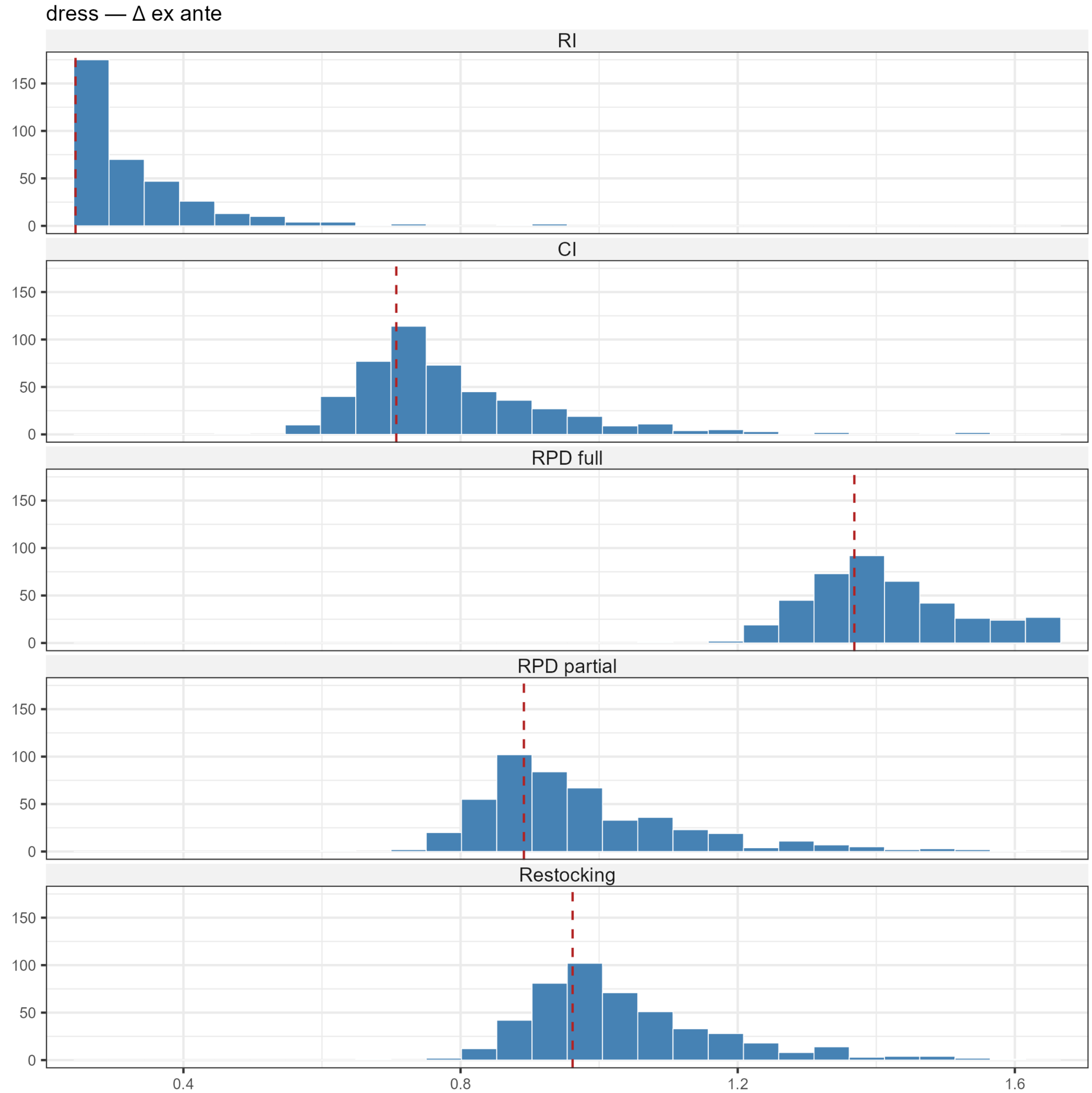

sweaters — Δ ex post
RI
CI
RPD full
RPD partial
Restocking
500
400
300
200
100
0
0
2
4
6

Tops

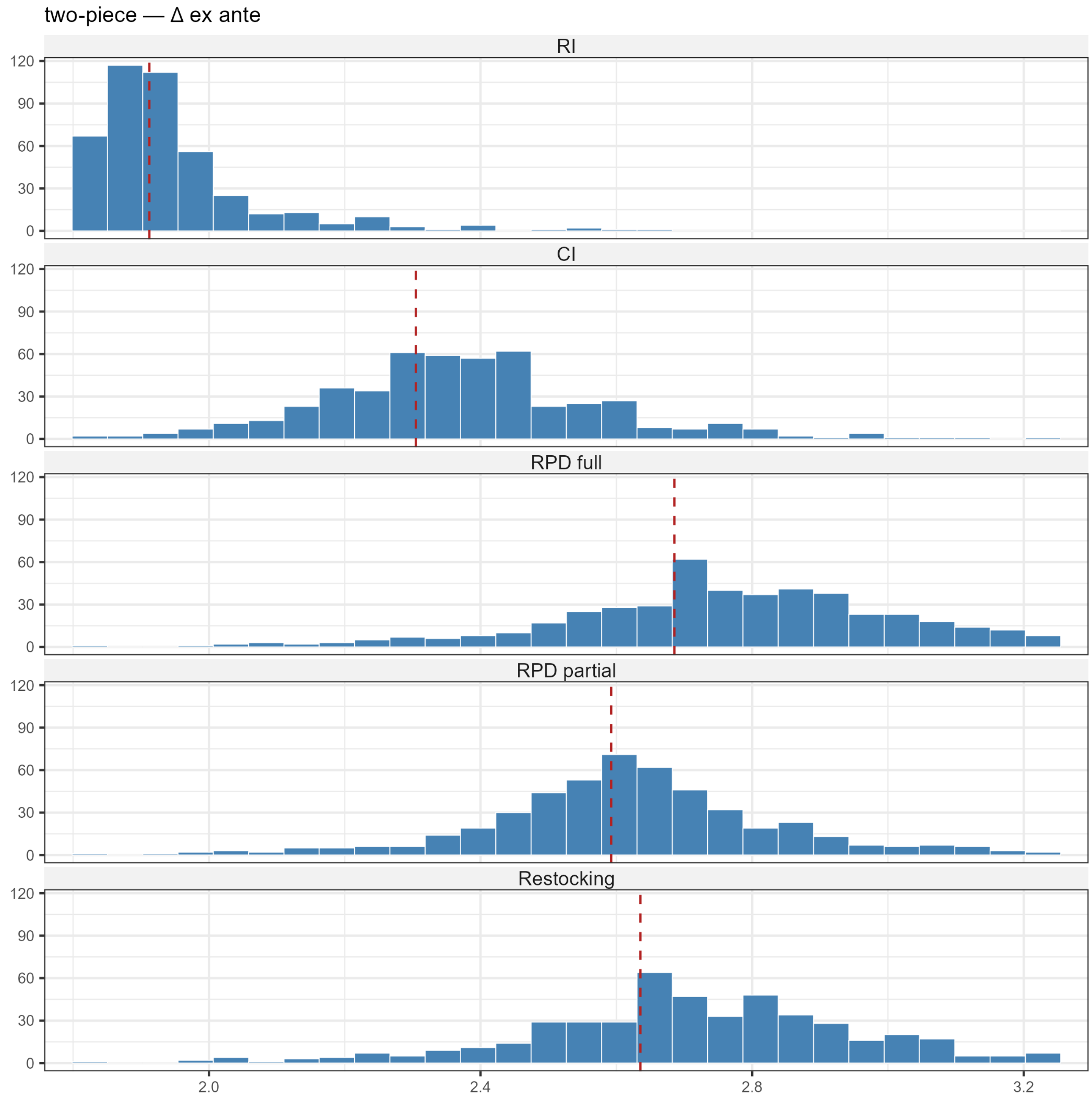

tops — Δ ex post
RI
CI
RPD full
RPD partial
Restocking
500
400
300
200
100
0
0.0
0.5
1.0
1.5

D.4 Δ ex-ante

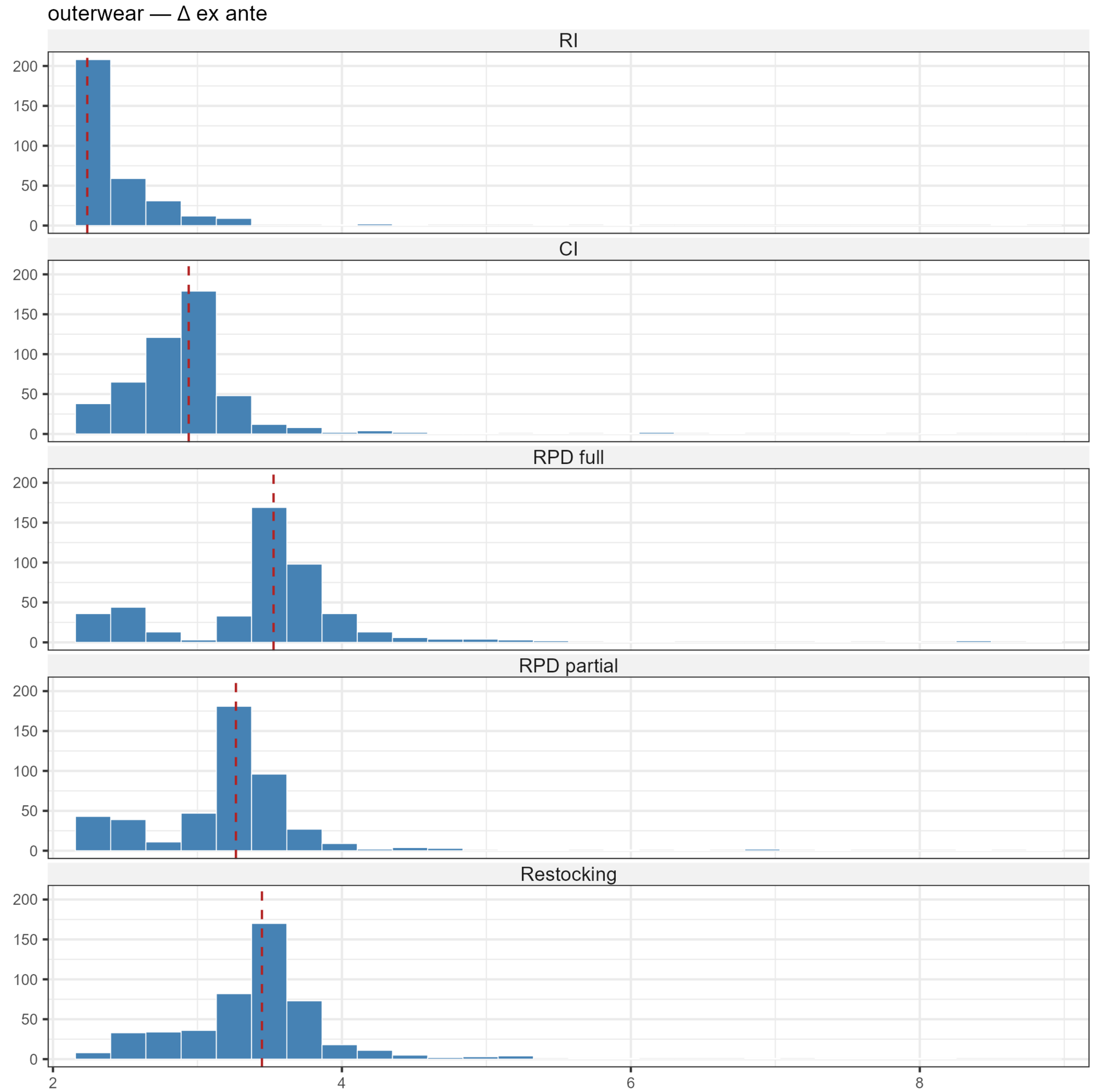
Dress



Two-piece

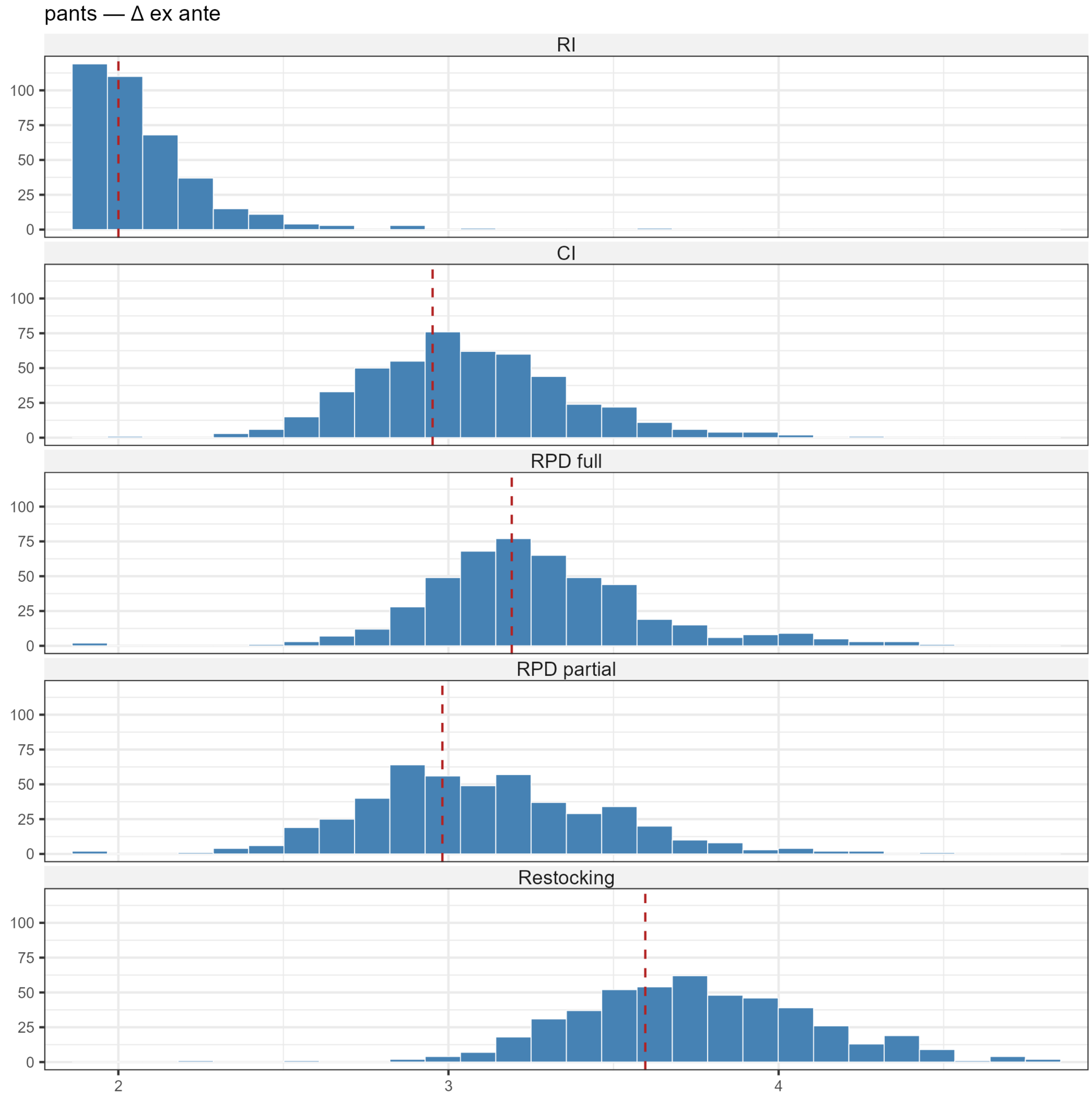

two-piece — Δ ex ante
RI
CI
RPD full
RPD partial
Restocking
120
90
60
30
0
2.0
2.4
2.8
3.2

Outerwear



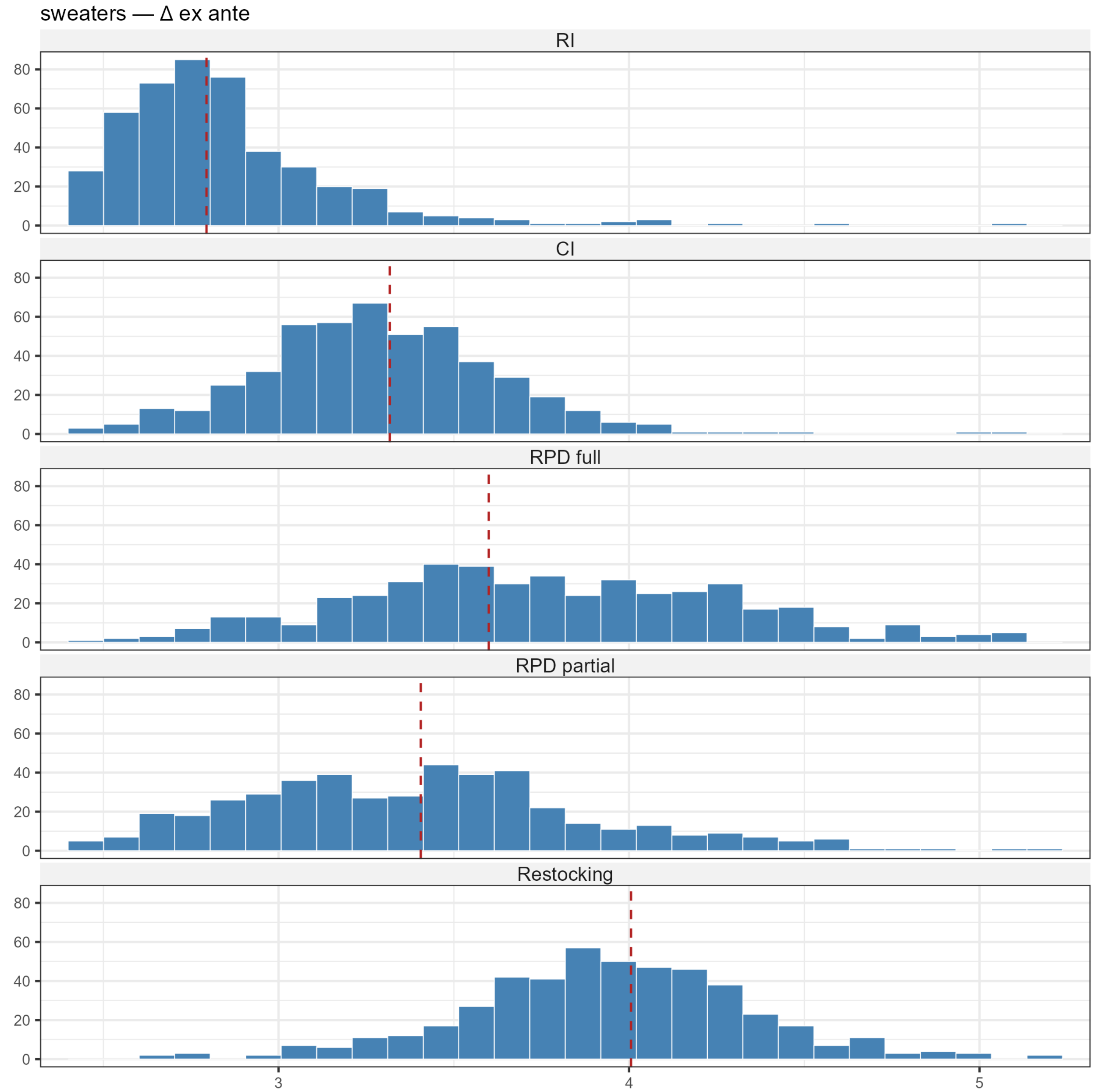
outerwear — Δ ex ante
RI
CI
RPD full
RPD partial
Restocking
200
150
100
50
0
2
4
6
8

Pants

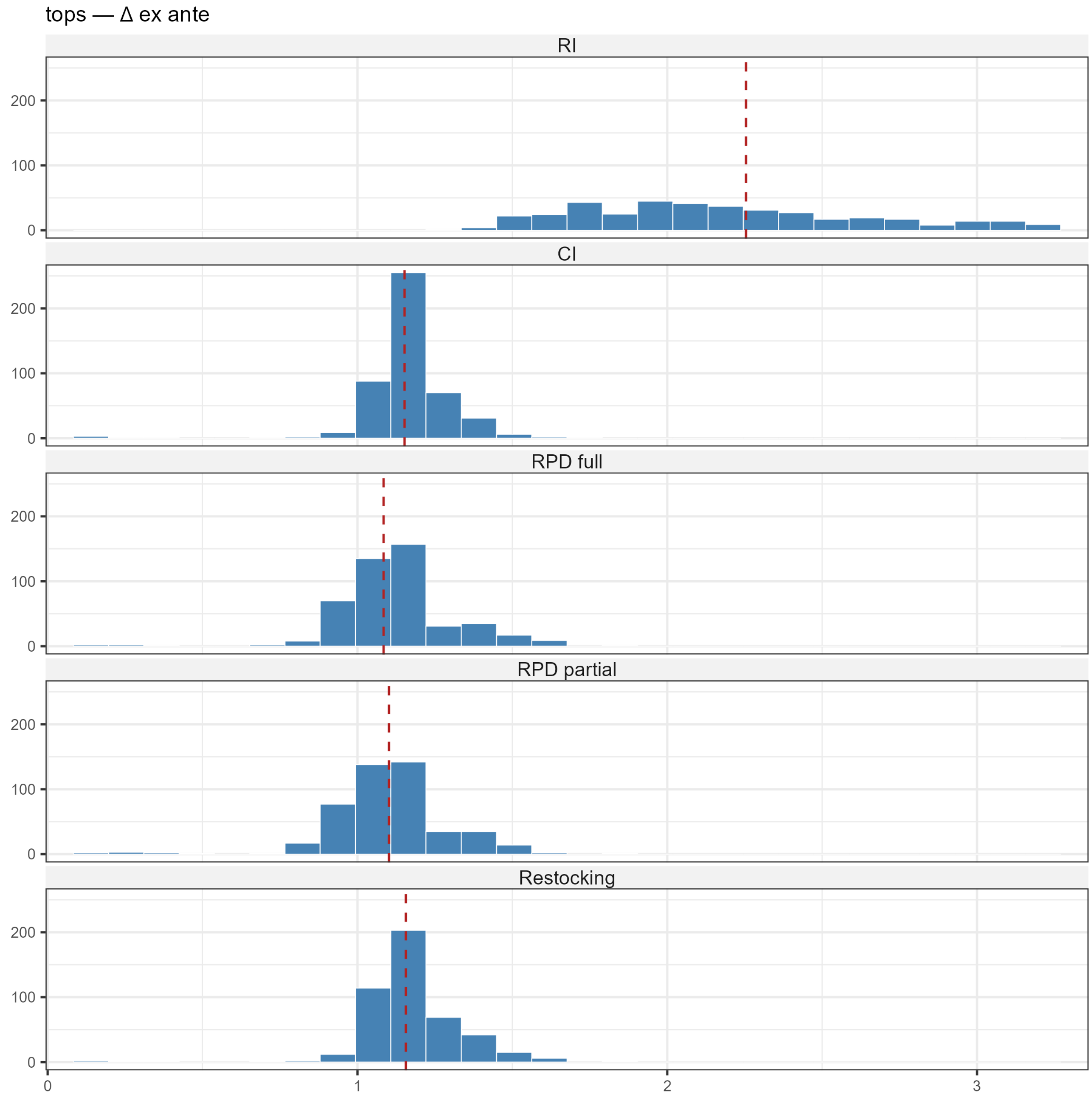


pants — Δ ex ante
RI
CI
RPD full
RPD partial
Restocking
100
75
50
25
0
2
3
4

Sweaters


sweaters — Δ ex ante
RI
CI
RPD full
RPD partial
Restocking
80
60
40
20
0
3
4
5

Tops



D.5 Pairwise win rates

Table 7: Pairwise bootstrap win rates for actual welfare.

	RI	CI	RPD full	RPD partial	Restocking
Dress					
RI	—	1.00	1.00	1.00	1.00
CI	0.00	—	1.00	1.00	1.00
RPD full	0.00	0.00	—	0.00	0.00
RPD partial	0.00	0.00	1.00	—	0.89
Restocking	0.00	0.00	1.00	0.11	—
Two-piece					
RI	—	0.00	0.00	0.00	0.00
CI	1.00	—	0.25	0.00	0.05
RPD full	1.00	0.75	—	0.00	0.00
RPD partial	1.00	1.00	1.00	—	1.00
Restocking	1.00	0.95	1.00	0.00	—
Outerwear					
RI	—	0.04	0.04	0.04	0.04
CI	0.96	—	0.30	0.05	0.10
RPD full	0.96	0.70	—	0.07	0.20
RPD partial	0.96	0.95	0.93	—	0.99
Restocking	0.96	0.90	0.80	0.01	—
Pants					
RI	—	0.02	0.02	0.02	0.00
CI	0.98	—	0.24	0.02	0.21
RPD full	0.98	0.76	—	0.01	0.25
RPD partial	0.98	0.98	0.99	—	0.98
Restocking	1.00	0.79	0.75	0.02	—
Sweaters					
RI	—	0.02	0.04	0.03	0.03
CI	0.98	—	0.04	0.03	0.04
RPD full	0.96	0.96	—	0.15	0.64
RPD partial	0.97	0.97	0.85	—	0.99
Restocking	0.97	0.96	0.36	0.01	—
Tops					
RI	—	0.00	0.00	0.00	0.00
CI	1.00	—	0.00	0.00	0.00
RPD full	1.00	1.00	—	0.62	0.50
RPD partial	1.00	1.00	0.38	—	0.56
Restocking	1.00	1.00	0.50	0.44	—

Notes: Each off-diagonal cell is the share of bootstrap draws in which the row policy has higher actual welfare than the column policy (non-missing pairs only). Diagonal cells are blank.